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OECD ECONOMIC SURVEYS

2002-2003

Finland

OECD

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF FINLAND (2001)

THE LAND

Area (1 000 km^2, 1999)	338.1
of which:	
Agricultural	27.0
Forests	229.4
Lakes	34.1

Major cities (thousand inhabitants, end 2001):	
Helsinki	559.7
Espoo	216.8
Tampere	197.8
Vantaa	179.9

THE PEOPLE

Population (thousand, end 2001)	5 195
Number of inhabitants per km^2 of land area	17.1
Net natural increase (thousand)	7.6
Net migration (thousand)	5.8

Labour force (thousand)	2 605
Employment (thousand)	2 367
Employment (% of total):	
Agriculture, forestry and fishing	5.7
Industry and construction	21.0
Services	73.3

PARLIAMENT AND GOVERNMENT

Composition of Parliament (number of seats):	
Social Democratic Party	51
Centre Party	48
National Coalition Party (conservatives)	46
Left Alliance	20
Green League	11
Swedish People's Party	11
Christian League	10
Other	3
Total	200

Government, number of ministers from:	
Social Democratic Party	7
National Coalition Party (conservatives)	7
Left Alliance	2
Swedish People's Party	2
Total	18

Last general elections: 21 March 1999

PRODUCTION AND PUBLIC SECTOR

Gross domestic product (billion EUR)	136.0
GDP per head (EUR)	26 175
Gross fixed capital investment:	
% of GDP	19.8
Per head (EUR)	5 171

Public consumption (% of GDP)	21.0
General government (% of GDP):	
Current and capital expenditure	44.3
Current revenue	49.2

FOREIGN TRADE

Exports of goods and services (% of GDP)	40.1
Main exports (% of total):	
Metals, machinery and transport equipment	27.9
Electrical and optical equipment	27.5
Wood, pulp and paper	26.6
Other goods	17.9

Imports of goods and services (% of GDP)	31.7
Main imports (% of total):	
Intermediate goods	39.2
Consumer goods	24.7
Capital goods	24.4
Energy	11.7

THE CURRENCY

Monetary unit: Euro

Currency units per USD, average of daily figures:	
Year 2002	1.061
December 2002	0.982

N*ote*: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of Member countries.

The economic situation and policies of Finland were reviewed by the Committee on 9 January 2003. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 21 January 2003.

The Secretariat's draft report was prepared for the Committee by David Turner, Philip Hemmings and Seija Parviainen under the supervision of Peter Hoeller.

The previous Survey of Finland was issued in December 2001.

Assessment and recommendations

A number of policy challenges need to be addressed to ensure continued strong performance

Finland's medium-term growth performance has been among the best in the OECD. However, to maintain this position a number of major challenges will need to be addressed. Firstly, population ageing in Finland will occur sooner and more rapidly than in most other OECD countries. This raises the question as to how stringent fiscal policy will need to be over the coming years to ensure the long-run sustainability of public finances. It also underlines the importance of raising employment, especially among older workers. Priority should be given to reducing unemployment more generally, which is still considerably above the OECD average. Assessment of these issues needs to take account of a wide-ranging pension reform that will begin to be implemented in 2005, as well as the desirability of reducing the tax burden on labour that is still high by international comparison. Secondly, the growth of the economy over the last decade has been increasingly dependent on the contribution from the ICT-sector, but there are major uncertainties about the outlook for this sector and a large contribution to future growth cannot be taken for granted. Various policy initiatives are therefore needed to enhance growth prospects as well as ensuring they are more broadly based. Finally, policies that foster sustainable development will need to focus on cost-effective solutions to achieve the ambitious environmental targets that have already been established.

The short-term recovery is more advanced than elsewhere in the euro area

Despite the severity of the downturn in 2001, Finland is likely to recover more quickly than most other euro area countries; indeed, it could be the only one that shows stronger growth in 2002 than in 2001. Inflation has remained close to the euro area average, though dipping significantly below

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it towards the end of 2002. The recent cut of 50 basis points in euro area short-term interest rates will provide additional support to demand, and be particularly helpful in bolstering demand from Finland's main export markets. Output growth is projected to strengthen from 1½ per cent in 2002 to around 3 per cent in 2003 as international demand gains momentum. Provided the pick-up in global activity continues, the boost to exports and a revival of investment should lead to an acceleration of output growth to nearly 4 per cent in 2004. Nevertheless, the continued presence of slack and a moderate central wage agreement for 2003 and 2004 should ensure that inflation remains subdued.

The major uncertainty concerns the strength of international demand

The short-term outlook hinges critically on a continued pick-up in export growth. Much will depend on the performance of ICT-based exports, which weathered the industry-wide downturn relatively well, with Nokia substantially increasing its share of the market in mobile telephone handsets. However, prospects for the industry in 2003 and beyond depend on a positive international reaction of consumers to third-generation mobile telephony. If the pick-up in international demand is delayed, there is a risk that employment will suffer with knock-on effects on domestic demand.

Current objectives for the government surplus should be retained, although undershooting due to cyclical weakness should be tolerated

In order to prepare for the fiscal implications of the imminent ageing challenge, Finland currently has an ambitious objective of running a central government surplus of 1½ to 2 per cent of GDP to reduce debt. In 2000 and 2001 this objective was comfortably achieved but in 2002 the surplus fell below 1 per cent of GDP. Much of this deterioration can be explained by the cyclical downturn, and most of the remainder by the loss of tax revenues related to the slump in asset prices. With the latter unlikely to recover significantly in the near future and continued weakness in corporate tax revenues, the surplus objective is unlikely to be met in 2003. Even by 2004, with output still likely to be below potential and with a prospective loss of revenues due to compliance with EU harmonisation of indirect taxes on cars and alcohol, the surplus objective may not be reached. In these circumstances, temporary undershooting of the surplus target should be tolerated insofar as it reflects cyclical

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weakness. If the target is not achieved even when output has recovered to trend, then the onus should be on further expenditure restraint rather than increased taxation. Indeed, a further objective should be sufficient expenditure restraint to provide room for also reducing taxes, particularly on labour. Although the recent pension reform is expected to improve fiscal sustainability, the uncertainties related to its effects argue in favour of keeping the 1½ to 2 per cent central government surplus target.

The pension reform has important elements that enhance sustainability…

The pension reform that will be implemented as from 2005 includes a number of striking features that should enhance the sustainability of the system. In particular there will be a sharp rise in accrual rates from the age of 63 and a mechanism to adjust the generosity of the system to increasing life expectancy and so contain cost pressures. There will also be an actuarially fairer basis for calculating pensions based on lifetime earnings rather than just over the last ten years of working life. Old-age pension reform is accompanied by the reform of the so-called "unemployment pipeline", whereby workers have been able to effectively retire at the age of 55, claiming unemployment benefit until they formally qualify for a pension. Together with earlier changes, the pension reform package is expected to raise the average age of retirement by 2-3 years.

… but higher contribution rates will still be required and potential flaws exist

While the reform is an important step towards enhancing the sustainability of the system over coming decades, more could have been achieved in this respect, particularly because some aspects of the reform appear overly generous. It implies, for instance, that the baby boom generation will not be strongly affected, in particular because the reform of the unemployment pipeline scheme applies only to those born in 1950 or later. In consequence much of the savings only become apparent after 2030. The package also includes a rise in accrual rates for those over 52, which serves no clear purpose insofar as these people would probably be mostly employed anyway. Pension entitlements will also accrue in the future during periods of non-employment (such as study). Overall, the average level of pensions is estimated to rise although the longevity adjustment will counteract this over time. With the expected

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lengthening of working life outweighing the impact of higher pensions, the pension reform is expected to halve the required increase in pension contributions by 2040. But they will still have to rise significantly (perhaps by more than 5 per cent of wages). Moreover, considerable uncertainty surrounds the estimated rise in employment rates that is crucial for the reforms to be effective. Of particular concern is that the abolition of early retirement schemes could lead to increased use of disability pensions, which could seriously undermine the reform. There is also uncertainty about the future rate of return on the pre-funded part of the pension system. Various proposals have been made to raise competition among the financial institutions that manage these funds, as well as to enhance their capacity to manage portfolio risks over a long-term horizon, but no decisions on these matters have yet been taken. Given these uncertainties as well as the rise in other ageing-related outlays, the pension reform package does not warrant a lowering of the current surplus in the social security account or a relaxation of the central government fiscal objective to at least the end of the decade.

Control of public expenditure should be improved…

The special chapter of this *Survey* focuses on government spending and identifies various areas, such as health and education, where significant efficiency gains could be reaped, suggesting scope for reducing real expenditure in the future. There has been slippage against the objective to hold central government spending constant in real terms at the 1999 level, and in 2002 real expenditure is expected to rise by a further 2 per cent. The authorities should consider reformulating the real expenditure target so as to avoid transparency problems, for example by explicitly agreeing how nominal government expenditures will be adjusted for inflation. Also interest payments on the public debt should be excluded from the target. More importantly, there is scope for improving the consistency between the annual budget process and the medium-term objectives, while spending control could be enhanced by mechanisms that have been fairly successful in other countries. In the United States, for example, increases in outlays for a particular programme could only be legislated once corresponding savings were identified elsewhere.

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… and efficiency gains reaped, notably in health and education

As in other Nordic countries, government spending is highly decentralised; municipalities are responsible for almost all health care and a large share of education. Outcomes are generally considered good in international comparison although there are concerns that it will be difficult to maintain quality at a relatively low cost in the future. There is generally good use of co-operative arrangements among municipalities although one of the most important systems – that used for hospitals – shows large variation in efficiency. In particular, municipalities need to be put into a stronger position as purchasers of hospital services, through a more focussed role and by ensuring they have greater expertise in the provision of medical services so as to strengthen their position in relation to the hospitals. Municipally provided public services would also be made more efficient if the government were to strengthen measures to encourage mergers between the many small rural municipalities as well as encourage greater use of co-operative arrangements to reap economies of scale. There is also scope for reductions in central government spending. For example, while educational achievements are very high, the time students take to complete tertiary education is often very long and this adds to the cost. Incentives to complete courses on schedule, while maintaining educational achievements, need to be strengthened.

Uncertainty for municipal funding should be reduced

Municipalities' independence is facilitated not only by considerable flexibility in how revenue is spent but also by some freedom to influence revenue itself. However, non-controllable fluctuations in tax revenue are large, principally due to the corporate income tax. This creates problems that are not helped, and even accentuated, by the equalisation scheme that itself operates with substantial lags that should be reduced further. Also, additional cuts in the share of the corporate income tax municipalities receive, compensated by larger block grants, would help to provide greater certainty for spending plans for those municipalities that are heavily dependent on this tax.

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Outsourcing has yet to be widely adopted and choice could be widened via vouchers

While Finland once led in the reform of public-management practices it is now visibly behind the leading countries in some areas, particularly with regard to outsourcing. The low level of outsourcing continues despite efforts by the authorities to ensure a level playing field between private-sector competitors and traditional "in-house" provision. The fact that outsourcing is not picking up suggests the authorities should continue efforts to overcome inertia in attitudes. The authorities should also follow-up on proposals to make greater use of voucher systems in elderly care as a means of enhancing both contestability and user choice and actively seek to widen the use of vouchers more generally.

More policy effort should be put into goal-setting and benchmarking in key spending areas

As elsewhere, Finland's ministries and departments have been encouraged to adopt modern management practices that entail making explicit statements about functions and policy goals so as to provide focus and clarity for spending objectives. While Finland has been more willing to develop these management techniques than some countries, the goal-setting exercises typically have neither explicit incentives or sanctions, nor much pressure coming from political or media attention. Benchmarking exercises that provide input and output indicators for public services that can be compared across providers are less developed than in some countries. Furthermore, there is reluctance to make the evaluations, such as those made for hospitals and schools, freely and widely available to the public. Benchmarking should be more systematic and regular in education and it should also be widened, particularly to health centres and other municipal services.

More jobs for the less-well educated are key for lowering structural unemployment

The Prime Minister recently set up a working group with a mandate to present, by the end of March, policies that would increase the employment rate to 75 per cent, which is 8 percentage points above the current level. Tax cuts in recent years and the pension reform are providing incentives to increase labour participation, although hiring incentives should also be raised. However achievement of the new employment rate target will require substantial further reforms. Most unemployment in Finland is structural, a major problem being a lack of jobs for the less skilled. This partly reflects the rigid and compressed wage structure, and

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high taxation and employer contribution rates. The high labour costs are reflected in a relatively small number of private sector service jobs. Total labour costs of the low skilled should better reflect market conditions, by achieving greater flexibility in the collective wage agreements and by lowering social security contributions. Active labour market policies need to promote mobility of labour to areas with better job opportunities instead of providing subsidised public sector jobs or training, which often do not lead to permanent employment in the open labour market. And such policies should put greater emphasis on enhancing skills in general and on improving training opportunities for older workers in particular. While net benefit replacement rates are relatively high compared with many other OECD countries, it is the long duration and minimal degressivity (*i.e.* the extent to which payments taper off over time) of unemployment benefits that probably most hinders further reductions in structural unemployment and should be addressed.

Enforcement of competition policy should be strengthened

The functioning of product markets has improved. Network industries were liberalised early in Finland, with substantial benefits in terms of higher output and employment. The Finnish Competition Authority has broad powers, largely based on the EU competition "toolkit". The Competition Council and the previous Market Court were replaced by a new body, the Market Court, that has stronger judicial powers. While the competition policy framework is basically sound, enforcement should be strengthened. Fines that have been set in the past have been small, with the courts often reducing the fines recommended by the Authority. The basic range of sanctions should be greatly expanded and liability or sanction on the individuals who are responsible for violations of the law should be introduced. Moreover, the implementation of a leniency programme would make it easier to detect restraints on competition. Stronger enforcement and great vigilance on competition matters are necessary as several indicators suggest that competition might be weak in a number of sectors including the network industries. In a welcome move, the Competition Authority has recently been restructured, with a new unit focusing exclusively on eradicating cartels.

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Privatisation should be stepped up and agricultural subsidies lowered

Partly state-owned enterprises still employ more than 10 per cent of business sector employees although the government shareholding has become very small in some cases. While the privatisation mandates were broadened in 2001, little progress has been made, partly reflecting unfavourable stock market conditions and full privatisation of many companies is still not on the agenda. The authorities should ensure privatisation proceeds rapidly as soon as the market situation allows these mandates to be used. The government has announced that share-holdings in state-owned enterprises will be centralised in one government body, thereby treating ownership issues on a more consistent basis and splitting ownership and regulatory functions which should improve the functioning of product markets. While subsidies are generally low, they are very high in the agricultural sector and include a sizeable national component on top of EU subsidies. This sector has been undergoing a major restructuring following EU membership in 1995, since when the number of farms has fallen by about a quarter. The further restructuring of the agricultural sector, where Finland generally does not have a comparative advantage, would release resources for more productive uses, raise living standards by lowering prices and help restrain government spending.

Greenhouse gas abatement should focus on cost-effective solutions

The energy-intensive nature of the economy has resulted in high and rapidly growing greenhouse gas emissions that the government is committed to curb. The decision to allow the construction of a new nuclear power plant will reduce the expected overrun of the Finnish Kyoto emission target by half. The energy conservation and renewable energy policies designed to achieve the remaining reductions need to be implemented as cost effectively as possible. Finland was a pioneer in taxing carbon and has gained considerable experience in developing tax instruments for curbing emissions. The difficulties faced in implementing this approach emphasise clearly the need for concerted action especially in the case of the sectors operating in international commodity markets. In this light, Finland could benefit from participating in the EU-wide permit trading scheme and thereby further increasing the cost effectiveness of emission abatement policies. There is nonetheless

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scope for better equalisation of carbon taxation across sectors. Notably the favourable treatment accorded to peat needs to be reassessed, taking into account the question of whether the long-term benefits from regenerating this source of energy significantly offset its high CO_2 content.

Market-based instruments should be implemented to reduce air pollution

Even though Finland will be able to fulfil its obligations through air pollution control measures already adopted or envisaged, there may be a potential for increasing the cost-effectiveness of policies through suitable economic instruments like a trading scheme for NO_x emissions. Other effective instruments to reduce particulate pollution would be to accelerate the replacement of existing diesel engines with new models that incorporate better filters and to ensure that the tax differential between diesel and gasoline adequately reflects externalities from, *inter alia*, air pollution and greenhouse gases. Imminent reductions in the tax on the purchase of new cars will facilitate a faster renewal of the fleet. At the same time road pricing should be considered in order to address the higher externalities from transport in urban areas.

Summing up

Following the severe downturn in 2001, a recovery is underway and Finland should be able to return to a high-growth trajectory over the medium term. While Finland has easily met the fiscal targets of the Stability and Growth Pact, it has recently slipped against its own more ambitious surplus objectives, although this largely reflects cyclical weakness and the loss of asset-related revenues. In these circumstances, temporary shortfalls against the surplus target should be tolerated. Nevertheless, once the economy has fully recovered, the central government surplus of 1½ to 2 per cent of GDP should remain a medium-term objective over the current decade to reduce the fiscal pressures from ageing over following decades. If the achievement of this objective should require further consolidation, the emphasis should be on expenditure restraint rather than tax increases. Indeed a supplementary objective should be that expenditure restraint is sufficient not only to achieve the surplus objectives but also to allow for cuts in taxation, especially on labour. Continuous slippage against the medium-term target for central government expenditure

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suggests that the annual budget process needs better integration with medium-term objectives, while in several areas, significant public sector efficiency gains could be reaped, which suggests scope for reducing public spending. The wide-ranging pension reform to be implemented in 2005 is an important step in the right direction by phasing out early retirement schemes and raising incentives to postpone retirement. Nevertheless, some provisions appear overly generous, undermining sustainability and there is a risk that the abolition of the early retirement schemes will lead to an increased use of the disability scheme. Actions should therefore be taken to ensure the eligibility criteria are sufficiently strict to avoid this risk. Other measures should also be taken to increase participation as well as to lower unemployment that is still above the OECD average. The measures should focus on lowering the tax wedge on labour, increasing the flexibility of wage structures, raising the employability of the low skilled and reducing regional and skill mismatches. Medium-term growth can also be boosted by product and financial market reforms. Continued restructuring of the agricultural sector should pave the way towards a substantial lowering of subsidies. Competition policy enforcement needs strengthening, while environmental policies should put greater emphasis on market-based solutions. With a focussed structural reform programme, Finland should be able to return again to a strong economic performance in the coming years, while also securing the sustainability of public finances in the face of population ageing.

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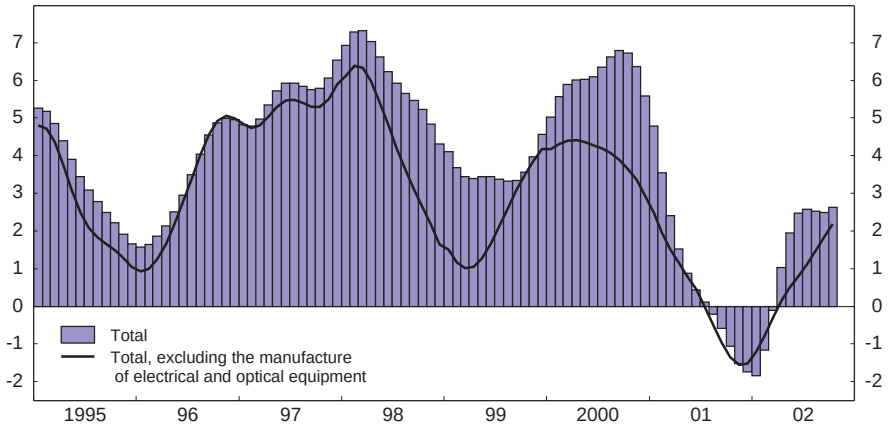
I. Macroeconomic developments, prospects and policy challenges

Finland was more severely affected than most other euro area countries by the global downturn in 2001, but has also recovered more quickly in 2002. Indeed it could be the only euro area country that enjoys faster growth in 2002 than in 2001. One factor underlying the more severe downturn was the global slump in demand for ICT (information and communication technology) goods and the importance of this sector in the Finnish economy, although more traditional exports particularly those of the forestry industry also experienced a sharp contraction. The ICT sector has also been prominent in leading the economy out of the downturn (Figure 1). Nevertheless, the outlook for this sector, in particular the

Figure 1. **Monthly output developments**

Per cent change over 12 months[1]



1. Monthly indicator of output (1995 = 100), trend.
Source: Statistics Finland, National Accounts.

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speed with which third generation mobile telephony will be adopted, represents a major uncertainty for the economy as a whole.

The experience of recent years has underlined the continued importance of global demand conditions and the contribution of the export industries to growth. Nevertheless, domestic demand, especially consumption, weathered the downturn relatively well. This is partly because firms were reluctant to shed labour, leading to a slump in productivity performance rather than a fall in aggregate employment which might have damaged consumer confidence more seriously, and fiscal policy has also been supportive. The unemployment rate has remained virtually stable in 2002 at about 9¼ per cent, but remains above the euro area average and well above that of the best performing OECD countries. It is essentially structural in nature, highlighting one important weakness in Finnish macroeconomic performance (Figure 2).

The current fiscal position is strong, Finland being one of the few euro area countries complying with the Stability and Growth Pact target for public finances to be close to balance or in surplus. To cope with the future ageing pressures, the government has set a medium-term target for the central government surplus of at least 1½ to 2 per cent of gross domestic product (GDP) which, given ongoing net asset accumulation in pension funds, corresponds to a general government surplus of 4 to 4½ per cent. However, recent slippage against this and other fiscal objectives, which is only partly due to cyclical weakness, raises the issue of how best to overcome the political economy problem of locking in fiscal surpluses that are judged necessary for long-term sustainability but which may appear tight in a short-term context.

Against this background, the following sections provide an overview of recent macroeconomic developments and review the fiscal stance. The chapter concludes with the prospects for 2003 and 2004, together with an assessment of both the risks surrounding these projections and the main macroeconomic policy challenges in the years ahead.

Recent economic developments

Output has been strongly influenced by international demand

Output growth averaged a robust 4.8 per cent over the long recovery from 1993 to 2000. But in 2001 output grew by only ¾ per cent, well below the euro area average for the first time since 1993 (Table 1). It was also the first time in a decade that the growth contribution of net exports was negative. Over the previous decade net exports added on average 1½ percentage points per annum to GDP growth, only exceeded in the euro area by Ireland and Luxembourg. Having grown at an annualised rate of 20 per cent between the beginning of 1999 and the third quarter of 2000, export volumes fell by 8 per cent over the subsequent year as

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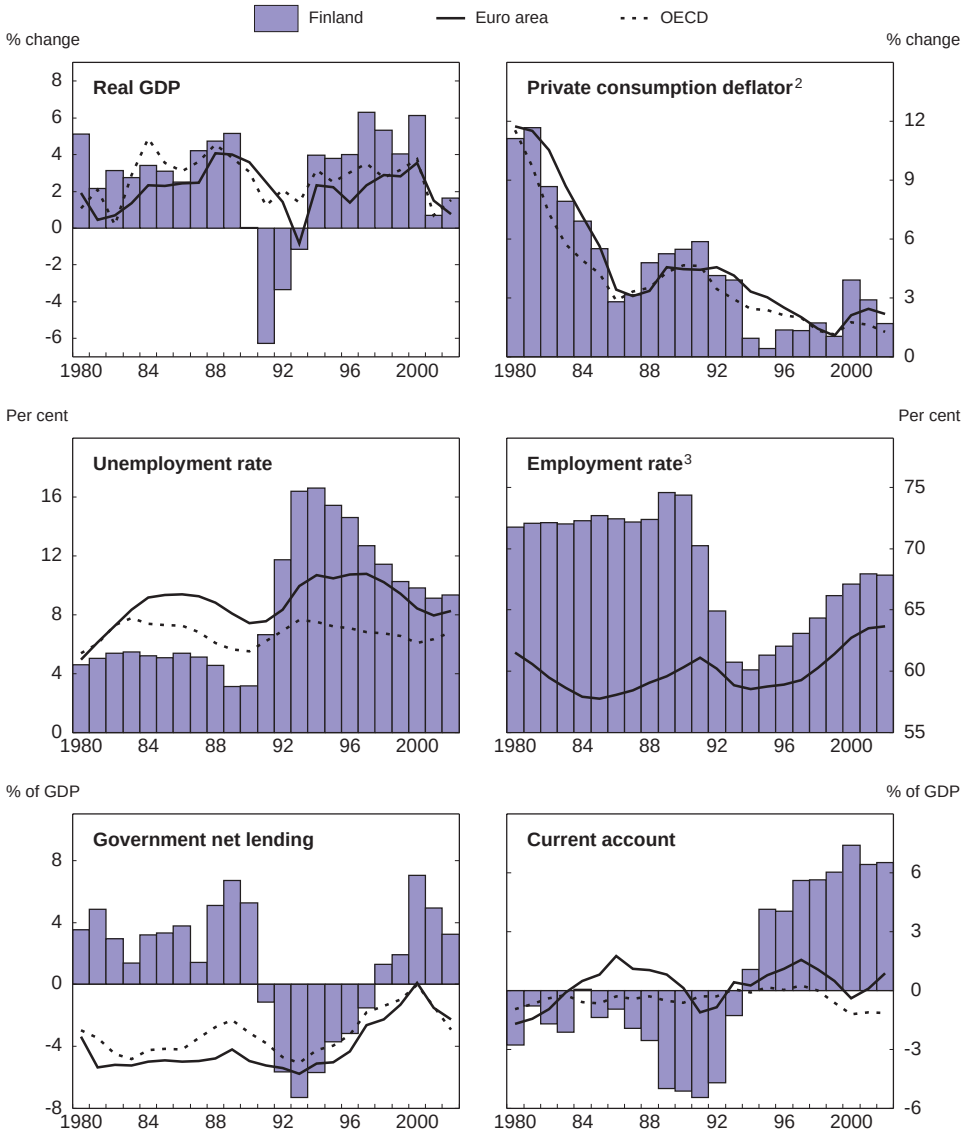
Figure 2. **Key indicators in long-term and international perspective**[1]



1. Estimated data for 2002.
2. OECD excludes high inflation countries.
3. Total employment as a per cent of working age population (aged 16-64).

Source: OECD (2002), *OECD Economic Outlook*, No. 72.

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Table 1. **Demand and output**
Percentage change, volume

	1999	2000	2001	2002 Q3 growth[1]	
				Over same quarter of previous year	Over previous quarter
Private consumption	4.0	2.6	1.1	2.9	–2.2
of which: Durable goods	7.9	4.6	–4.6	7.6	7.0
Government consumption	1.9	–0.2	2.1	2.0	1.7
Gross fixed capital formation	3.0	3.9	4.0	1.3	3.8
Public sector	–2.4	–5.4	0.8	2.2	–2.1
Private sector	4.0	5.5	4.6	1.2	4.8
Residential	12.7	3.6	–10.7	2.1	–3.5
Non-residential	1.0	6.3	10.2	0.9	7.3
Final domestic demand	3.3	2.2	1.9	2.3	–0.1
Stockbuilding[2]	–1.2	1.3	–0.8	–0.1	10.2
Exports of goods and services	6.8	20.1	–2.2	6.9	–4.2
Imports of goods and services	4.0	16.0	0.1	4.0	8.1
Foreign balance[2]	1.6	3.5	–1.1	1.8	–4.8
Statistical discrepancy[2]	0.7	–0.7	0.9	–1.5	–3.4
GDP	4.1	6.1	0.7	2.2	1.9
of which: Manufacturing	7.1	13.7	–0.7	3.3	6.7

1. Seasonally adjusted annual rate.
2. Contribution to GDP growth.
Source: Statistics Finland and OECD.

world trade slumped. The deceleration was most marked in the electronics industry, mainly Nokia (Box 1 and Table 2), where the effect of the general downturn in international demand was exacerbated by a lull in its product cycle; the value of exports of the electronics and electrical equipment industry fell by 15 per cent in 2001 after rising by more than 40 per cent in the previous year.

During the course of 2002 the economy recovered, although quarterly movements in GDP remained extremely volatile.[1] In the first quarter of 2002 GDP fell by 2.7 per cent at an annual rate, whereas in the second quarter it grew by 8.8 per cent, representing, respectively, the worst and best performance in the euro area. Even with modest growth in the third and fourth quarters, output will expand by 1½ per cent in 2002 as a whole, which is among the fastest in the euro area.

The recovery has been export led with a prominent role for the ICT sector; in the year since the trough in the third quarter of 2001, aggregate export volumes were up 6½ per cent and the output of the electronics industry, which is mostly exported, up 12½ per cent. This recovery confounds the recent air of pessimism

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Box 1. Nokia fact sheet[1]

Nokia is the world's leading producer of mobile phones. It expects to increase its world-wide market share to 40 per cent by the end of 2002, from 35 per cent in 2001 and 25 per cent in 1999.[2] Mobile phones account for about three-quarters of its net sales with networks accounting for most of the rest. It is estimated to have accounted for 1¾ percentage points of GDP growth in 2000, although this contribution was negligible in 2001 (Table 2). Nokia's share of total Finnish exports is almost one-quarter. Europe was its main market (49 per cent of its turnover) although its share has declined, whereas those for the Americas (25 per cent) and Asia-Pacific (26 per cent) have been increasing. The Finnish market accounted for only 1½ per cent of its turnover. In 2001 Nokia's share of total research and development (R&D) spending was close to one-third, and just under one-half of all private sector R&D. Taking into account Nokia's foreign R&D investments, R&D spending was about EUR 3 billion in 2001, compared to about EUR 3½ billion for total private R&D spending in Finland. While Nokia has a substantial impact on Finnish growth, exports and R&D its direct impact on employment is much smaller. In 2001, the number of Nokia employees in Finland fell marginally to 23 700, around 2 per cent of total employees in the business sector. Almost 60 per cent of its Finnish staff (and a third of its total staff) work in R&D, but the share of Finnish personnel declined to 41 per cent from 51 per cent of its total staff in 1998. Nokia paid EUR 0.7 billion taxes in Finland in 2001 (2 per cent of total taxes received by the general government), down by a third on the previous year. At the end of 2001, Nokia shares were 63 per cent of the market value of the Helsinki stock exchange and foreigners held 91 per cent. Nokia also co-operates in production and R&D with a network of numerous smaller firms in Finland. In 2000 there were about 300 such companies, with about 20 000 employees.

1. This sheet draws on material in Ali-Yrkkö and Hermans (2002).
2. *Financial Times* (2002).

surrounding the ICT sector in general, and telecommunications in particular, following delays and technical problems with third generation mobile telephony. It is based on continued product innovation by Nokia, with the introduction of new models including a range of camera phones, which have probably led to further market share gains in 2002. Developments in the other major export industries have been less spectacular (Table 2). There have been growing signs of a modest recovery in the forestry industry (including both paper and wood products) in 2002, with output estimated to be up by about 7 per cent in the year to the third quarter, after falling by 7 per cent in 2001. Output of the more traditional products of the metals industry (excluding electronics) is estimated to have fallen in 2002 given weak fixed investment in Europe.

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Table 2. **Contribution of manufacturing to growth**[1]

	2001, % of total		Percentage contribution to GDP growth				
	GDP	Exports	1998	1999	2000	2001	2002[2]
Wood, pulp and paper	5.1	26.6	0.2	0.2	0.3	–0.4	0.2
Electrical and optical equipment	8.0	27.5	1.4	1.4	2.4	0.1	0.7
of which: Nokia	..	..	0.8	1.0	1.8	0.0	..
Other metal products and equipment	6.0	27.9	0.2	0.0	0.4	0.1	–0.3
Other manufacturing	7.1	15.9	0.3	0.1	0.3	0.0	–0.1
Total manufacturing (%)	26.1	98.0	2.2	1.7	3.4	–0.2	0.5

1. Value added, gross, at fixed prices.
2. Estimate.

Source: Statistics Finland (2002), *National Accounts* 1993-2001; National Board of Customs; Ali-Yrkkö and Hermans (2002); and OECD.

A factor favouring Finnish exports over the last year is the geographic diversity of its markets. Although Finland's export markets are heavily dependent on other OECD European countries, accounting for 60 per cent of total goods exports in 2001, this share has fallen over the 1990s and is now lower than for any other euro area country apart from Greece. Conversely, Finland has a greater exposure to markets in Asia (which is especially important for electronic products) and non-OECD Europe (especially Russia and the Baltic States) than nearly any other euro area country, with each accounting for just under 10 per cent of Finnish goods exports.[2] Given that growth in trade in both Asia and non-OECD Europe has remained much stronger than in OECD Europe, this is another factor accounting for the stronger pick-up in exports over the course of 2002 relative to most other euro area countries.

Over recent years movements in export and import volumes have become increasingly correlated, partly reflecting the internationalisation of the production process (Figure 3).[3] This is most marked in the electronics industry, reflecting the importance of imported components in production as well as the outsourcing of production to other countries. Nevertheless, even similar percentage changes in export and import volumes are consistent with a substantial net export contribution to GDP growth given the large trade surplus in goods, equivalent to over 10 per cent of GDP in 2001 and higher than for any other euro area country except Ireland. Moreover, although Nokia will continue to outsource production of established models, it is likely to retain high valued-added activities such as research and development of new models in Finland.

Private consumption underpins domestic demand

Further evidence that the cycle is more advanced in Finland is that it is one of few euro area countries where consumption expanded faster in 2002 than in

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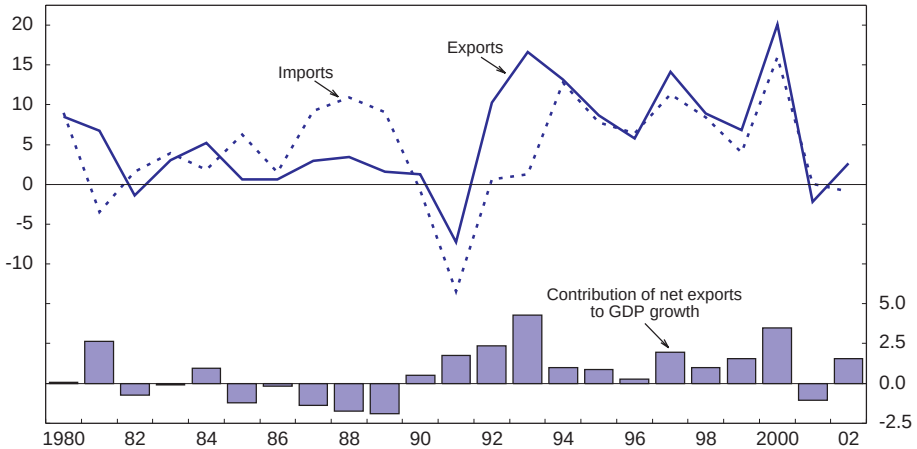
Figure 3. **Export and import volume growth**
Goods and services, per cent



Source: Statistics Finland and OECD.

2001, rising from about 1 per cent to an estimated 2½ per cent (Table 3). The main driving force has been continued growth in real disposable income due to continued growth in real wages as well as direct tax cuts. The absence of any significant increase in the unemployment rate, together with falling inflation and low interest rates helps to explain why there has only been a limited rise in the savings ratio despite a huge fall in equity prices.[4] The fall in the national stock market has been much steeper than in most other countries, falling nearly two-thirds between the peak in 2000 and November 2002, reflecting its higher exposure to the ICT sector. However, the real value of household financial assets has fallen far less (Figure 4), although it declined still by nearly one-quarter between 2000 and 2002. Most equities are held by foreigners and, even at their peak, equities only accounted for 30 per cent of household financial wealth with much of that held by relatively few individuals.

In common with many other OECD countries there has been a pick-up in house prices since the beginning of the year, possibly partly reflecting a switch in wealth holdings away from equities (OECD, 2002a). There are, however, regional differences, with house prices in the Helsinki area having risen by 11½ per cent in the year to the third quarter of 2002, whereas those for the rest of Finland have only risen by about 5 per cent over the same period. This differential partly reflects large migration flows to the capital area and tight planning restrictions that

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Table 3. **Household appropriation account**

	1996 Current prices, billion EUR	1997	1998	1999	2000[1]	2001	2002[2]
		Percentage changes					
Real disposable income	50.5	5.9	3.5	4.5	–0.7	3.6	2.6
Real consumption expenditure	51.9	3.5	5.1	4.0	2.6	1.1	2.5
Saving ratio (per cent)[3]	2.0	4.4	3.1	3.8	0.3	2.4	2.8
Disposable income	50.5	7.4	5.3	5.5	3.2	6.6	4.4
		Percentage points					
Contribution to disposable income							
Compensation of employees	49.4	5.2	7.2	4.9	6.1	6.1	4.2
Entrepreneurial and property income	13.4	2.4	1.2	1.7	3.0	1.4	0.3
Transfers received	21.9	0.1	0.4	0.8	0.4	1.9	1.5
Less:							
Transfers paid	16.2	0.8	1.9	1.4	0.9	1.6	1.1
Direct taxes	15.6	0.1	1.6	0.6	4.5	0.8	0.1
Interest on consumer debt	2.3	–0.5	0.0	–0.2	0.8	0.4	0.3

1. Heavily influenced by steep revenue increases due to stock options and capital gains which are not included in the national accounts measure of income, while taxes paid on these revenues are.
2. OECD estimates.
3. Ratio of household saving to disposable household income.

Source: Statistics Finland and OECD.

limit or slow down housing starts following an increase in demand for housing. Nevertheless, in real terms house prices remain below the levels of the early 1990s. The ratio of household debt to disposable income has continued to rise gradually since 1997, partly reflecting increased demand for mortgages, although it also remains well below the levels of the early 1990s. Residential investment, which fell more than 10 per cent in 2001, has levelled off in 2002 but has not yet responded to the significant pick-up in house prices.

Business investment remains weak

Although the average level of business fixed investment in 2001 was 10 per cent higher than in the previous year, this partly reflected carry-over effects from exceptionally strong growth in 2000. Business investment, while volatile, has remained weak since the second quarter of 2001 and, with survey evidence suggesting that many firms are operating below capacity, it is likely to have fallen in 2002. While the share of profits in business sector output has declined since the peak of 2000, it does so from a level, which is high by international standards (Citron and Walton, 2002). Stockbuilding fell substantially in the second quarter

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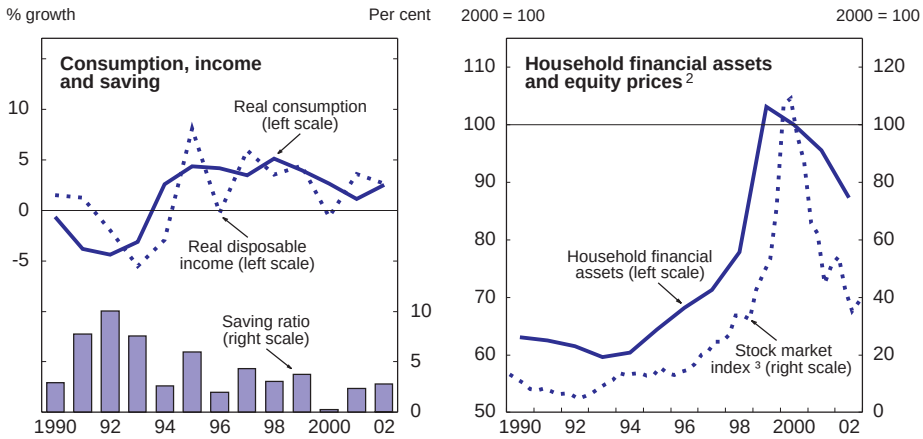
Figure 4. **Private consumption**[1]



1. Estimate for 2002.
2. Real prices; deflated by the consumer price index.
3. HEX all share index, quarterly average. Break in series in first quarter of 1997.

Source: Statistics Finland, Finnish Bankers' Association and OECD.

of 2002, despite the surge in output in the same quarter, and could contribute negatively (about ½ a percentage point) to GDP growth in 2002.

Employment has been stable, but unemployment remains high

Given the magnitude of the deceleration in output since 2000, it is remarkable that there has yet to be any significant fall in aggregate employment. This possibly reflects firms' reluctance to shed labour given earlier difficulties in attracting skilled labour during the upswing. The employment rate has stabilised in 2002 at over 67½ per cent. This is relatively high in relation to most European countries, but falls well short of the government's objective, specified at the beginning of its term of office, of 70 per cent (Table 4) and is also lower than in other Nordic countries. The unemployment rate also remained roughly stable between 2001 and 2002, although this follows seven consecutive years in which it fell.

Of greater concern is that the unemployment rate remains at over 9 per cent of the labour force, above the already high European Union average, and only exceeded by Spain and Greece. Unemployment is relatively high at both ends of the age spectrum (Table 5), although the ranking of the unemployment rate for "prime-age workers" (those aged 25 to 54) is only sixth among EU countries with Finland having a lower rate than Germany, France and Italy in 2001. The high

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Table 4. **Labour market developments**

	1996	1997	1998	1999	2000	2001	2002[1]
	Percentage changes						
Employment rate (%)[2]	61.9	62.9	64.1	66.0	66.9	67.7	67.7
Real GDP per worker	3.0	3.4	3.2	1.2	4.3	–0.5	2.3
Employment	1.4	2.0	2.4	3.3	1.7	1.4	0.0
Labour force	0.4	–0.2	0.9	2.0	1.2	0.7	0.2
	Per cent of labour force						
Unemployment rate	14.6	12.7	11.4	10.2	9.8	9.1	9.2
Persons unemployed for more than 12 months (% of total)[3]	30.1	30.5	30.2	28.1	27.7	27.4	26.6
Active labour market programmes *of which*:							
Job creation	2.6	2.5	2.3	2.0	1.7	1.5	1.6
Labour market training	2.1	2.3	2.1	1.9	1.5	1.3	1.4

1. Average of monthly data available for series in per cent of labour force. OECD estimates for percentage change data.
2. Total employment as a per cent of working-age population (both aged 15-64).
3. Registered unemployed.

Source: Ministry of Labour, Statistics Finland, Ministry of Finance and OECD.

Table 5. **Alternative international comparisons of the unemployment rate**

2001

	Finland	EU	Finland ranking in EU
Standardised rate	9.1	7.6	3
Age 15-24	19.9	13.9	3
Age 25-54	7.4	6.5	6
Age 55-64	8.9	6.4	2

Source: OECD (2002), OECD *Employment Outlook*.

unemployment rate among older-age workers is partly explained by the disincentives to work associated with the "unemployment pipeline" discussed in Chapter II.[5] The high rate of unemployment among younger workers is largely explained by the unusually large proportion of students who are counted as unemployed, which exhibits a strong seasonal pattern but still has a substantial effect on an annual average basis. The proportion of young adults attending education while at the same time being classified as unemployed is higher than in any other OECD country, while the unemployment rate of young adults not attending education is close to the OECD average.[6] Thus, excluding students reduces the standardised unemployment for Finland by more than in other OECD countries,

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from 9.1 to 7.5 per cent in 2002, and so brings it closer to the EU average. While arguably such an adjustment is warranted in terms of identifying the extent of the problem to be addressed by policy,[7] other adjustments that broaden the definition of unemployment (for example to include the effects of early retirement schemes, see Chapter II) would worsen Finland's ranking. Moreover, any comparison of the unemployment rate to the European Union average should not overlook the fact that the latter remains unacceptably high in relation to the best country experience.

Inflation has fallen gradually…

Inflationary pressures have eased during 2001 and 2002 as a consequence of excess capacity and the appreciation of the euro since the end of 2000.[8] Consumer price inflation, as measured by the harmonised index, has tracked the euro area average closely since early 2001, although in the third quarter of 2002 it dropped to just below 2 per cent, while the area-wide measure remained stubbornly above 2 per cent (Table 6 and Figure 5). On the other hand the margin with the euro area average has been consistently positive for service price inflation since 1998.

… but unit labour costs have risen sharply

With wages determined by the two-year central agreement, there has been no slowdown in wage rates, with increases of 4-5 per cent in 2001 and 2002

Table 6. **Prices and wages**

Percentage changes

	1996	1997	1998	1999	2000	2001	2002[1]
Consumer price index[2]	0.6	1.2	1.4	1.2	3.4	2.6	1.6
Harmonised index of consumer prices (HICP)	1.1	1.2	1.4	1.3	3.0	2.7	2.0
Core HICP	0.5	0.9	1.6	1.4	2.1	2.9	2.4
Wages							
Negotiated wages[3]	3.1	1.3	2.7	1.8	2.8	3.3	2¼
Wage drift[3]	1.1	1.1	0.8	1.0	1.2	1.2	1
Wage rate[4]	4.2	2.4	3.5	2.8	4.0	4.5	3¼
Wage rate[5]	3.3	4.1	4.9	2.7	5.6	5.3	4¼
Compensation rate[5]	2.1	2.8	4.4	2.3	4.1	4.9	4
Unit labour costs	–0.1	–1.0	1.7	0.6	–0.6	5.2	1½

1. Estimates.
2. Break in series in 2001 due to change in base year.
3. Annual average of quarterly data.
4. Total wage and salary earnings index.
5. Private sector, national accounts data.

Source: Statistics Finland, Ministry of Finance and OECD.

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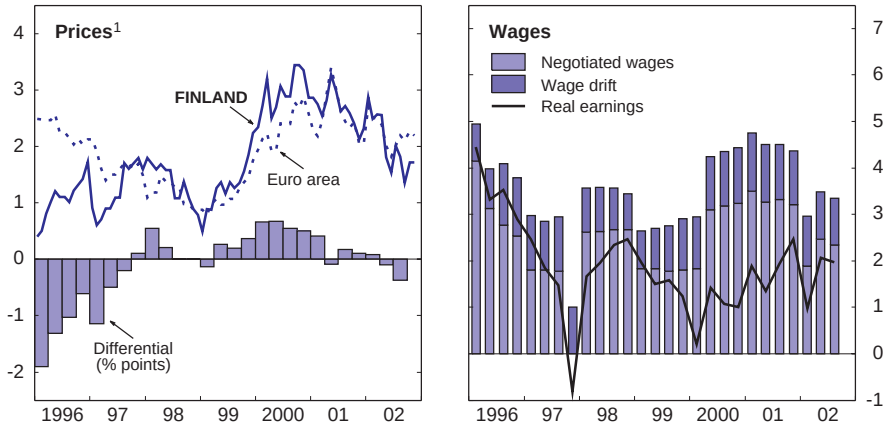
Figure 5. **Inflation**
Year-on-year percentage change



1. Harmonised index of consumer prices.
Source: Statistics Finland, Eurostat and OECD.

after allowing for wage drift (Figure 5 and Table 6).[9] In conjunction with the slump in productivity this has implied a substantial increase in unit labour costs, which for the whole economy rose by 5¼ per cent in 2001 and 1½ per cent in 2002, after increasing on average by ½ per cent per annum over the five-year period to 2000. The rise in unit labour costs has so far been absorbed by a fall in the profit share. Insofar as weak productivity is a cyclical phenomenon likely to be reversed, the rise in unit labour costs will not necessarily translate into future inflationary pressure.

The monetary stance is appropriate

One of the supposed drawbacks of belonging to a currency union is the problem of adjusting to country-specific shocks. Nevertheless, in adjusting to the ICT shock, euro area membership may have been helpful in promoting macroeconomic stability in Finland, particularly in avoiding gyrations in the exchange rate that might have been associated with large adjustments of equity portfolios (Figure 6). Moreover, the absence of any significant fall in the real exchange rate during the recent ICT downturn has not prevented the Finnish ICT sector from making substantial gains on market share.

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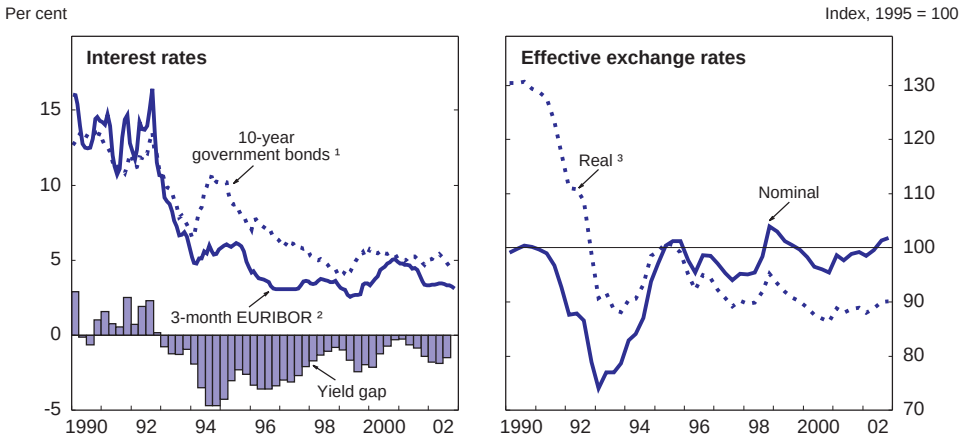
Figure 6. **Interest and exchange rate developments**



1. Ten-year government bond rate from January 1993 onwards, five-year rate for earlier period.
2. HELIBOR prior to January 1999. Greece included from January 2001.
3. Based on consumer prices.

Source: Bank of Finland and OECD, *Main Economic Indicators*.

The recent 50 basis point cut in euro area short-term interest rates will provide additional impetus to domestic demand. Perhaps more importantly, given growing evidence that demand in the core euro area countries is faltering, it will provide welcome support to Finland's main export markets.

Fiscal policy

Medium-term objectives

Finland has an exemplary record of maintaining overall fiscal discipline in the wake of the severe recession of the early 1990s. Over the course of the current cycle it will also be one of only a few euro area countries to meet the Stability and Growth Pact target for public finances to be close to balance or in surplus. Recent slippage against its own stricter surplus objectives can be largely explained by cyclical weakness and the loss of one-off revenues. These stricter fiscal objectives are judged appropriate in the light of earlier and more rapid population ageing in Finland than most other OECD countries. At the same time overruns on the medium-term expenditure target, suggest possible weakness in the framework of spending control, and carry the implication that in the absence of such overruns more could have been achieved in reducing the tax burden.

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When the government was elected in 1999 it adopted a number of medium-term objectives for fiscal policy:

- To maintain a central government structural surplus, later re-specified as an actual surplus of 1½-2 per cent of GDP together with a general government surplus of about 4½ per cent of GDP.
- To reduce central government debt to less than 50 per cent of GDP, later specified as excluding funds raised from privatisation.
- To reduce labour taxation by EUR 1.7-1.9 billion.
- And to keep central government expenditure, including interest payments at their 1999 level in real terms.[10]

Achieving the objective for the general government surplus mainly depends on hitting the target for the central government surplus. This is because employment pension funds, which have been in surplus of around 3 per cent of GDP, are included in the general government accounts and local government net borrowing has typically not exceeded a few tenths of a percentage point of GDP. Thus most of the variation in the general government surplus has been due to central government (Figure 7).

The fiscal goals were based on the assumption of average GDP growth of over 3 per cent, which has been (just) realised.[11] Initially as GDP growth was very strong the government surplus targets were substantially exceeded. Following exceptional GDP growth in 2000 both the general and central government surpluses peaked, at 7 and 3½ per cent of GDP respectively. Together with larger than expected sales of government-owned shares this led to a reduction in central government debt to 48½ per cent of GDP already in 2000, and it has remained comfortably below the original target since then (Table 7), although if privatisation proceeds are excluded this ratio would fall from 52½ to 49½ per cent of GDP between 2000 and 2001.

The deterioration in fiscal balances since 2000

With much weaker growth in 2001 and 2002, the general government surplus is likely to have halved to around 3½ per cent of GDP in 2002[12] (the OECD estimate is slightly lower at 3¼ per cent of GDP), with both the objectives for the central and general government surpluses being missed. Much of this fall in the surplus, perhaps as much as 2½ percentage points, can be attributed to the severity of the cycle. There was an additional loss of "one-off" revenues, equivalent to nearly 1 per cent of GDP, mostly associated with the massive swings in equity prices. Such revenues (direct taxes on household and corporate income from capital gains as well as direct taxes on household income from stock options) peaked at nearly 2 per cent of GDP in 2000. Tax revenues in many OECD countries are sensitive to movements in asset prices, although corporate tax revenues in Finland

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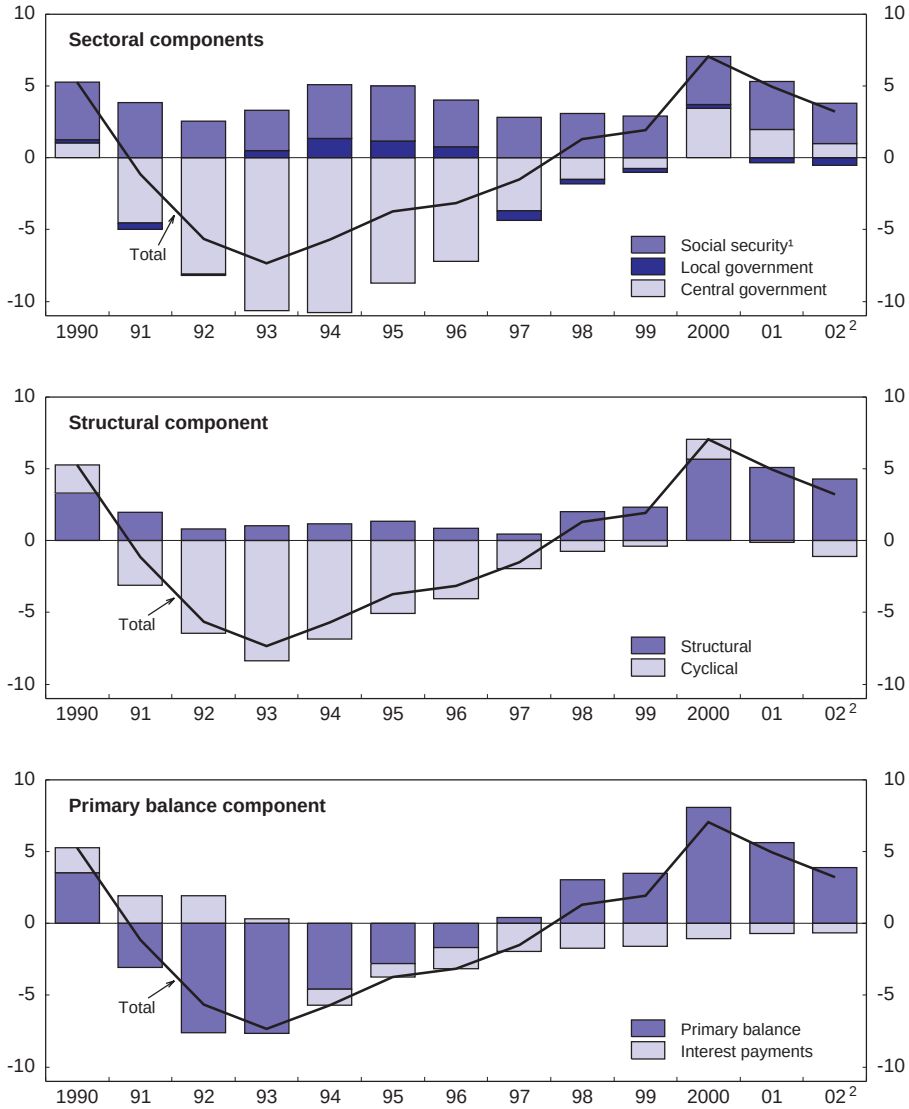
Figure 7. **General government net lending**
As a per cent of GDP



1. Including employment pension funds.
2. OECD estimate.

Source: Statistics Finland (2002), *National Accounts 1993-2001* and OECD.

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Table 7. **Public finances**
Per cent of GDP

	1997	1998	1999	2000	2001[1]	2002[2]	2003[2]
Expenditure	56.8	53.2	52.1	48.7	47.1	47.5	47.1
Revenue	55.3	54.5	54.0	55.7	52.0	51.3	49.9
Tax[3]	46.1	46.1	46.2	47.1	45.1	44.6	43.2
Other revenue[4]	9.2	8.4	7.8	8.6	6.9	6.8	6.7
Net lending	–1.5	1.3	1.9	7.0	4.9	3.8	2.7
Central government	–3.7	–1.5	–0.7	3.5	2.0	0.9	0.3
Local government	–0.7	–0.3	–0.3	0.2	–0.4	0.0	–0.4
Social security funds	2.8	3.1	2.9	3.3	3.3	2.9	2.8
Gross debt[5]	54.1	48.8	46.8	44.0[1]	43.4	42.5	41.9
of which: Central government	65.3	60.2	56.5	48.4[1]	45.4	43.2	41.7
Government bond yield (%)[6]	6.0	4.8	4.7	5.5	5.0	4.8	4.4

1. Preliminary data.
2. Ministry of Finance projections.
3. Including social security contributions.
4. Property income and transfers received other than social security contributions.
5. EMU definition.
6. Ten-year government bond rate.

Source: Statistics Finland, Ministry of Finance and OECD.

may be more sensitive than most.[13] Combining the effect of the cyclical downturn and the loss of one-off revenues, suggests that the extent of discretionary easing since 2000 has been negligible (Figure 8).

While the performance against the surplus objectives is broadly favourable – on average the objective of a general government surplus of 4½ per cent of GDP has almost been realised over the period 1999 to 2002 – there has been significant over-shooting of the medium-term expenditure objective. The rise in central government outlays measured against the original objective of holding them constant in real terms at the 1999 level has been considerable; after running about 1¾ per cent higher than the target in 2000 and 2001 there is likely to have been a further 2 per cent real increase in 2002. This is despite a fall in net interest payments by ½ per cent of GDP since 1999. They are included in the expenditure target, even though government has little influence over their development. Much of the rise in central government spending is explained by increased transfers to local government. Indeed, local government consumption rose much faster than that of central government, with the former rising by 13 per cent in nominal terms between 2000 and 2002 (equivalent to ¾ per cent of GDP), whereas central government consumption expenditure has increased 6 per cent (equivalent to a constant share of GDP). Spending increases were strongest for health care and education, with general government compensation of employees expected to

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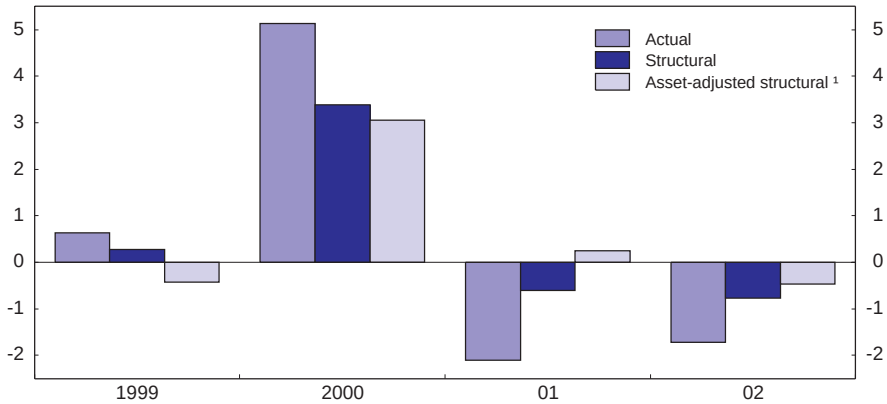
Figure 8. **Changes in different measures of the general government surplus**
Per cent of GDP



1. Structural surplus less direct tax revenues that are particularly sensitive to asset prices, namely those on households' stock options and capital gains, and non-financial corporations' capital gains.
Source: OECD.

have risen nearly 11 per cent between 2000 and 2002. Although the medium-term objectives for reducing labour taxation have been substantially exceeded and the aggregate tax ratio has fallen (Table 7), the implication is that in the absence of expenditure slippage, more could have been achieved.

The 2003 Budget: the surplus is projected to fall further below target

The Budget proposals for 2003 imply a further shrinking in the surplus relative to medium-term objectives, with the central government surplus falling to ½ per cent of GDP, and the general government surplus to 2¾ per cent of GDP. Central and local government spending is projected to remain stable as a share of GDP, but tax revenues are expected to fall by just over 1 per cent of GDP. Only about 0.3 percentage point of this loss in tax revenue can be attributed to the income tax cut (Box 2), including an additional reduction of around EUR 100 million announced in November 2002 following the central wage agreement for 2003 and 2004. Much of the remaining loss is due to lower corporate tax revenues, reflecting the lagged effect of the weak profit situation in 2002. Standard calculations of the change in the discretionary, as opposed to cyclical, component of the budget surplus suggest that the fiscal stance will be expansionary by around 1 per cent of GDP in 2003 (Ministry of Finance, 2002a).

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Box 2. **Summary of measures in the 2003 Budget proposals**

- All marginal tax rates on earned income will be cut by 0.3 percentage point and the scales' income brackets raised by one per cent. In addition both the deduction of work-related expenses and the earned income tax allowance for municipal taxation will be raised. The average tax rate on wage and salary income will decline by less than ½ a percentage point, slightly favouring the low paid.
- Energy taxes will be raised by 5 per cent.
- The proportion of the corporate income tax yield accounted for by central government will rise 3½ per cent, with a corresponding fall in the share of municipalities. This is part of a wider reform, which also increases central government transfers to municipalities, that aims to promote greater stability in local government finances.
- Central government transfers to the municipalities to meet the operating costs of social and health care will be increased by 1½ per cent (as an index adjustment).
- The use of labour market subsidies will be expanded as an active form of employment support.
- New infrastructure projects to improve road connections and harbours will be implemented.

Short-term projections

As global activity picks up next year, the boost to exports should ensure that output growth will reach around 3 per cent in 2003, in line with potential growth (Table 8). While the contribution of net exports to growth, at around 1 per cent of GDP, is likely to be below that achieved throughout much of the later 1990s, the recovery should also continue to benefit from robust domestic demand. The projections suggest that Finland could outperform the euro area again in 2003 and 2004 by a considerable margin. However, the volatility of output makes the strength and sustainability of the recovery going forward difficult to judge.

An agreement between the main trade unions and employers' organisations was reached in late 2002 for an increase in wage rates of 2.9 per cent in 2003 and 2.2 per cent in 2004. This agreement, once allowance is made for modest tax cuts and for wage drift that has consistently been about 1 per cent per annum, should underpin growth in real disposable income in excess of 2 per cent in 2003. With the recent decline in euro area short-term interest rates and continued low inflation, there should be no significant change in the saving ratio, implying growth in consumers' expenditure of about 2 per cent. However, with many firms still

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Table 8. **Short-term projections**
Percentage changes, volume

	2000 Current prices, billion EUR	2001	2002	2003	2004
GDP components					
Private consumption	65.2	1.1	2.5	2.1	2.4
Government consumption	27.0	2.1	2.0	1.8	1.9
Gross fixed capital formation	25.1	4.0	–2.3	–0.1	2.9
Public sector	3.3	0.8	1.3	0.0	0.0
Private sector	21.8	4.6	–2.9	–0.1	3.4
Residential	6.4	–10.7	–1.2	2.4	1.4
Non-residential	15.4	10.2	–3.4	–0.8	4.0
Final domestic demand	117.4	1.9	1.4	1.6	2.4
Stockbuilding[1]	1.0	–0.8	–0.6	0.3	0.3
Total domestic demand	118.3	1.0	0.6	1.9	2.8
Exports of goods and services	56.4	–2.2	2.6	7.0	8.7
Imports of goods and services	44.0	0.1	–0.9	6.7	8.2
Foreign balance[1]	12.4	–1.1	1.6	1.1	1.4
Statistical discrepancy[1]	0.4	0.9	–0.5	0.5	0.0
GDP	131.1	0.7	1.6	3.2	3.8
GDP *of the euro area*	..	1.5	0.8	1.8	2.7
Prices and wages					
Harmonised index of consumer prices	..	2.7	1.7	2.0	1.8
GDP deflator	..	3.0	1.4	2.1	2.4
Private sector wage rate	..	5.3	4.2	4.2	4.1
Labour market					
Employment	2 326[2]	1.4	0.0	0.5	1.4
Unemployment rate (level, % of labour force)	9.8	9.2	9.3	9.5	9.4
Current account balance (% of GDP)	..	6.4	6.5	6.5	7.6
Interest rates (%)					
Short-term	..	4.2	3.3	3.0	3.6
Long-term	..	5.0	5.0	4.7	5.0
Memorandum items					
Output gap (level, % of potential GDP)	..	–0.2	–1.8	–1.8	–1.3
Net lending (% of GDP)	..	4.9	3.2	2.9	3.6
Household saving ratio (% of disposable income)	..	2.4	2.8	3.0	2.6

1. Contribution to GDP growth.
2. Thousand persons.
Source: OECD (2002), OECD *Economic Outlook*, No. 72.

operating with excess capacity, business investment may not pick up much during 2003. On the other hand, residential investment should rise moderately in response to increasing house prices, and following the run down in stocks in 2002, stockbuilding may make some positive contribution to growth, of the order of a

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¼ percentage point. GDP growth may further increase in 2004 both as world trade and exports further accelerate and business fixed investment begins to revive.

Despite the pick-up in growth, unemployment may not decline by much – the flip side of labour hoarding during the downturn – and is likely to remain well above the euro area average. Inflation should remain low, although it may not fall much further, despite the continued presence of slack in the economy as indicated by a negative output gap. This is because the central wage agreement is likely to push up overall unit labour costs by about 2 per cent per annum in 2003 and slightly less in 2004. This would imply a sustained period in which unit labour costs increased faster, or are at least in line with, the GDP deflator, in contrast with much of the experience of the 1990s when the rate of change was clearly below that of the GDP deflator.

Even with growth recovering to above potential rates, the objective of a general government surplus equal to 4½ per cent of GDP is unlikely to be reached. A significant part of revenue losses since 2000 are unlikely to be recovered in the upturn because they were related to exceptionally strong asset prices. Moreover, continuing wage pressures in the public sector may raise the deflator for government consumption, which could be running well ahead of the GDP deflator. Finally, there may be significant losses in indirect tax revenue in 2004, perhaps equivalent to as much as 1 per cent of GDP, as a result of measures to comply with EU directives on harmonising indirect taxation on cars and alcohol (see Chapter IIII for details).

Substantial uncertainties surround the projections

A major concern is whether the recent pick-up in export growth will be sustained. Much depends on the performance of ICT-based exports, which have weathered the industry-wide downturn relatively well. However, prospects for the industry in 2003 and beyond depend on a positive international reaction of consumers to third-generation mobile telephony. If the pick-up in international demand is delayed there is a risk that employment will suffer, with knock on effects on domestic demand.

Main policy challenges

One of the main challenges facing macroeconomic policy-makers is preparing for the impact of population ageing on public finances, while at the same time ensuring that there is scope for further cuts in taxation. The effect of ageing on the budgetary position is recognised by the authorities as reflected in the ambitious setting of current fiscal objectives and as further demonstrated by the recent agreement on a wide-ranging pension reform. However, given that the overall effects of the reform are uncertain and may not become clear for many decades, as discussed further in Chapter II, a prudent approach would suggest

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that the general government surplus target of 4½ per cent of GDP should be maintained for the remainder of the decade. At the same time slippage against this target should be tolerated to the extent that it reflects cyclical weakness.

Over the next few years it would also seem desirable to at least hold general government expenditure constant in real terms at the level envisaged in the 2003 Budget proposal. Clarification of the nature of the real expenditure target (which should focus on primary expenditure) as well as reforms that bind the annual budget procedure more closely to medium-term targets would be helpful in achieving these objectives. Moreover, there is probably scope for reducing spending in some areas (Chapter III), which would allow for further tax cuts during the recovery as soon as the general government surplus returns to 4½ per cent of GDP. If the recovery was insufficient to return the surplus to target then further expenditure restraint should be the preferred course of action given that the tax burden is already too high.

Comparison with earlier experience and with other countries, suggests that there is scope both to raise participation rates and lower structural unemployment, and reforms could also lead to sizeable efficiency gains in many sectors, including the public sector (Chapters III and IV). As well as being worthy in their own right, progress in these areas would strengthen the economy against the impending demographic shock as well as provide some insurance against the possibility that the growth contribution of the ICT sector may not be as large as in the past.

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II. Ageing, pension reform and long-term public finances

Ageing occurs earlier and is more rapid in Finland than in most other OECD countries. Recognising the urgent need to act, Finland is among the few OECD countries to introduce a comprehensive pension reform to address one of the major fiscal pressures resulting from ageing. This chapter first highlights the major features of the reform and assesses its effects. It then considers it in the broader context of the ageing pressures on long-term public finances. In particular it addresses the question of how ageing and the pension reform should influence the setting of current fiscal objectives.

Demographic trends

Finland's population will peak in the early 2020s, at about 5.3 million people, but the labour force will already start to decline in 2004 by ½ per cent annually. The old-age dependency ratio (those over 65 years as a percentage of the working age population) will rise from 23 per cent currently to about 37 per cent by 2020, the fastest rise in the OECD area. Ageing will continue with the dependency ratio reaching 45 per cent around 2030, although thereafter the dependency ratio will plateau at just under 50 per cent over the period 2040-50 (Figure 9).

The current system and the pension reform

Currently, there are two old-age pension schemes: the national pension scheme and the earnings-related pension scheme. The national pension scheme guarantees a minimum income for resident persons who are not entitled to an earnings-related pension, or to those for which the latter is small. The earnings-related pension consists of all the pensions that have accrued from each employment contract and from self-employment. Benefits are based on the number of years in employment, the accrual rate and the "pensionable wage" which is based on the gross wage net of employee's pension contributions. The system is of a defined-benefit nature, based on tripartite agreements and governed by several pension acts. It is financed by employer's and employee's contributions and consists of a mix of a pay-as-you-go and a funded system. The employer's and

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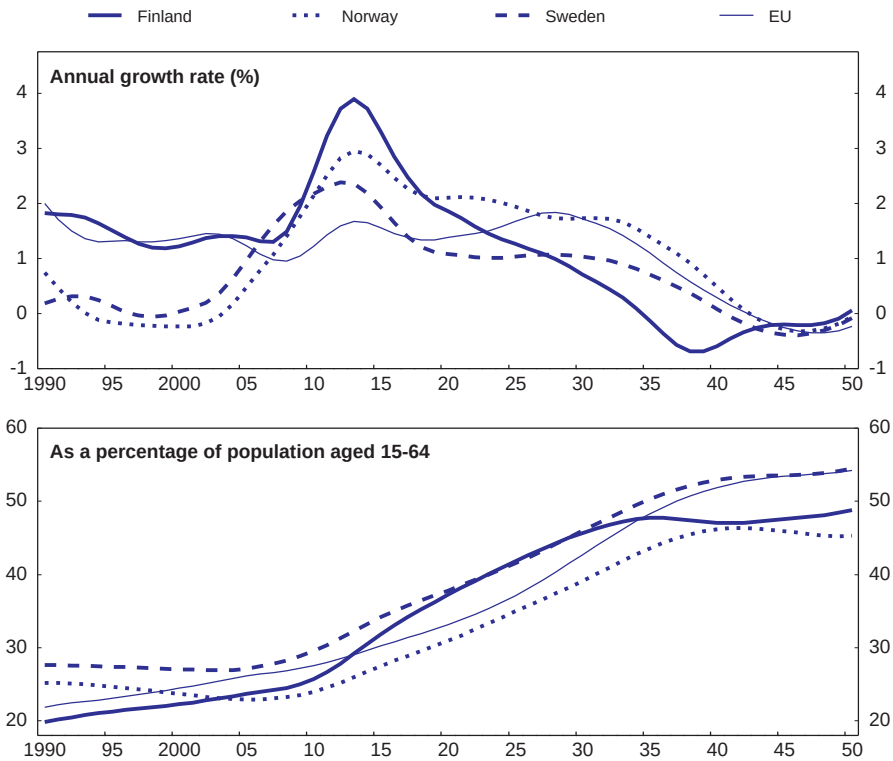
Figure 9. **The demographics of ageing in Finland**
Population aged 65 and over[1]



1. Data for the period 2001-50 are medium variant population projections.
Source: United Nations (2001), *World Population Prospects 1950-2050 (The 2000 Revision).*

employee's contributions are used directly to pay for current pensions and indirectly to fund future pension payments. Pre-funding is collective and, thus, has no effect on the size of the pension, but it affects the future path of contribution rates. The reform is focused on the earnings-related pension scheme and the various options to retire early.

The reform[14] was agreed in autumn 2002 and is expected to be voted by parliament in early 2003. It will affect pensions as from 2005. Agreement on the reform so far only concerns pensions for those working in the private sector, although reform of public sector pensions is likely to be agreed in 2003 and is expected to be along similar lines.

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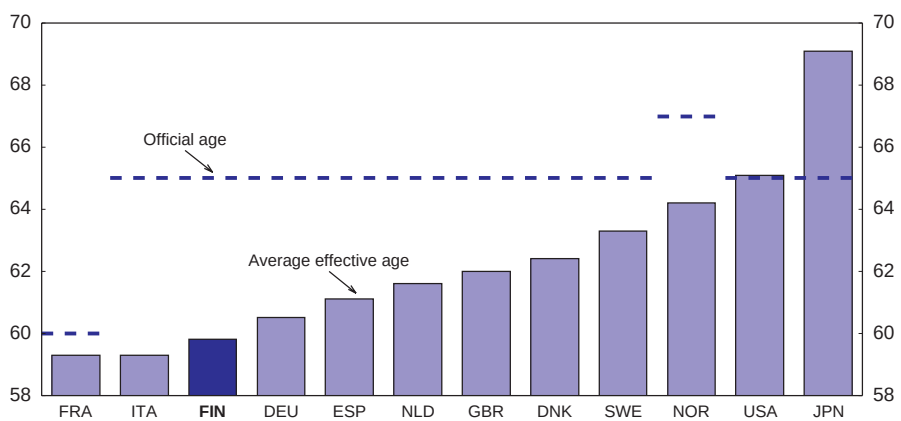
A major aim of the reform is to raise the average age of retirement which is currently only 59, compared to an official statutory age of retirement of 65. Eligibility for retirement in the current system is from the age of 60, with the level of pension payments subject to an actuarial reduction for every month below 65, although the proportion of pensioners taking this option is relatively small. Of much greater importance are various early retirement schemes, which accounted for about three-quarters of all new private sector retirees in 2001, the largest of which are the disability pension and the unemployment pension. While it is not unusual among OECD countries for the average age of retirement to be below the official statutory age (which is usually 65), Finland is among a minority of countries for which the average effective age is below 60 (Figure 10). The reform follows the broad recommendations of previous *Surveys*[15] by creating better incentives to stay at work longer and by reducing early retirement options. These two aspects of the reform are considered first, with details of the reform presented in Box 3.

Increasing incentives to retire later

Previous OECD work[16] has established that retirement decisions can be strongly influenced by fiscal incentives, which can be separated into two components. The first is the replacement rate – *i.e.* the pension received as a proportion

Figure 10. **Retirement age in selected OECD countries**
Males, 1994-99[1]



1. 1993-98 for Netherlands and Germany.
Source: P. Scherer (2002), "Age of Withdrawal from the Labour Force in OECD Countries", *Labour Market and Social Policy Occasional Papers*, No. 49, OECD.

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Box 3. The pension reform*

There are a number of important elements in the current reform package:

- A *flexible retirement age between* 62 *and* 68 *will be introduced, with a sharp rise in the accrual rate after* 62. From the age of 63 pensions will be calculated according to the accrued rights. Between 62 and 63 the pension will be reduced by 0.7 per cent for each month of early retirement prior to 63. The accumulation rate will remain at 1.5 per cent a year between ages of 18 and 52, increase to 1.9 per cent between 53 and 62 and then rise to 4.5 per cent between 63 and 67 (this compares to the current rates of 1.5 per cent between ages 53 and 59 and 2.5 per cent for those aged over 60). The ceiling on the maximum pension will be abolished.
- T*he pension base will be determined by earnings over the entire career* rather than over the last ten years of each employment relationship as at present. As well as being actuarially "fairer", this should promote labour market flexibility, as under the current system changing jobs is likely to penalise the final pension, particularly if wages were rising over the course of a career. Pensions will start accumulating after the age of 18 until the age of 68. They will also accrue during some non-working periods, *e.g.* for students who graduate for a maximum of five years and for parents who stay home to take care of children aged under three. For both groups the accumulation is based on an income base of no more than EUR 500 a month.
- T*he system will adjust to future increases in life expectancy* from 2009 by applying a "life expectancy coefficient" to the calculation of pensions. This coefficient is a means of taking into account that people will tend to live longer in the future than today and so provide a mechanism for keeping total pension costs in check.
- T*he method of indexation will change*. There are two indices in the pension system. The first adjusts past earnings to the present level when computing the pension at the time of retirement. This "wage multiplier" puts a weight of 80 per cent on wages and 20 per cent on the consumer price index. The other index aims at keeping the purchasing power of pensions intact. This index will have a weight of 80 per cent on consumer prices and 20 per cent on wage and salary earnings.
- T*he structure of employees' pension contributions will change*. Employees aged 53 or over have to pay a 27 per cent higher employees' pension contribution than those aged under 53. Currently, the contribution is 4.4 per cent of earnings irrespective of age.
- P*ension funding will be strengthened from* 2003 *onwards* so that additional funding of 7.5 per cent of the insured wage sum will be available by 2013.
- O*ptions for early retirement will be further curtailed*. The most important changes are to the unemployment and disability pension, which accounted for 5 and 25 per cent of all pensioners in 2001, respectively, and are discussed further in the main text. In addition the minimum age for a part-time pension, which accounts for 2½ per cent of all pensioners, is to be raised from 56 to 58, while the amount of old-age pension accumulated during part-time retirement will be halved.

* Ministry of Finance, the Central Pension Security Institute and the Finnish Pension Alliance TELA.

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of working income prior to retirement. The higher the replacement rate, the greater the incentive to retire. The second component is the change in net pension wealth from working an additional year and so forgoing an extra year of pension and paying a further year of contributions. If as a result of working an extra year net pension wealth remains constant then the system is neutral, but if it falls then the system imposes an implicit tax on working.

For Finland, the net replacement rate under the current system for a worker receiving an average production wage in his sixties considering retirement is about 60 per cent.[17] This corresponds roughly to the median across a range of 15 OECD countries shown in Figure 11, and is certainly less generous than in some of the major European countries. Given that under the present pension system, retirement before or after the official retirement age of 65, is penalised or rewarded at a rate calculated to be actuarially fair, it might be expected that the system is broadly neutral as regards the additional pension wealth from working an extra year.[18] However, this is not the case because there is a ceiling on the maximum pension that can be earned and a worker with a full career history is likely to run into this ceiling in his mid-sixties.[19] Thus, under the current system there is progressively an implicit tax on pension wealth for a typical worker in the mid-sixties – *i.e.* pension wealth begins to fall as a result of working an extra year – that will increase the incentive to retire before the age of 65.

Preliminary OECD calculations illustrate the impact of the pension reform in terms of its effect on these incentives (Figure 11). Mainly as a consequence of abolishing the ceiling on the maximum pension and the higher accrual rates earned from the age of 63, the change in pension wealth for workers continuing to work in their sixties will be much more positive. Indeed, for a worker in the early sixties the implied change in pension wealth from continuing to work for an additional year will be among the most favourable of the 15 OECD countries considered in Figure 11. Most importantly, the absence of a ceiling will avoid the current situation where continuing to work can lead to a sharp fall in pension wealth. In terms of the effect on the replacement rate, because old-age pension can no longer be drawn at the ages of 60 and 61 there will also be little incentive to retire on an old-age pension before the age of 62. From the age of 62 the reform is likely to raise replacement rates, which will, to some extent, counteract the incentive to continue working. Estimates suggest that the reform would lead to an average 15 per cent increase in pension levels (Central Pension Security Institute, 2002), although in these calculations the estimated (positive) effect of longer working careers is included, but the negative effect of the life expectancy adjustment is not (see also Annex I).

The net result of these conflicting effects on incentives is difficult to judge.[20] It is likely that the "larger" reduction in the implicit tax on pension wealth and zero replacement rate at ages of 60 and 61, will dominate the "smaller"

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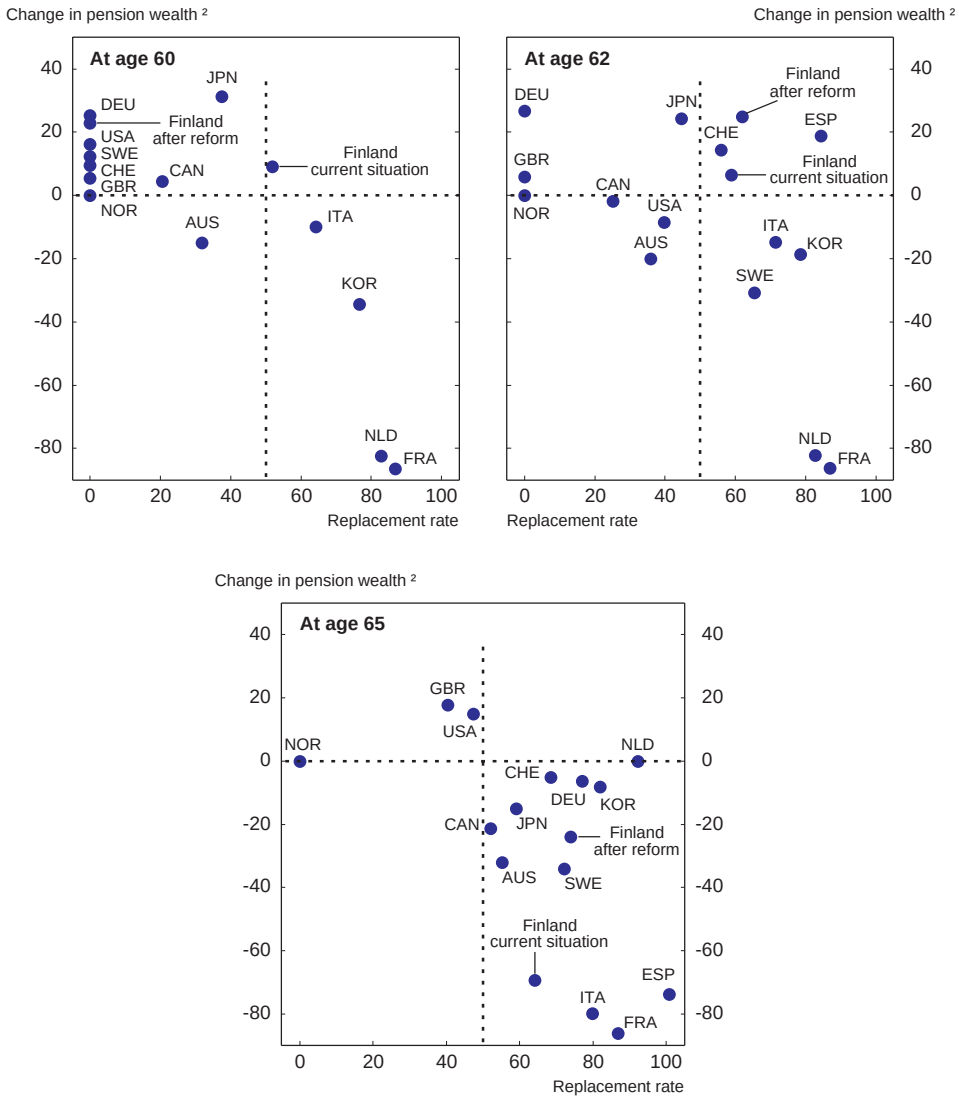
Figure 11. **Financial incentives to retire under regular retirement schemes**[1]

Per cent



1. Calculations are for a full-career worker with average earnings. The higher the replacement rate and the greater the fall in pension wealth (*i.e.* the more south-easterly the co-ordinate) the greater the incentive to retire.
2. Changes in pension wealth as a percentage of net annual earnings.

Source: OECD (2002b) and OECD calculations for effect of reform.

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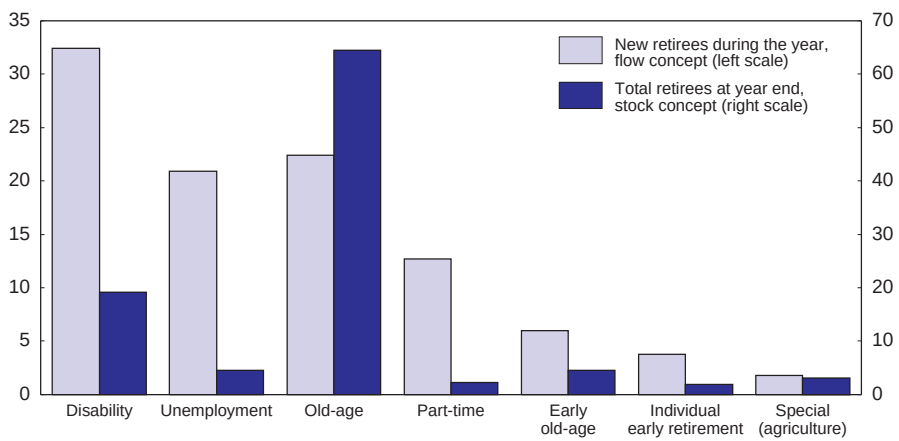
increase in the replacement rate from age 62 (where the relative size of these movements is judged according to the existing range of country experience represented in Figure 11). However, for these incentive effects to translate into a higher average age of retirement it is essential that alternative pathways to early retirement are also curtailed.

Reforming early retirement schemes

Recent work across a range of OECD countries has found that, in general, old-age pension systems considered in isolation do not usually provide strong incentives to retire early,[21] but that other early retirement schemes, such as disability, unemployment-related or other special early retirement schemes, often do. The operation of early retirement schemes is particularly important in explaining the relatively low retirement age in Finland. During 2001 only 22 per cent of all new retirees retired at the age of 65 to take the full standard old-age pension, with a further 6 per cent retiring early to take a reduced old-age pension (Figure 12). The remainder retired under various early retirement schemes, the most important being disability and unemployment pensions, which accounted for 32 and 21 per cent of all retirees, respectively.[22] The fiscal incentives to use unemployment and disability benefits as an alternative pathway to early retirement are quite strong (Figure 13). A person aged 55 receiving an average wage who stops

Figure 12. **Retirement by pension scheme**[1]
Per cent of total, 2001



1. Public and private sector, including both earnings-related and national pensions.
Source: Central Pension Security Institute.

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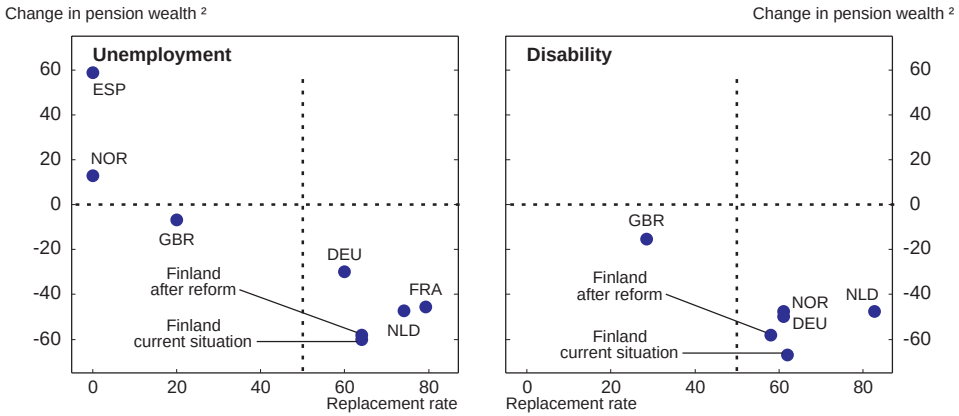
Figure 13. **Financial incentives to retire under unemployment and disability schemes**[1]

For a 57 year-old, per cent



1. Calculations are for a full-career worker with average earnings. The higher the replacement rate and the greater the fall in pension wealth (*i.e.* the more south-easterly the co-ordinate) the greater the incentive to retire.
2. Changes in pension wealth as a percentage of net annual earnings.

Source: OECD (2002b) and OECD calculations for effect of reform.

working, either as a result of disability or unemployment, will face a quite high initial replacement rate[23] of about 60 per cent, and the change in pension wealth from an additional year of work is strongly negative. Moreover these results generalise across all older ages up to the official age of retirement at 65 (OECD, 2002b). An important component of the reform package is, therefore, a range of measures to curtail the use of early retirement schemes. The two main early retirement schemes, based on unemployment and disability, are described below together with the effect of the reform.

Early exit from the labour market is currently facilitated by the so-called "unemployment pipeline", whereby from the age of 57 the standard unemployment insurance period is followed by an extension to age 60, finally leading to an unemployment pension. An unemployment pension is paid to those aged 60 to 65 who have been long-term unemployed and received unemployment benefits for the maximum period, which is 500 days. This means that the "pipeline" leading to an unemployment pension already starts at the age of 55. Evidence of the distortionary effect this has on incentives can be seen from the sharp rise in age-specific unemployment rates from the age of 55 (Rantala, 2002). Moreover, the incidence of unemployment is calculated to be about twice as high for those who are old enough to qualify for the "pipeline", compared to those who are immediately below the age of qualifying for it (Ilmakunnas and Rantala, 2002).

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Following the reform the minimum age for qualifying for the "unemployment pipeline" will rise to 57, and the unemployment pension will be abolished and replaced by an extension of unemployment benefits from the age of 60 up to 65. However, these changes only apply to those born after 1950, slowing the effect of the reform. The main effect of the reform should be to lower the incentive to stop working and claim unemployment benefit for those aged 55 and 56. The effect of the reform on those aged 57 and over is less clear cut. On the one hand, unemployment will typically imply a similar net replacement rate to the unemployment pension and so incentives to retire may be little changed. On the other hand the change in pension wealth from continued work is likely to be less unfavourable as a result of the reform.[24] The overall effect is likely to be some reduction in the use of unemployment as a pathway to early retirement, although for those aged over 57 the size of the effect may be marginal.

There are currently two forms of disability pension. The main disability pension is awarded to any person aged 16 to 64 who is certified as being incapacitated for work. In addition there is an individual early retirement pension that is available to those aged 60 to 64, for which the medical eligibility conditions are less strict. The reform will phase out the individual early retirement scheme, which is in line with previous *Survey* recommendations. Nevertheless, there is still a concern that the curtailing of other paths to early retirement (including the "unemployment pipeline") may put further pressure on disability pensions. In particular there has been a link between disability and unemployment pensions in Finland during the 1990s, with the number of disability pensions declining while the number of persons in the unemployment pension pipeline increased (Gould and Nyman, 2002). If this link also works the other way round, the effects of the reform on raising both the average effective retirement age and aggregate employment could be significantly dampened. Indeed there are a number of OECD countries, perhaps Norway, Netherlands and Sweden being the most relevant examples, where possible abuse of disability pensions have effectively become a form of unemployment benefit with an adverse effect on labour market performance.

More could be done to raise the demand for older workers

While much of the preceding analysis has focussed on the supply-side financial incentives of older workers, it is also important to ensure that demand for older workers is not impaired. Wages need to be flexible to adjust to productivity, in particular to avoid widespread occurrences of labour demand falling because declining productivity with age is not matched by appropriate wage adjustment. In this respect, a positive aspect of the reform is that, because pensions will be calculated on the basis of lifetime earnings, they will be less closely linked to wages just prior to retirement, suggesting that resistance to continue working at a lower wage might be reduced. Training of older workers to maintain marketable

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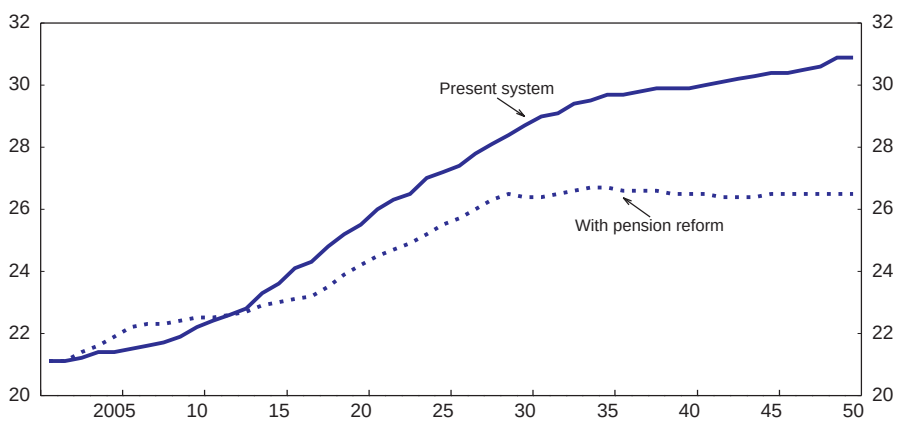
skills is also important. The tapering off with age in the incidence of job-related training, is less marked in Finland than in most other OECD countries (OECD, 1999), although it is greater than in other Nordic countries. However, the recent setting up of the National Programme on Ageing Workers has raised awareness of these and other issues related to an ageing workforce. Perhaps the most obvious weakness of the existing pension system in terms of its effect on the demand for older workers, is that employers' social security contributions rise with the age of their employees to finance increased risk of unemployment and disability.[25] The previous *Survey* recommended pooling this risk by spreading the increased costs across the contribution rates of all employees and so removing an important disincentive to hire or retain older workers. The reform will not, however, address this problem. Indeed it could exacerbate it, because employee contributions will be higher for older workers and it is unclear where the effective incidence of such increases will finally lie.

Pension contribution rates will still need to rise substantially

Overall, the reform is officially estimated to halve the required increase in contribution rates expected by 2050 (Figure 14), with almost two-thirds of the reduced pressure on pension costs being attributable to the life expectancy adjustment (Table 9). Nevertheless, contributions will still need to be significantly

Figure 14. **Pension contribution rates**[1]
Per cent of wages[2]



1. Employment pension contribution of wage earners.
2. Annual sum of gross wages.
Source: Central Pension Security Institute.

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Table 9. **Components of pension costs**
Per cent of wages[1]

	2002	2020	2030	2050
Without reform	17.7	27.8	32.3	35.7
With reform	17.7	26.3	30.1	31.4
Difference	..	–1.5	–2.2	–4.3
of which:				
Life expectancy coefficient	..	–0.2	–0.9	–2.5
1.9% accumulation rate	..	0.5	0.8	1.1
Non-wage periods (studies, etc.)	..	0.8	1.5	2.7

1. Annual sum of gross wages.
Source: Central Pension Security Institute (2002).

higher, by an estimated 5½ per cent of wages on average over the period 2030 to 2050, than at present. Moreover, the effects of the reform over the next thirty years are more modest – for example contribution rates after the reform are estimated to increase by 5¼ rather than 7½ per cent of wages by 2030 – because the phasing in of various components of the reform means that the largest cohorts (the baby boom generations) are not strongly affected. A major reason why the expected increase in the age of retirement does not have a greater effect on lowering the required increase in contributions is that average pension levels are expected to rise. Not only might this have an adverse effect on incentives to extend working life, as discussed earlier, but it also directly counteracts the objective of putting the system on a sound financial footing. In this light some elements of the reform seem overly generous, especially given the uncertainty about the effects on employment rates of older workers that will be required to (partially) offset these higher costs. In particular, increasing the accrual rate from the age of 52 and extending the accumulation period to non-paid periods (such as periods of study), will probably have little effect on incentives to work longer, but does have a significant cost. Indeed, without these two elements of the reform three-quarters of the expected increase in pension costs (equivalent to 3¾ per cent of total wages) between 2020 and 2050 would have been avoided (Table 9).

Long-term public finances

Ageing will clearly have a major impact on public finances. Recent comparative exercises, which did *not* take into account the pension reform, suggested that the rise in total old-age-related spending (as a percentage of GDP) in Finland is expected to be the fourth steepest in the OECD (OECD, 2001a). Moreover, calculations using the same spending projections as an input suggest that, at least according to one measure of fiscal sustainability (measuring the immediate and permanent adjustment of the primary surplus required to avoid further fiscal

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policy changes), Finland did have one of the weakest positions in the OECD (Frederiksen, 2001).[26]

To consider the broader fiscal impact of ageing it is useful to examine recent long-term scenarios by the Ministry of Finance and Bank of Finland, both of which incorporate the effect of the pension reform. Both scenarios are broadly consistent with the current objective of keeping the general government surplus at 4½ per cent of GDP on average over the course of this decade, although thereafter there is some deterioration in the fiscal position. In the baseline scenario by the Ministry of Finance (2002b) public pension outlays (flat-rate national pensions and earnings-related employment pensions) are expected to rise by nearly 4 percentage points of GDP, to 14½ per cent of GDP in 2050 (Table 10). At the same time, more health and long-term care services for the elderly could lead to an increase in outlays by a further 4 percentage points. The projections illustrate the fiscal consequences, when holding total taxes (including social security

Table 10. **Public finances in the long run**
Per cent of GDP

	2000	2010	2020	2030	2040	2050
Total expenditure	46.7	45.6	47.6	50.4	52.0	53.1
Pensions	10.7	11.7	13.3	14.7	14.6	14.4
Health care	4.6	4.9	5.2	5.6	5.7	5.8
Long-term care	1.6	1.9	2.2	2.7	3.3	3.3
Interest payments	2.8	2.1	1.9	2.4	3.4	4.6
Other	27.0	25.0	25.0	25.0	25.0	25.0
Total revenues	53.7	49.6	50.5	51.3	51.4	51.6
Taxes and statutory social security contributions	46.9	42.3	42.3	42.3	42.3	42.3
Employment pension contributions	7.2	8.3	8.8	9.3	9.3	9.2
Other	6.8	7.3	8.2	9.0	9.1	9.3
Financial balance	7.0	4.0	2.9	0.9	–0.6	–1.5
Primary balance[1]	8.1	2.5	–0.1	–2.3	–2.7	–2.5
Gross debt level	52.5	35.4	31.1	40.5	58.2	77.8
Net financial assets	4.0	32.5	52.9	52.7	35.8	16.2
Central and local government	–41.0	–22.5	–18.3	–27.7	–45.4	–65.1
Pension funds	45.0	55.0	71.2	80.4	81.2	81.1
Memorandum items						
Assumptions (in per cent)						
Labour productivity (% change)	2.8	2.2	1.8	1.8	1.8	1.8
Employment rate	66.9	68.7	71.2	72.5	72.8	73.2
Unemployment rate	9.8	7.0	7.0	7.0	7.0	7.0
Real interest rate	2.0	4.0	4.0	4.0	4.0	4.0
Inflation (% change)	3.4	2.0	2.0	2.0	2.0	2.0

1. Financial balance before net interest income.
Source: Ministry of Finance (2002b).

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contributions) and other expenditures stable as a share of GDP. The general government surplus would fall steadily below the current medium-term objectives from around the end of this decade, with the primary balance going into deficit from around 2020. After that date the combined net debt of central and local government would rise continuously from 18 to 65 per cent of GDP by 2050, while pension fund assets would stabilise at about 70-80 per cent of GDP.

The Bank of Finland (Kinnunen, 2002) has conducted a similar exercise, although in this case it was assumed that tax rates would adjust to avoid central government finances going into deficit and to maintain the surplus of the social security funds in the range of 2 to 3 per cent of GDP. In this case the pension reform would offset only about one-third of the tax increase needed to cope with ageing, with the long-run increase in the tax ratio being 4 rather than 6 percentage points.

Long-term fiscal sustainability is very sensitive to the future rate of return that pension funds earn on their assets (Box 4). The Ministry of Finance scenario assumes a long-term real rate of return of 4 per cent. If this is reduced by 1 percentage point then the general government financial balance would be more than ¾ of a percentage point of GDP lower over the period between 2020 to 2030 and around 1 percentage point lower over the period to 2050. This in turn would imply an earlier and sharper increase in the debt ratio that would be increased by about 25 per cent of GDP by 2050.

There are also considerable uncertainties about the effect of ageing on health care costs. A variant of the Bank's scenario shows that modest public sector productivity growth in ageing-related public services (mainly health care) would improve long-term fiscal sustainability by more than the pension reform. On the other hand more capital intensive health care or increases in the relative wages of health care workers could substantially raise costs.

An increase in the employment rate will be key to containing the long-term fiscal pressures that arise from an increasing dependency ratio. The Bank of Finland scenario suggests that the government's long-term goal of a 75 per cent employment rate could be reached by 2030, while the Ministry of Finance projects the employment rate will reach 73 per cent by 2050, compared to the current rate of 67½ per cent. On the basis of international comparisons, there is clearly scope for raising the employment rate of older workers (especially men), but estimates of the overall effect of the pension reform on the aggregate employment rate are relatively modest. Those from the Bank, Ministry of Finance and Central Pension Security Institute suggest an increase in the employment rate by around 1 to 2 percentage points over the coming decade and 2 to 4 percentage points by 2050. To realise an increase in the employment rate by more than this probably requires far-reaching labour market reforms on top of the change to the pension system.[27]

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Box 4. The institutional set-up and pre-funding of the earnings-related pension scheme

Part of the earnings-related pensions is pre-funded. The funds are managed by a large number of private pension institutions. The schemes for private sector employees are run by insurance companies, company pension funds and industry-wide pension funds, with the insurance companies dominating with a market share of about 80 per cent. The market value of private sector pension fund assets is currently close to EUR 50 billion (a third of GDP) and that of the public sector pension fund nearly EUR 20 billion. These assets are booked as general government assets, even though the institutions that manage them are mutuals and thus "owned" by the employers, who are the customers. As a counterpart, the contributions are also recorded as government receipts.

Pre-funding has no effect on the size of pensions, but it affects the future path of contribution rates, which also crucially depends on the rate of return on the funds. Concerning the latter, the institutional framework, in particular, the prudential rules and degree of competition among the institutions managing the funds is important.

Various proposals have been made to raise competition among the institutions managing the pension funds. Currently the Ministry of Social Affairs and Health sets a yield requirement annually, following proposals by the insurers, for all pension institutions, while the solvency level is calculated based on a portfolio model. If an institution does not reach this rate of return, its solvency capital decreases. If yields are better than required, profits are used to raise solvency or to pay premium refunds to the employers. In principle, the size of premium refunds signals the performance of the institutions managing the pension funds and should lead to competition among them. However, if an employer changes the pension institution, refunds are lost, which makes switching institutions generally unprofitable, even if the new provider could obtain better returns on the invested assets. One proposal has thus focused on increasing the transferability of assets among institutions. Also changes to the common yield requirement have been discussed. A high yield requirement does not necessarily ensure high benefits for future pensioners, because it also makes solvency requirements tighter, thus reducing the possibility to invest in riskier assets. A consequence of the current framework is that portfolios of the insurers are quite similar. The Ministry of Finance has proposed the use of a minimum yield requirement that would not change frequently, which would make it easier to pursue alternative investment strategies. But the success of this approach will depend on there being sufficient competition among the pension institutions. No decisions for changing the current set-up have yet been made.

In both scenarios an underlying rate of productivity growth of 1¾ per cent is assumed. Although this does not appear optimistic in the light of the performance over the last decade, as pointed out in Chapter IV, this has been heavily

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dependent on the success of the ICT sector. As demonstrated in an earlier exercise by the Ministry of Finance (2002b), even temporarily slower growth over the medium term can imply a very substantial worsening of the fiscal position over the longer term. In particular, a temporarily lower growth rate, averaging 2 percentage points less than the baseline over the period 2003-06, leads to a decline in the public sector surplus of more than 4 percentage points of GDP by 2010 which steadily continues to worsen thereafter. This occurs despite an assumed acceleration in growth thereafter that leaves the level of GDP unchanged from 2020. This variant usefully illustrates the sensitivity of long-run sustainability computations to "initial conditions", and more generally underlines the benefits of ensuring that the fiscal position is strong going into the next decade.

Summary

The pension reform includes several bold elements – particularly the higher accrual rates for older workers and the abolition of the pension ceiling – that together with the further curtailment of early retirement options are likely to increase incentives to work longer. Official estimates suggest that the current reform proposals will raise the average retirement age by 1½ years by 2050 (Takala and Uusitalo, 2002). Taken together with the effects of the earlier reforms the rise could be double that and so consistent with the government's long-term aim of raising the retirement age by 2-3 years. Other components of the reform package are also to be welcomed. In particular the longevity adjustment, which bolsters the sustainability of the system against one of the major demographic uncertainties, and the move to assessing pension rights on the basis of earnings over the whole career, which should improve labour market flexibility. Nevertheless, the phasing in of the reform and the overall increase in generosity will limit the improvement in the sustainability of the system. It also remains to be seen how far tightening access to some early retirement schemes and discontinuing others will put more pressure on the remaining schemes, which could seriously undermine the beneficial effects of incentives to work longer.

These uncertainties about the longer-term effects of the pension reform suggest caution in modifying the present fiscal objectives. Both of the long-term scenarios, produced by the Ministry of Finance and Bank of Finland, show that while any major deterioration in the fiscal position may be contained by the pension reform, significant tensions remain. In particular, the Ministry of Finance projection implies a stable aggregate tax rate (but with a steady deterioration in the debt ratio) and the Bank projection implies a rising tax rate, whereas significant improvements in the labour market are likely to require a *falling* tax rate. While such long-term scenarios are inevitably sensitive to a number of difficult assumptions, they do underline the merits of sticking to the current central and general government surplus objectives over the remainder of the decade as well as

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continued restraint on public expenditure. At the same time there is also scope for positive surprises. For example, it is difficult to assess what the net effect of the pension package will be on effective retirement age and the aggregate employment rate. If the combined effects of the pension reform and other measures were to lead to a faster increase in the employment rate than currently anticipated, then this would suggest a need to re-examine the fiscal objectives in a positive light.

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III. Enhancing the effectiveness of public spending[28]

Finland is committed to high quality and extensive public services, many of which are provided equally to rich and poor, stemming from a strong egalitarian ethic that it shares with other Nordic countries (Figure 15).[29] This ethic also manifests itself in a high degree of income re-distribution but necessitates a heavy tax burden that is becoming increasingly difficult to sustain due to tax competition and the need to harmonise certain taxes with other EU countries. Pressure on taxation implies a need for adjustment of expenditure. Despite the strong fiscal discipline that was exercised following the surge in public spending during the recession of the early 1990s and the subsequent pruning of expenditure as a share of GDP back to the euro area average (Figure 16), several issues need to be tackled to ensure the long-term sustainability of the public accounts:

- *First*, while acknowledging Finland's record of maintaining overall fiscal discipline in the wake of the severe recession of the early 1990s, the government has recently missed some of its medium-term fiscal targets. This partly reflects the focus on annual budgets which does not give enough prominence to the medium-run targets that are laid out in the Government Programme at the beginning of the term of office.

- *Second*, it is important that spending pressure due to ageing is properly anticipated and contained by reform. In particular, as is emphasised in Chapter II, the September 2002 agreement on pension and unemployment benefit reform to lengthen working lives and better target expenditure, needs to be fully implemented and followed-up with monitoring and, if necessary, additional action.

- *Third*, while Finland was among the first countries to change public-sector management practices, it has not followed through to the same extent as the leaders in this regard. Uneven application of good management practices and other features of public service provision mean that some spending programmes contain little incentives to contain costs and demand.[30] Also, the private sector continues to play only a small role in providing

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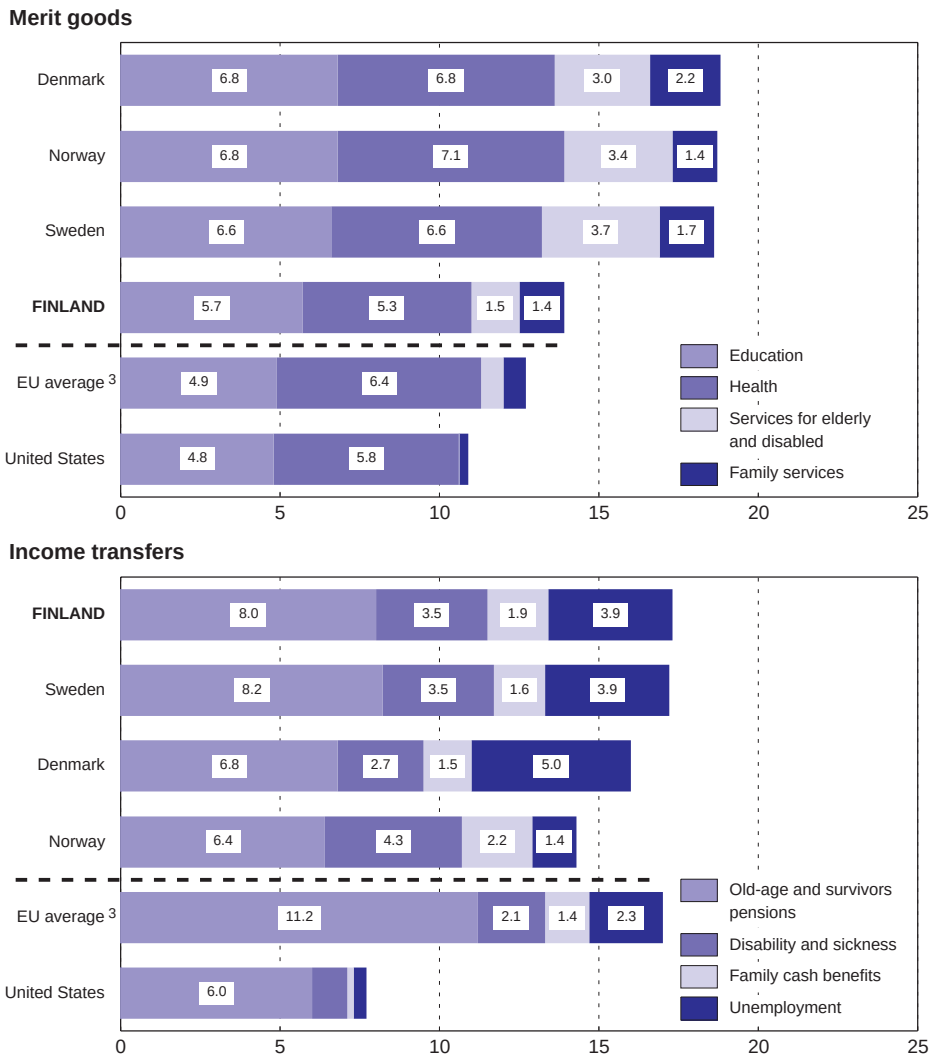
Figure 15. **Major current government outlays**[1]
Per cent of GDP, 1999[2]



1. Denmark, Finland, Norway and Sweden are broadly comparable because transfers are taxed whereas for most of the EU and the United States they are not. Removing the effects of different approaches to taxing transfers reduces overall spending ratios by 4 to 6 percentage points.
2. Or 1998 when not available. Education data always concern 1998.
3. Weighted average based on 1995 GDP and purchasing power parities; excluding Luxembourg.

Source: OECD (2001), *Social Expenditure Database* and *Education at a Glance*.

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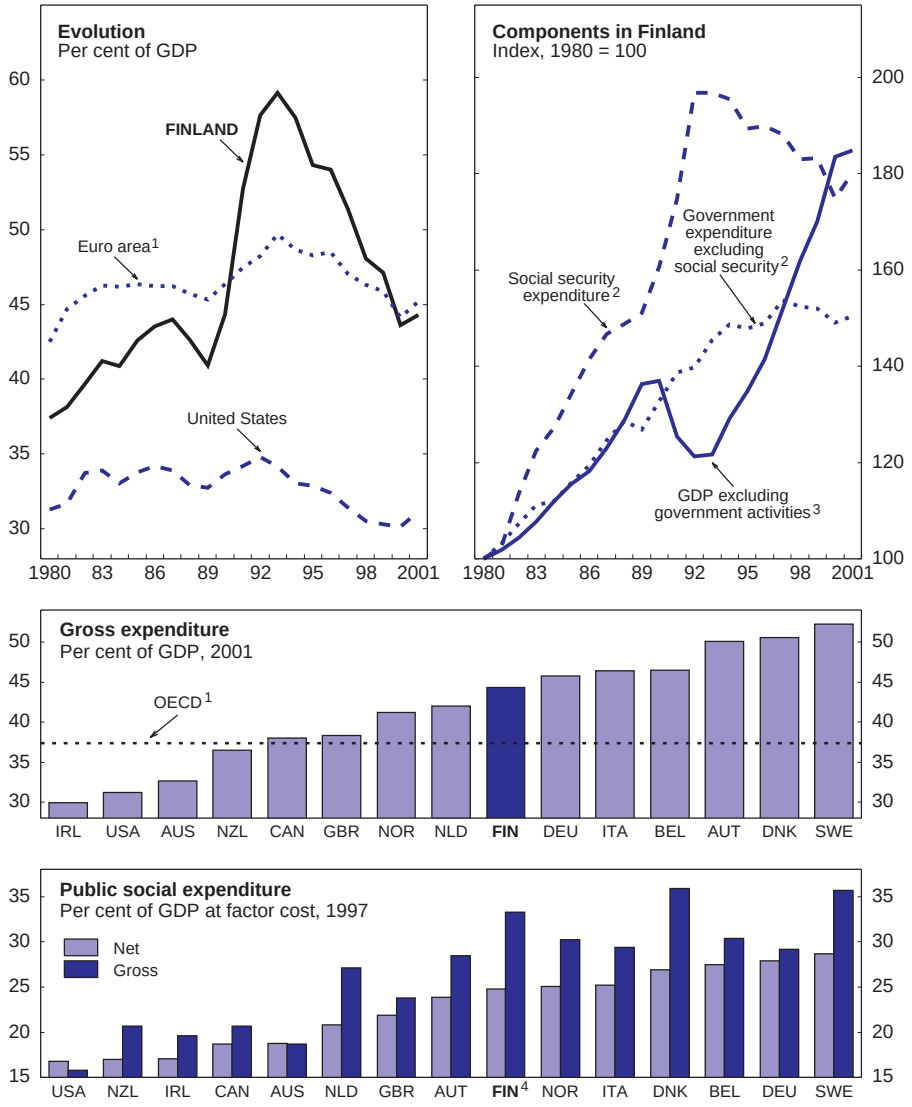
Figure 16. **Public spending in international perspective**



1. Weighted average.
2. Nominal data adjusted to 1995 constant prices using the deflator for government final consumption expenditure.
3. Gross value added at fixed prices for total production less government activities.
4. Net expenditure excludes the church tax paid by the majority of Finnish taxpayers.

Source: Statistics Finland and OECD.

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services, reflecting a reluctance of government entities to explore alternatives to in-house provision and resulting in weak contestability.

The first section of this chapter highlights the major forces shaping public spending in Finland in recent years. The second section then discusses various ways of improving aggregate fiscal discipline. This is followed by an examination of public expenditure issues in local government, including an assessment of the potential gains from further mergers between municipalities. The chapter then discusses the key issues with regard to transfers and examines efficiency issues in health, elderly care and education. It concludes with a wrap-up section which provides a set of specific policy recommendations.

Forces shaping public spending developments

Strong fiscal discipline after the 1990s crisis has generally prompted efficiency gains

The severe recession of the early 1990s involved both a large fall in GDP and dramatic increases in public spending due to rising claims on both national insurance schemes and municipality-based social assistance. This surge in public spending in turn prompted not only changes to social transfer policy, but also cutbacks in other areas so that growth in non-social-security spending flattened considerably from the mid-1990s (Figure 16). Currently, total public spending as a share of GDP is some 15 percentage points below the peak of the early 1990s, but still slightly above the pre-recession level and well above the OECD average. Overall, the strong spending restraint is frequently seen as having been reflected more in efficiency gains than in significant cuts in the quantity or quality of services.[31] However, the cuts were largely across the board, rather than targeted, so that some imbalances have emerged, in particular tensions over pay in certain high-skill areas. Spending pressures have also intensified since the budget moved into surplus in 1998. Further expenditure restraint will require increasingly profound changes as the more easily realised savings have already been made.

Spending pressures due to ageing are rising

As discussed in Chapter II, ageing-related spending will soon start to rise rapidly. Growth in the population aged 65 and above will begin to increase significantly and Finland could experience one of the most dramatic switches from surplus to deficit compared with other countries in coming decades. The recently legislated pension reform will mitigate, but not erase, the future hike in pension payments. Moreover, increases in other areas of ageing-related expenditure such as health care and long-term care for the elderly, may also be large unless significant improvements in cost efficiency are made.

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Pressures to reduce the tax burden

Economic and political pressures to reduce taxes are providing additional impetus to restrain spending. Finland has one of the most highly taxed economies in the OECD with tax revenue being equivalent to more than 45 per cent of GDP, making it more susceptible to tax-related inefficiencies and more vulnerable to tax competition.[32] Increasing tax-base mobility has already resulted in strong competition for some tax bases, which is likely to have been one consideration leading to the introduction of the dual income tax system. Tax competition also adds to the argument that labour-income tax and employer contributions should be reduced. In addition, tax harmonisation and freedom in the movement of goods in the European Union will continue to force reforms. In particular, there is pressure to reduce the tax levels on alcohol and a recent decision from a case brought to the EU court relating to the taxes on car purchases will also result in reduced revenue.[33]

Maintaining aggregate fiscal discipline

Fiscal discipline and policy goals concerning public expenditure are principally achieved through medium-term spending rules set out in the Government Programme, a document issued at the beginning of the term of office. However, there appears to be a problem in ensuring that the fiscal rules of the Programme are adhered to in the annual budget process, as exemplified in the slippage that has occurred in recent years (Chapter I). As explained in Box 5, the annual Operating and Financial Submission documents include medium-term plans by spending ministries but little use is made of them, except for the first year. A move towards a more genuine, permanent medium-term rolling budget should be considered to augment the fiscal guidelines provided by the Programme (OECD, 2002c).[34] The architecture of government financing might also be altered to deal with the problem of lack of flexibility and co-ordination in spending across ministries – so-called "stovepipe" government. While this is a common and not easily eradicated issue, it is often amplified where there are coalition governments due to increased rivalry between ministries. There is some recognition of this by the government. In 2001 a ministerial steering group produced a report which advocated stronger cross-administrative capacities, suggesting that programmes cutting across ministries could be placed in the Prime Minister's Office and run by specially appointed ministers (Ministry of Finance, 2001a).[35] Another budgeting issue is that efforts should continue to introduce more purpose to the first round of parliamentary discussions in the annual budget process (Box 5).

Moreover, the Government Programme's spending rules should be enhanced. First, fiscal discipline could be strengthened by a rule that stipulates that, in general, spending decisions within a budget year need to be offset by spending cuts elsewhere – rules like this have proved successful in other countries, notably the United States. Second, the inclusion of interest payments in the

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Box 5. The annual budget process

The central government's annual budget process consists of a series of interactions between the Ministry of Finance and the spending ministries that begin in January and end in September when the budget proposal is presented to Parliament for discussion. Parliament generally passes the budget in December ready for the start of the financial year in January (Table 11). A key event in the budget calendar is the late-August Government Budget Session when budget decisions and documentation are finalised in a meeting between the Minister of Finance and spending ministries. Parliament tends to pass many amendments to the proposed budget but they tend to be minor.[1] By convention, there are at least two supplementary budgets over the course of the year. According to the Ministry of Finance, to-date these have not been a source of significant relaxation of fiscal discipline and typically perform a mechanical role, such as the correction of estimated appropriations to account for changes in economic developments.

The spending rules laid down in the Government Programme need to be strongly applied to the annual budget process to ensure that medium and long-run considerations are taken on board. Although the initial *Operating and Financial Submissions* made in January outline spending for the three years beyond the budget year, they do not play a significant role and are changed for each budget with little reference or reconciliation with the previous year's figures.[2]

Concrete parliamentary involvement in the budget process effectively occurs only once, when it assesses the budget document in the autumn. Although there is a discussion of budget policy in April, no decisions are made and it serves more for parliamentarians to make a case for additional spending. The somewhat inactive role of these discussions suggests that maintaining two ministerial submissions, one in January and one in May, is perhaps unnecessary (see OECD, 2002c). After a vote in 2000 concerning the annual aggregate spending limit decision, the spending limits in the budgets from 2001 onwards have been discussed on the basis of a so-called Prime Minister's announcement, without a vote of confidence.

1. The government's budget proposal is assessed by the parliamentary Finance Committee, which conducts hearings and then makes a report which forms the basis of the final parliamentary discussions prior to the vote on the budget.
2. The determination of the state budget one year at a time is written into the constitution (OECD, 2002c).

target to maintain government expenditure constant at the 1999 level has implied leeway to spend more on primary outlays than initially intended as spending on interest payments has declined. Finally, and maybe most importantly, numerous areas are identified below where spending should be scrutinised carefully, implying that the government target to cap real spending could have been more ambitious in the past and should be more so in the future.

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Table 11. **The annual budget process**

Central government	
December-January	Spending ministries send preliminary budget proposal, plus plan for the following three years to the Ministry of Finance, referred to as the Operating and Financial Submissions.
January-February	Discussions between Ministry of Finance and spending ministries.
March	*Government spending limits discussion* – Government decides spending limits for each ministry.
April	Parliamentary discussion of budget policy.
May	Spending ministries submit budget drafts to Ministry of Finance.
July	Draft budgets by spending ministries and first draft of the total budget prepared by the Ministry of Finance released publicly.
Mid-August	Bilateral budget discussions between Minister of Finance and other ministers.
Late August	*Government budget session* – Final budget proposal decided.
Early September	Budget proposal presented to Parliament.
1 January	Start of fiscal year.
Parliamentary timeline	
September to early November	Parliamentary Hearings on the budget proposal, Sectoral Committee analysis.
November	Finance Committee sections draft and submit reports.
End November	General Finance Committee debate and amendments. Completion of Finance report.
Early December	Parliament amends and passes budget.

Source: Ministry of Finance.

There is also room for greater clarity in the technical details of the spending rules. For example, details on the deflator used to evaluate whether real spending has remained on course are not readily available and there is no explicit commitment to consistency in the deflation method over time. This criticism aside, central government accounting generally follows good practice. Accrual accounting has been widely adopted to avoid the misperceptions of budgetary situations that can arise in cash accounting. And long-run budget simulations have been extensively used to assess the consequences of ageing (Chapter II). These simulations incorporate generational accounting techniques that highlight the impact of shifts in the demographic structure on revenues and expenditure (Ministry of Finance, 2002c).

Public expenditure issues in local government

Similar to the other Nordic countries, government in Finland is highly decentralised, with local government delivering a large share of public services. A total of 448 municipalities are responsible for all public health care services, primary and secondary education, as well as amenities such as water, sewerage, local transport and roads (Table 12). Municipalities are also responsible for providing

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Table 12. **The distribution of spending responsibilities between central and local government**

Activity	Provision	Comment
Municipality responsibilities		
Health and education		
Primary health care	Municipality-run clinics sometimes with co-operation between clinics. General practitioners and dentists employed by the municipalities.	Payment of general practitioners is based on a national compensation scheme.
Hospital care	Municipalities purchase hospital services from a joint municipal board ("Hospital district").	The 20 hospital districts are owned by municipalities. Competition between the districts for customers is limited.
Primary and secondary education	Municipality-run schools. Vocational schools are usually run by joint municipality boards. In addition there are some private and state schools.	The state pays about half of the expenditure of primary and secondary education in the form of government transfers (state grants), which are calculated by differentiated unit prices per student.
Polytechnics[1]	Polytechnics are mainly municipally run or private. They are run by one municipality (7), joint municipality (11), limited companies (8) or foundations (3).	The polytechnics receive their core funding as state grants (which consist technically of two parts: the state share and the share of local governments).
Social services		
Child care	Municipality-run day-care centres and municipality-organised (and subsidised) "family" day care.	Child care provision is governed by national rules about charges for child care. Parents can take a cash payment in lieu of municipality-provided care but only a very small minority of households choose this option.
Elderly and disabled care	Municipalities are responsible for the majority of elderly care outside of hospitals.	. .
Other social services	Socially targeted housing and general social assistance.	. .
Other services		
Water, waste and energy	Municipalities purchase services from providers.	Municipalities are commonly the (sole) owners of the water provision service. Recent changes to regulations mean that water charges can no longer be treated as general income by municipalities.
Transport	Municipalities purchase services from providers.	Competitive tendering for bus services is common.
Infrastructure	Supervision of land use and building activities, road maintenance and construction (except national highways).	. .

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Table 12. **The distribution of spending responsibilities between central and local government** (*cont.*)

Activity	Provision	Comment
Central government (and, KELA, the Social Insurance Institution) responsibilities		
Health	Reimbursements for certain medical treatments are administered via KELA.	Reimbursements are financed by employers, employees and also by central government.
Universities[1]	All 20 universities are owned by the state.	The universities receive their core funding from the state budget. About 30 per cent of their income is derived from commercial sources or academic projects outside the universities.
Student financial aid	Paid and administered by KELA.	Student financial aid consists of a study grant, housing supplement and government-guaranteed loan. It is payable for students in upper secondary and tertiary as well as adult education.
Social insurance	Unemployment, sickness, disability, family benefits schemes are administered by KELA and unemployment insurance funds.	. .
Public-employment services and active labour market policies (ALMPs)	The public employment service incorporates the administration of ALMPs and is run via a nationwide network of offices.	The majority of the subsidised jobs which form part of ALMP policy are channelled towards municipalities.

1. Tertiary education consists of two pillars: polytechnics and universities.
Source: OECD.

"bottom rung" social assistance for those who do not qualify or who have used up their rights to the benefits provided by the Social Insurance Institution of Finland, KELA.[36] Although there is a layer of regional government, its role is confined to co-ordinating regional planning and aid.[37]

Municipality financing: some aspects of the system require attention

Municipalities have considerable freedom in how revenue is spent and, in principle, reasonably strong influence on the size of revenue through their right to set the flat rate local income tax but also by attracting corporate tax base (Table 13). The current funding system was put in place as part of major reforms in 1993, prior to which central government had a more direct influence on spending via earmarked grants.[38] About half of municipal revenue comes from taxes, a quarter from charges for services (mainly water, sewerage and electricity) and about one-sixth from central government block grants (Figure 17 and Table 13).

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Table 13. **Financial resources of municipalities**

Resource	Comment
Income from taxation (about 50 per cent of revenue on average)	
Household income tax	Municipalities receive directly the revenue from a flat rate tax levied on households that they can set at any level. A large majority of municipalities charge between 18 and 19 per cent.
Corporate income tax	Municipalities receive (in 2003) 19.75 per cent of the corporate income tax revenue within the municipal boundary. This can vary significantly over time, creating problems for financial planning.
Property tax	Property taxes are subject to upper and lower limits set by law.
Tax equalisation	If a municipality's potential tax revenue per capita (the tax-base per capita times the national average rate of municipality tax) is: – < 90 per cent of the national average, then it receives the difference (around three-quarters of municipalities in this group). – > 90 per cent of the national average, then it pays 40 per cent of revenue in excess of the 90 per cent threshold.
Block grants from central government (about 15 per cent of revenue on average)[1]	
Ministry of Social Affairs and Health	For health the grant is a percentage of estimated health costs based on population characteristics including morbidity, and for social affairs the grant is a percentage of estimated social welfare costs based on population characteristics including the level of unemployment.
Ministry of Education	The grant is a percentage of estimated costs based on student numbers.
Fees and user charges (about 25 per cent of revenue on average)	
Water and sewerage	From 2002 a separate account for water and sewerage services must be maintained.
Energy	. .
Other	Includes fees for primary health care. Charges for hospital care go to the hospital districts and so do not enter directly into the municipal accounts.
Other budget resources	
Bank loans	Municipalities are supposed to maintain a medium-term balance between current receipts or disbursements. Many borrow to finance building projects while borrowing to help cover running expenses is rare.
Off-budget resources	
Subsidised jobs	Municipalities are the major clients for the central government's subsidised jobs.

1. The annual block grants are determined by a special committee comprised of representatives from the municipalities and central government.

Source: OECD.

The main tax revenue comes from the flat tax on household income. Municipalities also receive a share of the local corporation tax generated. Although the share itself is fixed by central government, revenue nevertheless varies widely across

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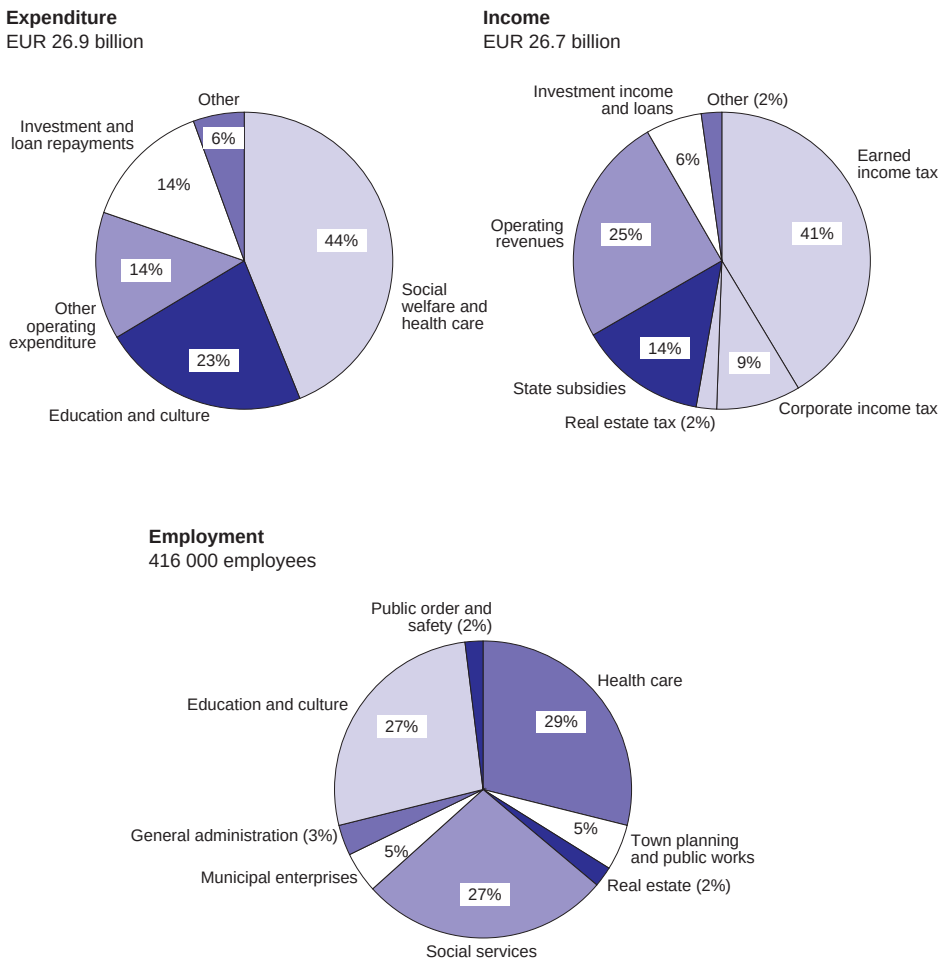
Figure 17. **Municipality income, expenditure and employment**
2001



Source: The Association of Finnish Local and Regional Authorities.

municipalities as well as over the business cycle due to variation in the tax base. A large share of revenue can be treated in a general account by the municipality, thus facilitating account management and flexibility in resource allocation; exceptions are revenue from water, sewerage and electricity charges which since 2002 must be in separate accounts. A potentially significant source of budgetary abuse

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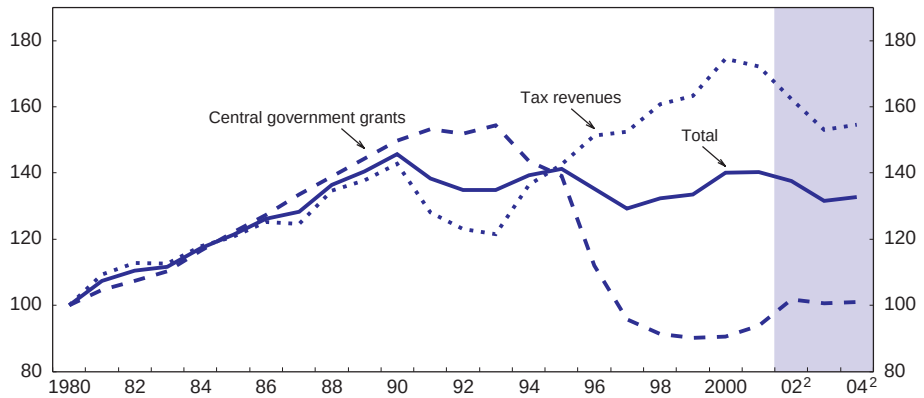
by municipalities lies in the moral hazard created by the use of discretionary central government support to those municipalities in financial difficulties. Although there have been no signs of widespread abuse of this system as an implicit financial guarantee by municipalities, the authorities should remain vigilant to adverse developments.[39]

Although the municipal financing arrangements are *prima facie* laudable for advocating a strong role for local government in public expenditure, the system faces certain practical problems and in reality does not always work well. One problem is that municipalities can encounter considerable fluctuations in municipal revenue, mainly due to variations in the tax base over the business cycle. Indeed the Ministry of Finance has calculated that about 40 per cent of the decrease in corporate tax revenue that occurred between 2000 and 2002 is attributable to weakening macroeconomic conditions. Over the 1990s municipal tax revenue rose strongly and was countered by cuts in central government grants and this resulted in relatively stable income for municipalities (Figure 18).[40] Although some Ministry of Finance documents seem to suggest that this adjustment was fully intended (Ministry of Finance, 2001b), there is no automatic mechanism via the block grant funding formula, or official discretionary policy to protect municipalities from cyclical variation in revenue. However the fact that this "policy" is not

Figure 18. **Municipality revenue: the roles played by tax and central government grants**

Volume indices, 1980 = 100[1]



1. 1995 constant prices using the deflator for government final consumption expenditure.
2. Estimates for 2002 and projections for 2003-04.

Source: Statistics Finland (2002), *National Accounts 1993-2001* and The Association of Finnish Local and Regional Authorities.

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explicit and widely broadcast to municipalities may add uncertainty to municipal decisions and lead to inappropriate contingent budgeting by them. A need to increase the predictability of public funding is acknowledged in a recent Ministry of Finance report (Ministry of Finance, 2002c). The authorities have also recognised the need to reduce the exposure of municipalities to corporation tax revenue, which is one of the main sources of fluctuating income. In the latest of a series of cuts, the municipality share of 23¼ per cent of corporate tax revenue will be reduced to 19¾ per cent in 2003.

The tax equalisation scheme (Box 6) is frequently criticised. One problem is that there is a degree of uncertainty about the level of equalisation payments and they are made with some delay, both of which can amplify the problem of unpredictable and fluctuating revenue.[41] Even after equalisation, important revenue differences remain between many municipalities; most significantly between the large urban municipalities and the rest of the country. For example, before equalisation, tax revenue in Helsinki was nearly EUR 3 900 per capita in 2001 compared to around EUR 2 200 in one of the larger recipient municipalities, such

Box 6. The cross-municipality tax equalisation scheme

Tax equalisation plays an important role in municipality finances. In systems that rely heavily on tax to fund municipalities, there is far too much variation in tax bases for municipalities to be capable of providing similar standards of public services at similar tax rates. For this reason and for broader equity concerns, tax equalisation schemes are often applied. In the Finnish equalisation scheme municipalities are evaluated on the basis of the level of their potential tax revenue per capita (PTR) which is the tax revenue per capita they would get if they applied the national average municipal tax rate. The scheme compensates all municipalities whose PTR is below 90 per cent of the national average with a per capita amount which brings tax revenue up to the 90 per cent threshold. Municipalities whose PTR is above this threshold, make a contribution to the equalisation scheme equal to 40 per cent of potential tax revenue in excess of the 90 per cent benchmark (Table 14). Until 2002, the maximum contribution was limited to 15 per cent of a municipality's tax revenue. This cap has subsequently been removed although there has been some compensation by increases in grants for social and health care.* Despite the removal of this limit, Finland's scheme equalises local government revenues to a much lesser extent compared with the equivalent schemes operating in Denmark, Norway and Sweden.

* The equalisation scheme does not have significant implications for the government budget as the total amount given by contributing municipalities roughly equals the amount paid out to recipient municipalities.

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as Joensuu. Equalisation only changes the figures to around EUR 3 250 for Helsinki and EUR 2 350 for recipient municipalities (Table 14). These differences in tax revenue largely reflect differences in tax base given that tax rates are fairly uniform across municipalities. It can be argued that while some incentive for municipalities to strengthen their tax bases is welcome, the large number of factors influencing location decisions that are largely beyond their control suggests the incomplete tax equalisation scheme allows some municipalities to enjoy above-average tax income and, in principle, below-average tax rates, which largely reflects happenstance rather than active policy. On the other hand, despite their relatively generous tax revenue, the large municipalities such as Helsinki and Espoo are popularly viewed as having more stretched public services compared with rural areas. It is not clear whether this is due to misperception or genuinely large cost, demand or efficiency differences between urban and rural areas, which outweigh the tax "advantage" of urban municipalities.

Table 14. **Evaluating the impact of the tax-equalisation scheme**[1]
Based on 2001 municipality income figures, in EUR

Municipality[2]	Population (thousands)	Tax revenue per capita before equalisation	Tax revenue less estimated per capita equalisation	Estimated equalisation per capita
Contributors				
Helsinki	555	3 880	3 267	–614
Espoo	213	4 187	3 451	–736
Tampere	195	2 871	2 661	–210
Vantaa	178	3 162	2 836	–326
Turku	173	2 668	2 539	–128
Oulu	121	3 109	2 804	–305
Other contributors[3]	22	2 770	2 601	–169
Recipients				
Kuopio	87	2 325	2 347	21
Joensuu	52	2 263	2 347	84
Mikkeli	47	2 160	2 347	187
Rovaniemi	35	2 258	2 347	88
Jyväskylän mlk	32	2 173	2 347	174
Savonlinna	28	2 213	2 347	134
Other recipients[3]	5	1 827	2 347	519

1. The estimated equalisation is based on calculating the "potential" tax revenue per capita (PTR) which is the tax revenue per capita the municipality would get if it applied the municipal tax rate at the national average (17.68 per cent in these calculations). If the municipality's PTR is over 90 per cent of the national PTR then it contributes 40 per cent of the difference. If the municipality's PTR is less than 90 per cent of the national PTR then it receives the difference.
2. The individual municipalities listed are the largest in terms of population size.
3. Unweighted average.

Source: Ministry of the Interior.

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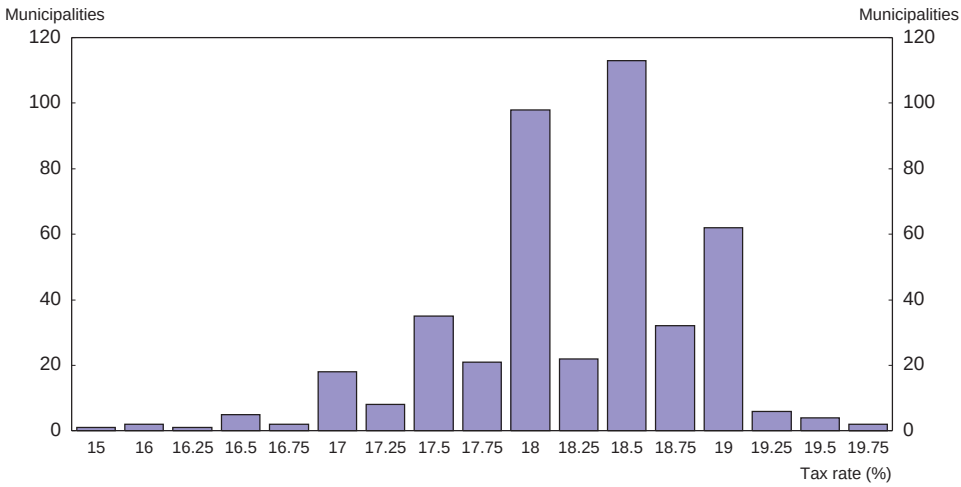
Finally, the municipalities do not appear to exercise their revenue-raising and spending powers as might be expected. Particularly puzzling is their tax rate-setting behaviour. The vast majority of municipalities set their income tax rates between 18 and 19 per cent (Figure 19). This is surprising given municipalities' freedom in rate-setting and that factors are operating other than voter preferences and inter-municipality competition as is commonly suggested in the literature on fiscal federalism.[42] One possible explanation is tacit agreement not to use the municipality income tax rates to compete for household tax-bases because aggressive competition between municipalities may compromise fruitful economic co-operation in other areas, such as the combined provision or purchase of services.[43] Also, municipalities may find competition over corporate tax revenue more fruitful, given the political risks and economic trade-offs of changing the household tax rate; the large number of business centres set up by municipalities perhaps bears testimony to this. Finally, it may also be a strategic advantage for municipalities to present a relatively united front of income tax rates in their annual collective negotiations with the central government to determine grant levels.

The authorities may be able to alleviate the problems associated with tax revenue and equalisation by further cuts in the municipalities' corporate tax share. Indeed, it is instructive that both Sweden and Norway have abandoned altogether having local authorities get a share of company tax revenue. Also,

Figure 19. **The distribution of the flat-rate municipal income tax**
2001



Source: Ministry of the Interior.

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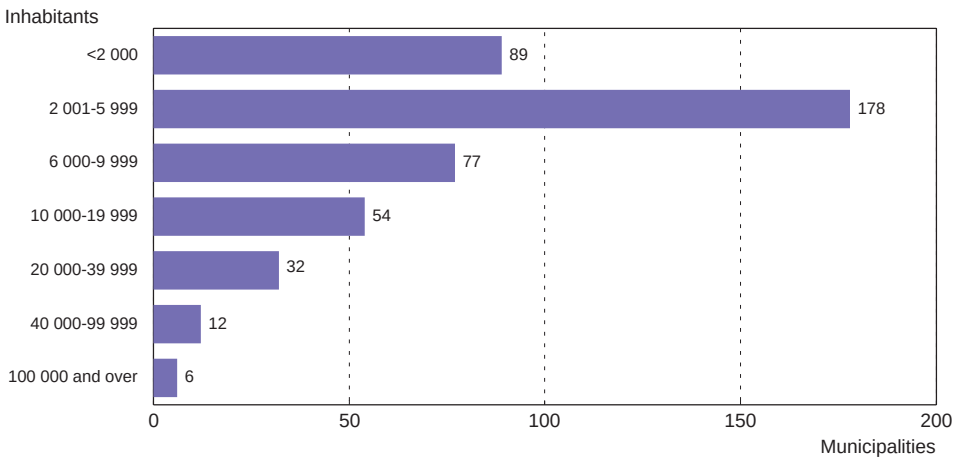
reductions in the delays in equalisation payment might be considered, perhaps by using an "immediate" equalisation system similar to that in Denmark. Investigation as to why some of the large urban municipalities with relatively high per capita revenue apparently have more difficulty in coping with the pressure of demand for some public services compared with rural areas may reveal a need to review the production structures of public services and possibly the block-grant funding formula.

Evaluating the potential gains from merging municipalities

Rural depopulation and economies of scale in many public services raise the question of whether many of the smaller municipalities should be merged, a question that has arisen in other Nordic countries. Finnish municipalities vary enormously in population, and over half of them, mostly in rural areas, have less than 6 000 inhabitants. At the same time, there are a handful of urban centres with populations of more than 100 000, notably the municipality of Helsinki with over half a million inhabitants (Figure 20). Therefore, at the risk of the loss of some degree of local representation, mergers among the small municipalities would make for a more balanced structure and to some extent save on administrative

Figure 20. **Municipalities by population**[1]
At end December 2001



1. Based on data for 448 municipalities and 5.2 million inhabitants.
Source: The Association of Finnish Local and Regional Authorities.

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overheads and generate other economies of scale.[44] This has long been recognised and a system of state support for mergers has been in place for 15 years. However, this scheme has not resulted in many mergers to date. The state is to increase the incentives to merge beginning in 2003, using non-earmarked per capita grants.[45] Given the slow pace of mergers among municipalities, the authorities should ensure other forms of administrative restructuring as well as further use of joint provision are fully exploited. At present joint provision is mainly used in the health sector and to some extent in education. The authorities should advocate other forms of joint provision; for example, there may be room for exploiting telecommunications technology to make economies of scale in local authority administration.

However it should be noted that administrative restructuring cannot be expected to resolve all the difficulties of providing public services in remote areas and what may be considered appropriate policy for urban areas may not be so for dispersed rural communities. Transport considerations and the size of markets and populations in rural areas can severely limit the feasible number of producers so that measures to increase choice and reap efficiency gains from competition between public-service producers are less effective. For example, choice in primary and secondary education in rural communities is often extremely limited or effectively non-existent and entrenched monopolies in the service sector are likely more common in rural than urban areas. In such circumstances there are risks in granting a large degree of autonomy to municipalities and pitfalls in some of the other reforms that are commonly recommended as improving public-sector efficiency.

A *general need for more benchmarking and accountability in municipalities*

Decentralised responsibility for public services can be more responsive to local tastes and conditions than centralised public administration because decision-makers are on-site and therefore better informed.[46] However, the gains from localised provision can only be realised if central government properly oversees that municipalities are providing accepted standards of service efficiently. Insufficient oversight can risk municipalities being poorly co-ordinated and having weak attention to efficiency, too much oversight and the system becomes decentralised by name only. In this regard, the relationship between central and local government in Finland generally seems well balanced and conforms to the notion of "subsidiarity" advocated by the European Union. Nevertheless, central government needs to play a more active role in ensuring municipalities focus on the right goals and produce services efficiently. At present output-oriented goal setting has not filtered down strongly to municipalities. And comparisons across producers, whether municipalities themselves or other entities, is underdeveloped, as also is outsourcing. This issue is discussed further in the sections on health and education spending below.

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Transfers: key reforms are underway

Transfers are an important component of Finland's commitment to a system of public support that ensures relatively high minimum living standards. As in other countries, there is room for improving existing transfer schemes with regard to issues such as targeting, poverty traps, dead-weight loss and so on. In particular, Chapter IV stresses that some of the transfer schemes to the unemployed require attention. At the same time, while transfer schemes can have problems, in some circumstances they can be preferable substitutes to public service provision. In particular Finland could make more use of vouchers, for example in childcare and elderly care, so as to put choice more directly in the hands of the user and generate competition between producers.

General observations aside, the most immediate issue in the sphere of transfers is the need to address the fiscal implications of ageing. The authorities have long recognised this and the sharpness and imminence of the demographic transition in Finland has made these issues urgent. In September 2002 agreement was reached on a package of reforms targeted principally at increasing the incentives for older workers to remain in employment (see Chapter II for further detail).

Ensuring efficiency in delivering public services

General issues

Outsourcing to private producers remains low

Despite the absence of legal impediments and a good measure of encouragement, the use of private producers for public services remains low in Finland, although in a recent survey municipalities claim to be working to change this.[47] To help level the playing field between public and private producers, the Finnish Competition Authority is pushing for greater exposure of government entities to the legal demands of the Competition Act. To date, areas of public service such as health and education have generally been regarded as beyond the scope of the Act because they are not primarily commercial in character. However, with the increasing use of market-based production, especially of ancillary services such as catering and cleaning, the case for applying the Competition Act has strengthened.[48]

Also adding to pressures for change, a working group on public provision set up by the Ministry of Trade and Industry issued its final report in 2001. The report broadly supports the current arrangements for the provision of public services by municipalities but recognises a need to further encourage outsourcing and, more radically, advocates the use of voucher systems (Ministry of Trade and Industry, 2001). Among the detailed proposals of the working group, are suggestions that municipalities be required to develop systems of cost accounting for their own production so that alternative arrangements can more easily be evalu-

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ated and as a check against cross-subsidisation in public provision. The working group also recognises a need for more joint production by municipalities in order to deepen customer bases and exploit economies of scale. A specific problem with the introduction of vouchers in Finland is that, depending on their purpose, they can be counted as taxable income. This is disadvantageous to municipalities and households as it involves a loss of resources to the central government and this could impede the introduction of voucher systems.

It remains puzzling to both policymakers and other observers that the level of outsourcing has remained so low. A recent report by the Finnish Competition Authority (FCA) concludes the principal problem lies in rigid attitudes and a lack of professional experience and understanding of outsourcing by municipality staff. The Authority also reports that municipalities have often blamed a lack of private sector producers as a reason for not using the open market to provide services. Of course there could be reasons beyond those found in the FCA's findings: in certain circumstances differences in value added tax treatment between public and private producers may be influencing municipal decisions.[49] Also, the predominance of permanent contracts in the public sector and problems in transferring from public to private sector pension arrangements may also be making outsourcing either politically unpopular for municipalities or simply too expensive.

Clarifying and motivating policy objectives: outcome-orientation, goal setting and benchmarking

The government of Finland, along with that in many other OECD countries, is increasingly formulating explicit goals and targets, using outcome indicators and conducting benchmarking exercises.[50] The degree to which these activities have been developed varies widely across the different areas and levels of government. In some cases explicit objective-setting is well-developed. For example, the Ministry of Labour has specific strategic guidelines, such as offering 95 000 workplaces and training opportunities in active labour market schemes for the unemployed. Similarly, the Ministry of Education's *Development Plan* identifies a series of goals from pre-primary education through to adult learning. In contrast, some ministries have only conducted cursory goal-setting exercises.

Although there is good reason not to rush in output-oriented goal setting given the difficulties of developing good indicators and incentive structures, the Finnish authorities are nevertheless overly timid. In particular, there are seldom substantial penalties or rewards in relation to reaching targets. One exception is in the funding for tertiary education where the funding formula for central government grants to universities includes the number of graduations.[51] By contrast, however, the Ministry of Labour's strategic guidelines, for example, make no explicit reference to any consequences for those responsible in the case of failure to reach targets. In addition the targets, although included in budget documenta-

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tion, have not become common currency among politicians and the media. Therefore, unlike countries such as the United Kingdom, political accountability for the achievement of quantitative targets has not yet emerged. Wider diffusion and an increased profile of the targets may be beneficial.

Moreover, benchmarking is not fully exploited in Finland. The potential for making informative comparisons in costs and outcomes across a wide range of public services is large, especially in light of the strongly decentralised system of public expenditure. Although there are a number of both regular and occasional comparative evaluations of public services and general guidelines for evaluating public-sector activities have been developed, there is something of a taboo in the Finnish public service against allowing evaluations to inform the public, including statistics comparing performance.[52] Indeed access to the evaluations is positively discouraged by a policy of charging for research reports. The central government should take much stronger initiatives to introduce league tables that compare outputs as well as costs, perhaps taking as models the initiatives in Denmark, Norway and Sweden to improve information for citizens (see 2002 *Surveys* of Denmark, Norway and Sweden). There are signs that change is underway in this regard; a recent report by the Ministry of Finance (Ministry of Finance, 2002d) acknowledges the need to improve the openness of administration to citizens and increase the use of cost and output indicators.[53]

Developments in e-government

E-government is an increasingly important means of promoting public-sector efficiency, particularly in areas such as outsourcing, procurement, and administrative activities such as tax collection. According to indicators on the development and use of e-government, Finland is one of the leading countries in this area.[54] A step towards making more use of information technology in the interface between government and the public was taken in spring 2002 with the opening of a common portal for the whole public sector.[55]

However there is plenty of room for catch-up to best practice, particularly in the application of e-government in municipalities.[56] Technical, legal and social considerations also affect the application of e-government. Notably, the privacy and information-security laws are quite strict in Finland, which, for example, can make the application of e-government to health care complicated. Also, consideration must be given to those with difficulties in accessing or using the electronic services, so that the complete abandonment of traditional means of service delivery is a long way off.

Developments in the policy towards public employees

The Finnish public service is moving slowly away from a traditional model of remuneration for public-sector employees, which includes, for example, auto-

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matic steps for seniority.[57] Of the more notable reforms, a new wage system is gradually being introduced for central government employees aimed particularly at rewarding young and highly educated staff. The wage level of each employee consists of a "task" component and a "personal" component (Box 7). In some variants of the scheme the personal component can, in theory, reach half of the final wage level. While the reform process is making gradual progress within the central government, full implementation of the system remains limited.[58] Similar wage reforms for municipal employees are still at the planning stage. Development away from an overly rigid remuneration scheme is welcome, but needless to say these reforms cannot be regarded as universally applicable across the very wide range of civil-service activities, or as a panacea to morale and recruitment issues.

As in many other OECD countries, Finland's public-sector employers are likely to face increasing pressure to make pay and conditions more attractive in the coming years.[59] In part this is because large numbers of public sector employees are coming up for retirement, about one third of municipal employees are due to retire by 2010 (Vuorento, 2001). At the same time, there are shifting demands across the public service, with increasing pressures in areas such as health alongside areas of declining demand, particularly for child-care and teaching, thus

Box 7. **Customs officers' pay: an example of the new wage system for central government employees**

The Finnish customs officers entered the new salary system in 1999-2001. The core of the salary consists of a task-related component based on an assessment of what tasks an individual performs. This is then augmented in a number of ways:

- An *individual component* that is based on an assessment of individual performance. The maximum size of this part is 30 per cent of the level of task-related pay.
- An *experience component*. After the first year in service the experience-related pay is 5 per cent of the task-related component, rising to a maximum of 10 per cent after the third year of service.
- A *guarantee component*, which (if necessary) brings the new salary up to the old salary level, so that nominal salaries are not reduced with the introduction of the new salary scheme.
- *Working conditions compensation* of up to 4 per cent of task-related pay.

Individuals are regularly assessed through a process of target setting and evaluation of performance conducted interactively between management and employees. The administrative set-up costs of the system included the assessment and valuation of tasks performed in more than two thousand posts as well as additional management training.

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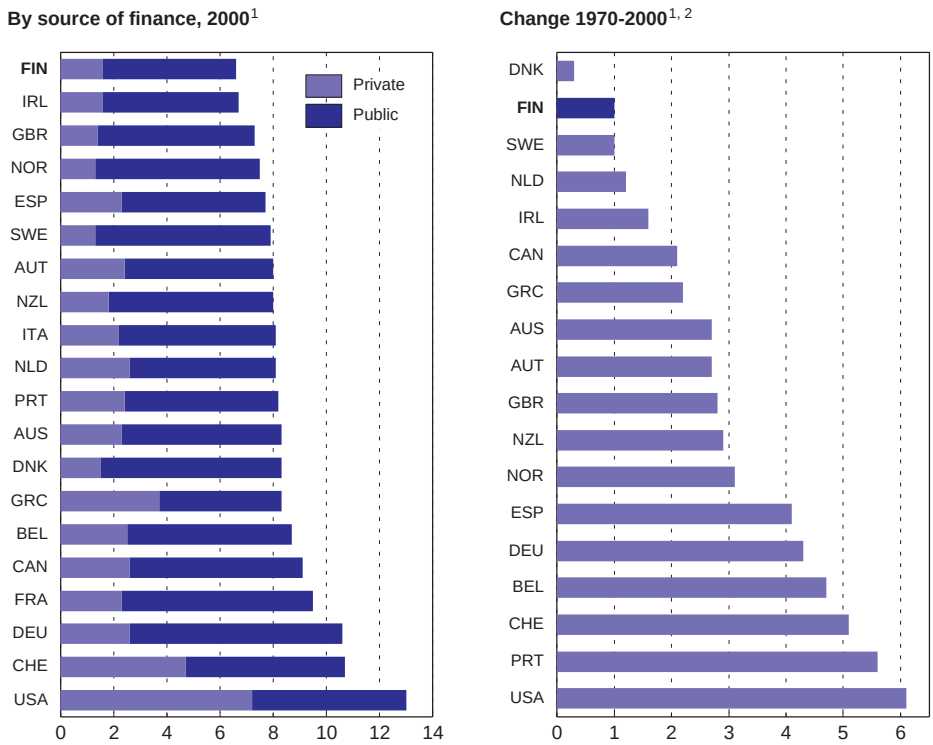
underscoring the need to avoid an across-the-board approach to cuts (or increases) in employment levels.[60] Signs of demand pressures are already evident, particularly in the health sector, where the employers are reverting to offering permanent contracts after a prolonged period of favouring temporary contracts.[61] The authorities should recognise that a widespread increase in employment security in order to make jobs more attractive could lead to inflexibility and inertia in the future.

Efficiency issues in public health expenditure

Finland devotes approximately 7 per cent of GDP to health services, three-quarters of which is via public expenditure (Figure 21). By international

Figure 21. **Health expenditure in international perspective**
Total expenditure in per cent of GDP



1. 1998 for Sweden instead of 2000.
2. Instead of 1970: 1971 for Australia and Denmark, 1972 for Netherlands.
Source: *OECD Health Data 2002*, 4th edition.

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comparison the public resources devoted to health are relatively modest, which partly reflects the budgetary austerity of the 1990s (OECD, 2001b). Nevertheless, a special review of health care in the 1998 *Survey* judged the health system to be broadly in good shape. The long-standing tradition of local health centres in Finland has resulted in very good infant and child-care and high levels of immunisation. Also a strong emphasis in health education on issues such as nutrition, physical exercise, reproductive health and smoking have had a positive impact. Diagnosis and treatment also rank favourably in international comparisons (1998 *Survey*). However, there are now upward pressures on expenditures that are set to continue due to population ageing. Shortages of certain categories of health personnel have strengthened wage demands, as exemplified by the 2001 doctors' strike.[62]

Public health care is largely the responsibility of municipalities (Figure 22). Local medical centres provide primary care, with some shared provision among small municipalities.[63] The services offered by health centres include: outpatient care, inpatient care, preventive services, dental care, maternity care, child health, school health care, care for the elderly, family planning, physiotherapy and occupational health care. Hospital care is provided by 20 regionally based joint municipality organisations ("hospital districts").[64] Primary health care is financed directly by the municipalities from their general budget. In secondary care, hospitals, in effect, pass on bills for care to the municipalities (see below). Under this funding arrangement for hospital care a potential problem for small municipalities is that the payment for treatment of high-cost medical conditions can potentially place heavy demands on their finances. The main mechanism for dealing with this is an arrangement in which ceilings are set to the amount a municipality has to pay for any single patient, with costs in excess of the ceiling being shared between municipalities in each hospital district. The discretionary government grants system is also sometimes used to help municipalities who have been facing heavy burdens due to health care costs.

The government has recently embarked on a reform of the health system, run by the Ministry of Social Affairs and Health and known as the National Health Project. Key aims of the reform are to ensure a more comprehensive availability of health care across the country, reductions in waiting times and improvements in staff training. The state grants for municipalities' welfare and health care will be increased as of 2003 by a total of EUR 223 million. Of this a total of EUR 57 million will be spent on the project to secure the future of health care. Funds for individual projects to improve services will also be made available, amounting to EUR 8 million in 2003 and rising to EUR 30 million a year between 2004 and 2007. The percentage of state funding to municipalities will be increased in subsequent years depending on progress in implementing the project. It is estimated that the Project will have made economies in health care equivalent to EUR 200 million by 2007. In addition to project funding, the state will fund the development and implementation of a new national electronic database for patient records.

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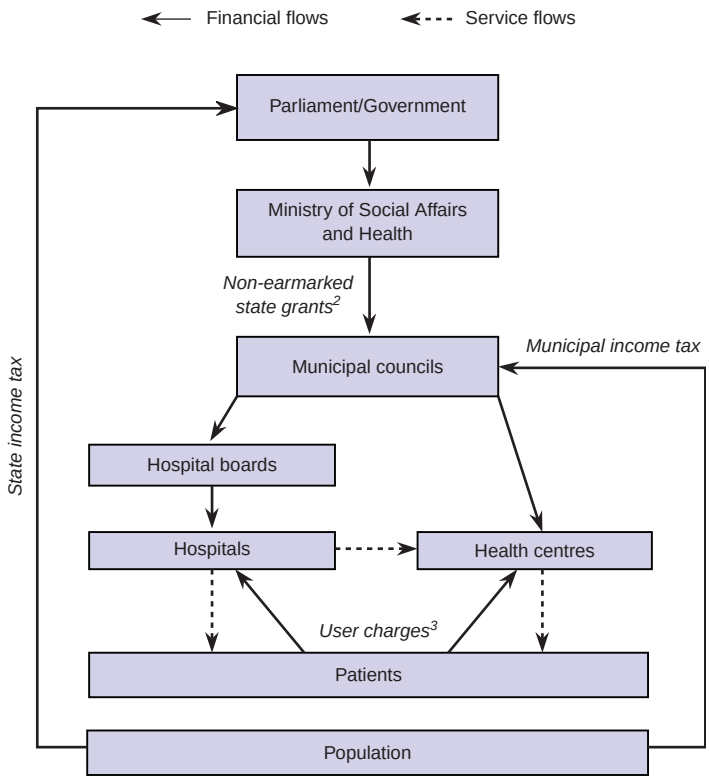
Figure 22. **The financing and provision of public health care**[1]
Since reforms in 1993



1. Does not include care funded by national health insurance and/or employers (private sector and occupational health care).
2. Based on a capitation formula.
3. User charges decided by municipalities with a ceiling fixed by government.

Source: OECD (1994), *The Reform of Health Care Systems: A Review of Seventeen OECD Countries,* Health Policy Studies, No. 5.

The hospital district system is a particular source of inefficiency

The current set-up of the hospital district system is recognised as being one of the weaknesses in what is generally regarded as a sound secondary health care system. One problem is a lack of division between purchasing and providing roles as the municipalities acquire services from hospital districts they are themselves responsible for running. Another problem arises from the relatively advan-

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tageous position of hospitals in relation to the hospital boards. Hospital management often has an information advantage, especially in relation to small municipalities, in the specialised knowledge needed to understand the demand for health services, treatment options, costs and so on (Järvelin, 2002 and the 1998 *Survey*). And the role of hospitals as large local employers also influences their relationship with municipalities. As a result, municipalities often feel powerless to influence the costs and provision of hospital care. In effect, hospitals face a relatively soft budget constraint, presenting municipalities, via the hospital districts, with a bill for services.[65]

To improve cost efficiency in hospital care the central government could play a more active role in ensuring the hospital system produces comparable accounts and other performance indicators, as at present it is extremely difficult to compare services and prices. A recent assessment by the Ministry of Social Affairs and Health recommends the establishment of a monitoring system on the national availability of services, the effectiveness of treatment, costs and productivity (Ministry of Social Affairs and Health, 2002a). A second approach is for the central government to encourage efficiency by requiring additional national minimum standards of service without fully compensating with additional funding. This would appear to be a key element in the current government thinking, especially with regard to waiting lists. The recent health care review by the Ministry of Social Affairs and Health recommends the implementation of a national waiting list system with the setting of detailed provisions on the maximum waiting periods for treatment and a requirement that if a board cannot provide the service in time then it must purchase the service from outside (Ministry of Social Affairs and Health, 2002a). The report also recommends stepping up efforts to ensure uniform waiting-list criteria are used throughout the country.

In recognition of the need for change, there is some welcome experimentation with variants of the current system. The Pirkanmaa hospital district, for example, is experimenting with a financing arrangement which gives greater freedom to municipalities to act individually as purchasers of medical services (Järvelin, 2002). More fundamentally, hospital efficiency could be improved by strengthening the distinction between purchasers and providers by, *inter alia*, removing responsibility for running the hospital districts from municipalities (leaving their role to being exclusively that of purchasers); ensuring that municipal officials have sufficient expertise to carry out this role; and giving municipalities the right to purchase from any hospital district and encouraging them to exercise this right (*i.e.* to shop around).

The advantageous position of the hospitals also engenders other inefficiencies. The Competition Authority has recently been investigating various cases where hospitals have been alleged to be using unfair pricing strategies when competing with the private sector in certain areas of health and ancillary services. For

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example, the Authority found the laboratory services offered by the Pirkanmaa Hospital District failed to comply with competitive neutrality and has also investigated allegations of cross-subsidisation in health-care laundry services (FCA, 2001). And, more broadly, the separate administration of primary care (mainly health centres) and secondary care (hospitals) also presents challenges for efficiency, in particular with regard to overlap and co-ordination. The recommendations of a recent review of health care frequently refer to a need for greater co-ordination and exchange of information (Ministry of Social Affairs and Health, 2002b).

A need to widen choice and promote alternatives to public provision

The public health care system does not give users much choice in terms of the medical centre or hospital they attend, or by whom they are treated. In primary health care, patients are assigned to doctors on the basis of location of residence and in secondary care patients cannot choose where they will be treated and only seldom have the possibility to choose a specialist (Ministry of Social Affairs and Health, 1999). At the same time, in certain areas of health care, private sector alternatives have developed quite extensively. Most notably this has happened for doctors' services; about one third of all doctors (both general practitioners and specialists) operate practices which the state supports by partially refunding the fees charged (Järvelin, 2002). There are relatively few large-scale private-sector operations, such as hospitals.

The current government recognises the need for more choice in public health care and a number of proposals have been made in this regard. Notably, the Ministry of Social Affairs and Health is advocating the use of vouchers for home-help services (Ministry of Social Affairs and Health, 2002b). In a recent evaluation of voucher systems the Ministry underscores the need, *inter alia*, for: equal treatment of different provider options, quality guarantees, and the inclusion of the municipality's own provision in the voucher system. It is proposed that the minimum value of the vouchers should be based on existing income-related charges, with municipalities being free to make supplements.[66] The Finnish Competition Authority has also voiced support for greater use of voucher schemes (FCA, 2002).

Goal-setting and benchmarking should be strengthened to improve efficiency

The health sector in Finland exemplifies a broad tendency of too little systematic benchmarking. To their credit, the authorities are making progress in the hospital sector. The National Research and Development Centre for Welfare and Health (Stakes) has developed a system of annual evaluation across hospital districts which now covers 95 per cent of hospital care in Finland although, as elsewhere in the public sector, dissemination of the evaluations is restricted (Linna

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and Häkkinen, 1999). At the same time, however, there have only been periodic comparisons made across health centres. Efforts to deepen and broaden the scope of benchmarking should continue, not least because evidence suggests there are persistent differences in productivity levels across health-service providers, and that the adoption of best practice standards could significantly reduce costs. For example, a recent study shows considerable variation in cost efficiency among primary health care centres and that technical efficiency could be improved by 7 to 10 per cent on average (Räty *et al.*, 2002).

Pricing issues in health care

In general, there has been a welcome tendency towards increasing user awareness of health costs through the use of fees, although the price system, built up from numerous acts and decrees over time, can be criticised for being too complicated and opaque.[67] Over the 1990s there has been an increase in out-of-pocket payments in the health system. User charges for municipal services were increased over the 1990s, tax deductions for drug and other medical expenses were abolished in 1992 and there have been reductions in the reimbursement of pharmaceuticals by the national insurance scheme (Järvelin, 2002). At the same time, together with many other OECD countries, Finland still faces strong growth in pharmaceutical costs.[68] The government has responded with a number of measures over the past few years including: stricter assessment of retail outlets, greater assessment of the therapeutic and cost effectiveness of new drugs, special prescription arrangements for expensive drugs, retail price regulation that reduces incentives to sell more expensive drugs and a programme aimed at changing doctor's prescribing practices. However, these measures do not appear to have made any substantial impact on the rate of growth of pharmaceutical costs and further measures are needed (Järvelin, 2002). In December 2002 the Parliament passed a bill requiring pharmacies to use cheaper generic drugs, even if the doctor prescribes a more expensive product. The new law will come into force in April 2003.

The current system of remuneration for doctors' services may require attention. Any licensed physician in Finland can, in theory, set up a private practice and large proportion of specialists with public-sector posts in hospitals and health centres do so. One problem with this system is that it encourages a division of effort by specialists between their public-sector duties and private practice that makes it particularly difficult to establish an effective referral and gate-keeping system in the public sector (Järvelin, 2002). One influence on this problem is likely to be the incentives generated by the system of public-sector remuneration for specialists. Another consideration is that the division between public and private care is influenced by the system of reimbursement by the national insurance system. It is possible that this aspect of the system needs to be scrutinised as well as specialists' incentives.

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Efficiency issues in elderly care

Care for the elderly currently accounts for about one per cent of GDP and is set to absorb an increasing share of resources in coming years.[69] Some problems with the current fee structure need to be tackled as it can lead to a type of "poverty trap" in which income net of taxes and service charges can decline as gross income increases. At the same time, ceilings on user charges mean the relative cost of services is not reflected in the fee structure. This is particularly the case for high-cost institutional care (OECD, 2000). Despite this specific bias towards institutional care, the authorities have in general conducted a successful strategy to reduce live-in care over the 1990s, principally through greater use of housing with special care facilities ("service housing").[70]

A broad challenge for efficiency in elderly care arises because the wide range of services, from home help to intensive institutional treatment, requires a wide range of professionals and often spans across several providing institutions. As a result, ensuring that resources are directed to where the marginal benefit is greatest is particularly difficult. In this regard the Ministry of Social Affairs has been attempting to improve co-ordination across elderly care services by encouraging municipalities to formulate strategic plans and to follow guidance on the quality of services.[71] Plans by the Ministry to have indicators of elderly care compiled by the research institute Stakes should be fully implemented and the current strategy should be backed by more forceful measures if there is inadequate improvement in elderly services.

Reports that municipalities are reluctant to welcome new elderly residents, especially those in need of institutional care services, should be further investigated as such behaviour may be detrimental both socially and economically. The shift to urban areas by the working-age population is also triggering an increasing demand to move by the elderly wishing to live closer to their families. However from a fiscal perspective, municipalities are understandably resistant to inflows of elderly persons as they can place increasing demands on expenditure. If there is reasonable substance to these reports, the negative spillover across municipalities may even have a significant impact on the amount of family-provided care for the elderly, thus increasing the overall need for publicly provided services. This issue should be monitored and countered by an appropriate policy response, either in the form of regulations to prevent exclusion by municipalities or, ultimately, by greater central government involvement.[72]

Efficiency issues in education

As in other Nordic countries, compulsory education in Finland generally starts when children reach seven years of age, somewhat later than in most other OECD countries, although most children attend school even before that age.[73] The vast majority of students complete either upper secondary or vocational training

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and the entry rate to tertiary education is high.[74] Primary and secondary education (including upper-secondary education) is the responsibility of municipalities. Tertiary education consists of two pillars, polytechnics and universities. Polytechnics are mainly municipality-run while all universities are state owned. Fees are not charged at any stage in the education system, and those in upper secondary and tertiary education are also entitled to means-tested living allowances.

The quality of the Finnish education system in terms of outcomes is generally excellent. Comparative studies, such as the OECD's PISA report, find Finland to rank very highly in terms of student abilities at secondary-school level, although the Finnish government has expressed concern for a number of years about skills in mathematics. Standards in tertiary education are maintained by a comprehensive review system and a relatively high percentage of students in Finland take scientific or technology oriented degrees (OECD, 2002d).[75] Thus from a public expenditure perspective, the quality and mix of "output" in education in Finland is less of an issue compared with ensuring the system is efficient.[76]

In the non-tertiary sector benchmarking is weak and alternatives to public education are underdeveloped

A recent FCA report underscores that flexibility for private-sector schools is limited. There are about 50 private schools in Finland, mainly located in Helsinki, which are tightly monitored so as to ensure certain minimum standards. Also they are not normally allowed to charge fees and financing is based solely on grants from the government, further restricting the leeway for offering education that differs from the state system.[77]

As with municipalities' responsibilities in other areas, there is reluctance to provide comparisons across primary and secondary teaching establishments. Evaluations of teaching became compulsory in 1999. The evaluation process involves both specially appointed boards and municipal authorities. However, there is no explicit ranking of schools and a reluctance to disseminate the results of the evaluations.

Less time should be spent in acquiring university (and polytechnic) qualifications

Finland devotes a high percentage of GDP to publicly funded tertiary education compared with other countries, and the expected number of years the population spends in tertiary education is also high (Figure 23). In part these figures stem from a positive aspect of the system: high enrolments in tertiary education. Also a relatively large share of students take their studies to a higher level compared with other countries; the majority of university students complete with masters rather than bachelor's degrees. At the same time however it takes many students longer than necessary to graduate which adds to the public cost through additional fees and income support; only 5 to 15 per cent of students complete

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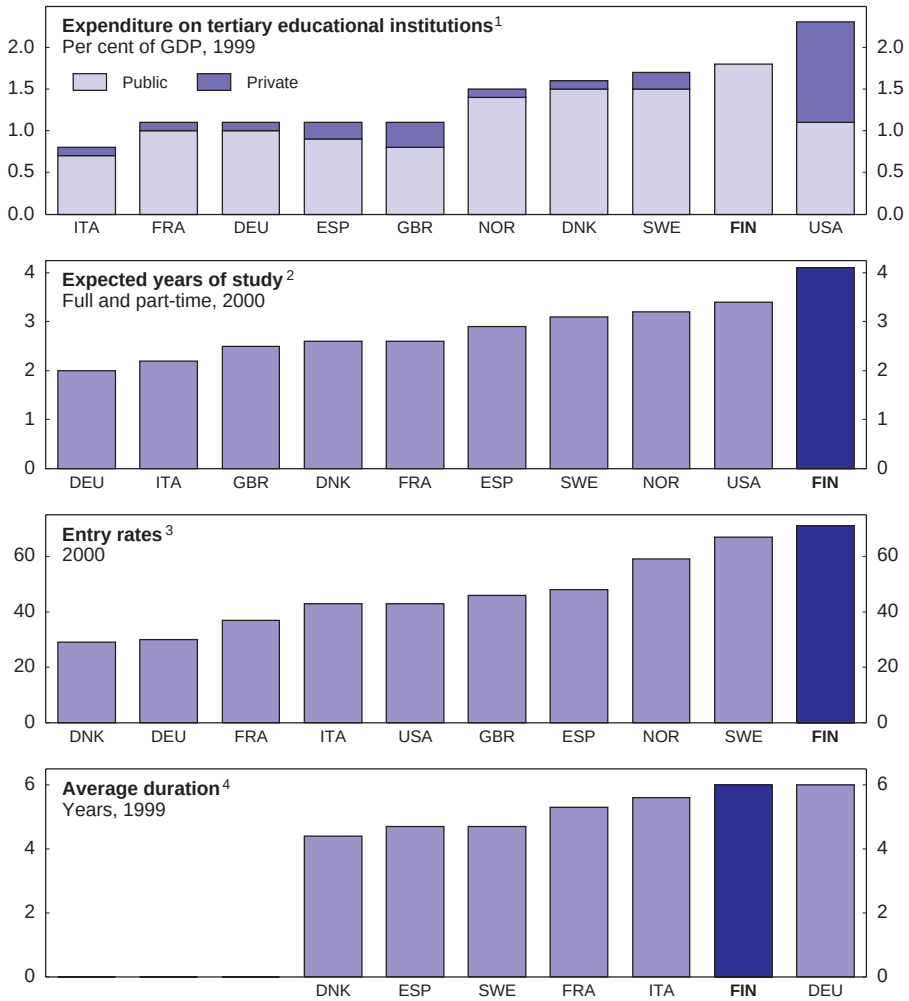
Figure 23. **Tertiary education: expenditure and participation**



1. Private expenditure is net of public subsidies for educational institutions whereas public expenditure includes those to households (including direct expenditure on educational institutions from international sources).
2. All tertiary education (type A, B and advanced research programmes) in public and private institutions. The expected years of study are calculated by adding the net enrolment rates across age groups.
3. Sum of net entry rates in tertiary type A education in public and private institutions.
4. For tertiary type A and advanced research programmes. Duration estimated using either a chain method or an approximation formula. No data available for Norway, United Kingdom and United States.

Source: OECD (2002), *Education at a Glance*.

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their degree within 5 years (Ministry of Finance, 2002e). Students also start tertiary education relatively late: the median age of entrants is 21½ years, while it is under 20 years in many other countries. The late start, combined with the long enrolment means most master's students do not graduate until they are in their late 20s. From a broad economic perspective, this late graduation represents valuable losses in high-quality labour supply.

The Ministry of Education's *Development Plan* for 1999-2004 acknowledges the need for improvement in the pace of degree studies and aims for 75 per cent of students to complete their degrees without significant delay: three years in the case of bachelor's degrees and five years in the case of master's degrees. To achieve these targets the state funding formula for universities was adjusted. This performance-linked formula includes penalties for institutions which are lagging behind the targets. It was first introduced in 1997 and will be fully operational in 2003 (Ministry of Education, 2000). There is also recognition that the financial incentives for students under the current system needs to be altered. One suggestion has been to adopt a system of support with sanctions, such as that in the Netherlands where failure to complete studies within a given period results in financial aid being converted into a repayable loan (Ministry of Finance, 2002e). Also, if students are made to bear more of the costs of education in the longer and more academic courses, this may induce more individuals to choose the shorter vocational courses and thus assist the government's campaign in this regard.[78] Whatever approach is taken, the authorities should overcome their reluctance to substantially alter the current system of public assistance via fees and living expenses given the substantial evidence that the private returns to tertiary education are usually large and typically exceed the public returns (Blöndal *et al.*, 2002). Reforms to public assistance for higher education should take into account that Finland's strongly progressive tax system can be viewed in part as a recuperation of the costs of higher education given the connection between earnings and education. Therefore reforms that entail substantial reductions to direct assistance for higher education should perhaps go hand-in-hand with tax reform.

Wrapping-up

The fundamental framework guiding public expenditure in Finland is sound and in many areas of public activity the country compares very favourably internationally. At the same time, slippage in fiscal discipline needs to be addressed and it will be important to implement, and if necessary follow through on, reforms of pensions and early retirement arrangements so as to ensure long-run sustainability in public finances. In addition, there are important efficiency issues with significant potential rewards that should be tackled. There is a need for a deeper commitment to reforming public-management practices and to this end it would be particularly useful if benchmarking were more widespread and

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Box 8. Recommendations for reforming public expenditure

Aggregate fiscal discipline

- Strengthen the influence of the spending rule of the Government Programme in the annual budget process. For example, move towards a more genuine, medium-term rolling budget by making greater use of the four-year budget projections that are included in the annual budget documents.
- Enforce spending limits and aim at lower aggregate spending as a per cent of GDP to allow further tax cuts.
- Consider introducing a fiscal rule in the Government Programme that stipulates that additional spending during the year needs to be offset by spending cuts elsewhere.
- Exclude interest rate payments from the expenditure rule in the Government Programme.
- Aim for greater clarity in the technical details of the spending rule.
- Continue efforts to get parliamentary commitment to overall spending limits earlier in the budget process and continue to look for ways to reduce financial inflexibility across ministries ("stovepipe government").

Municipal structure and financing

- Continue encouraging both the merger of small municipalities and the use of co-operative arrangements to exploit economies of scale.
- Reduce the role of direct tax revenue in municipal financing by, at least, further reductions in the municipalities' share of corporation tax. Make up for the lost revenue with increased block grants and evaluate the funding formula.
- Remain vigilant to municipalities abusing the system of discretionary funding the central government uses to assist municipalities in financial difficulties.

Transfers

- Monitor and if necessary follow up the recent measures to tighten transfers and to increase the incentives to remain in the workforce for longer.

Public services

- Ensure goals and targets are accompanied by stronger sanctions and rewards so that they play a greater role in political and media discussions.
- Make greater use of benchmarking, particularly across municipal-based public services. Make the results of benchmarking more widely available, for example by publishing league tables of input and output indicators.
- Continue to take measures that increase contestability in the provision of public services, including steps to increase outsourcing and the role of private-sector providers. Consider expanding choice, for example via voucher systems.

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Box 8. **Recommendations for reforming public expenditure** (*cont.*)

- Alter or replace the hospital district system so as to improve incentives for cost containment by, *inter alia*, ensuring that the municipal authorities are only the purchasers of hospital care and not also involved in the process of provision.
- Monitor and, if necessary, follow up on the initiatives to reduce study-time in tertiary education. Consider moving away from universal unconditional payment of fees. The system of support for living expenses for tertiary students should also be reformed with the aim of speeding up the pace of study and making sure that the public support to individuals' education is appropriate.
- Continue to develop e-government services and promote best practice among municipalities to improve overall implementation.
- Continue to develop and implement the new wage system for public-sector employees. Be careful in using permanent contracts as a means to making public-sector jobs more attractive.

more in the public eye. Finally, there is a general reluctance to step away from traditional-style in-house provision of public services and current government efforts to overcome this should be continued. Box 8 makes specific recommendations as to how these broad weaknesses can be tackled as well as recommendations as to how to deal with some of the more specific areas of concern in the way public spending is organised, most notably in the hospital sector.

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IV. Policies to boost potential output growth

Finland's consistently high ranking in survey-based measures of competitiveness[79] as well as the relative rapidity of the recovery from the current cycle, are testament to a number of long-standing structural strengths that have been underpinned by key reforms in certain areas. The advantageous bias towards science and technology, as seen in areas such as the education system and the high level of R&D spending, has helped the successful specialisation in the ICT-producing sector and contributed to technological literacy among consumers. Also privatisation, liberalisation and reductions in state aid to many sectors carried out in recent years mean that market mechanisms now operate more effectively to provide a more efficient allocation of resources.

At the same time, policymakers face a number of pressing challenges. Lowering the high level of unemployment and raising employment rates is a priority to reduce the pressures from population ageing and the resulting increase in the dependency ratio. The pension reform, discussed in Chapter II, will help in this respect. But substantial progress will also require a close examination of the weaknesses in the current wage determination, tax and benefit systems, as well as in the range of active labour market policies. There is also room for improvement in a number of areas of product-market policy. Privatisation has been slow and competition policy could be strengthened further, and agriculture is subsidised at a high cost.

There are, inevitably, uncertainties over which policy can only have a small influence. In the case of Finland it is clear that in the coming years much will depend on the commercial success of the ICT sector. In this regard, the current policy approach of enabling private-sector research, rather than attempting to lead research or pre-empt the next area of technical specialisation is probably wise. Perhaps most important in this context is for policymakers to consider whether the high rates of taxation pose a threat given the reliance on "knowledge-based" industries employing highly educated, highly paid, but also very mobile workers. At the same time, it is clear that greater diversification is desirable, which serves to further emphasise the merits of fostering new growth poles by a general improvement of framework conditions in labour and product markets.

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This chapter also examines policy towards climate change, air pollution and natural resources as part of the OECD's initiative to examine sustainability across a broad range of environmental, economic and social issues. Generally, Finland is taking environmental responsibilities seriously. It has made substantial efforts to reduce carbon emissions, even though the cost-effectiveness of some measures is in doubt. It also remains committed to internationally agreed reductions in other air pollutants, although, as in many other countries it perversely maintains a tax advantage for diesel fuel. The vast forest cover is increasing due to a more rapid rate of commercial planting in relation to harvesting. At present, policy to preserve and promote bio-diversity in forests is based mainly on regulation and, as elsewhere in environmental policy, greater use of economic instruments should be considered to enhance effectiveness.

The chapter begins by examining productivity issues and is followed by a review of various labour market policies. Product and financial market issues are discussed next and the final section highlights sustainable development issues. Table 15 provides a summary of the recommendations for further structural reform.

The strengths in productivity are considerable but not well diversified

Finland has a record of relatively strong but also volatile productivity growth compared with the major European economies as well as other Nordic countries (Figure 24).[80] Since 1980, productivity growth has averaged 2.6 per cent compared with 1.8 in neighbouring Sweden and 1.5 per cent for the euro area. It has also been one of the OECD countries with a particularly strong surge in productivity growth during the 1990s. The manufacturing sector has always played a key role in generating productivity growth and in the 1990s the ICT-producing sector has been particularly important (Figure 25).

Many factors are likely to have boosted productivity growth. Indeed, as is often found in analysis of productivity growth, much remains unaccounted for after allowing for changes in the physical and human capital stock, with the residual "multifactor" productivity growth making a large positive contribution.[81] Assessments of the Finnish economy point to a number of factors that may be generating this high multifactor productivity growth (*e.g.* Ministry of Finance, 2002e). A key positive influence is the strong bias towards science and technology. This can be seen across a wide range of indicators. According to the OECD *Science, Technology and Industry Scoreboard*, Finland not only ranks among the top countries in terms of R&D spending at about 3½ per cent of GDP but also in a broader indicator of investment in knowledge that accounts for higher education.[82] Business R&D expenditure in the ICT sector in 2000 represented over one per cent of GDP, the largest share of GDP of any OECD country. Similarly, the share of ICT-related innovations in total patent applications is the highest across the OECD (OECD, 2001c).

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Table 15. **Recommendations for further structural reform and actions taken**
Based on previous and current *Surveys* and action taken

Previous and new recommendations	Actions taken since previous *Survey*
LABOUR MARKET AND SOCIAL SECURITY SYSTEM	
Reduce incentives for early retirement and reform the pension system	
Increase further the minimum age for the unemployment pension; shorten "unemployment pipeline to pensions"; reduce incentives to retire early.	Early retirement schemes will be phased out gradually. Unemployment pipeline age limit will be raised from 55 to 57 for those born in 1950 or later. Higher accrual rate for those over 63 increases incentive to postpone retirement. However, there is a risk that take-up of the disability scheme will rise.
Reduce impediments for hiring older workers. Reduce, for instance, the contribution of the last employer to disability and unemployment pensions.	No action.
Lift the pension ceiling of 60 per cent of the pensionable earnings in order to make work above the age of 60 attractive. Abolish old-age pension accrual during early retirement. Allowing early retirement only from the age of 62 years would reduce the distortion significantly.	Recent pension reform will lift the ceiling, raise accrual rates in steps from age 52 and will allow flexible retirement for old-age pension. Individual early retirement and unemployment pensions will be abolished. The accrual during part-time pension will be only half the earlier size.
Smooth the future impact of ageing by increasing funding of the pension system now.	Recent pension reform will raise funding.
Introduce a properly funded pension scheme for central government employees.	The central government pension system will be reformed in 2003.
Put early retirement schemes on an actuarially sound footing. More remains be done, especially concerning the unemployment pipeline to pension.	Early retirement schemes will be phased out and minimum age for part-time pension will rise from 56 to 58. Minimum age for the reformed unemployment pipeline (without right to unemployment pension) will be raised from 55 to 57 for those born in 1950 or later.
Reform unemployment and related benefits	
Shorten maximum unemployment benefit duration to increase job search intensity. Phase down unemployment benefits during the first 500 days of unemployment. Make eligibility criteria for social security benefits stricter. Enforce eligibility criteria for unemployment benefits.	Unemployment benefits increased in March 2002.
Increase regional mobility of the unemployed. Lift the condition that placement involving relocation is suitable only if the vacancy cannot be filled locally.	No action.
Introduce medical re-examination of disability pension beneficiaries.	Individual early retirement, which was one form of the disability pension, will be abolished. The medical requirements for the remaining disability pension are higher than they were in the individual early retirement scheme.

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Table 15. **Recommendations for further structural reform and actions taken** (*cont.*)
Based on previous and current *Surveys* and action taken

Previous and new recommendations	Actions taken since previous *Survey*
Make active labour market programmes more effective	
Pursue better job counselling, for instance, reach the target of one contact per month with every unemployed person. Consider going further with the introduction of a general contestable placement market.	The contacts are made more efficient by increasing co-operation between social and labour administration through an experiment that is taking place in 2002 and 2003.
Reduce the number of subsidised jobs offered by local government.	The number of subsidised government jobs has been reduced and those in the private sector raised.
Step up ALMPs for older workers. Apply sanctions in case older unemployed refuse training or a suitable job.	National Programme for Ageing Workers has helped in raising participation.
Evaluate the effectiveness of all active labour market programmes and end those with poor results. Given limited effects, an overhaul of some ALMPs is needed.	Evaluations have been carried out, but there was little change to the ALMP schemes.
Expose the public employment service to competition as successfully done in other OECD countries.	The pilot reform started at the beginning of 2002.
Increase employment and labour cost flexibility	
Reform employment protection and working time legislation to encourage job creation, especially by small and medium-sized enterprises.	No action.
Support greater wage differentiation in wage agreements, to promote the employability of low-skilled workers, while maintaining moderate average wage rises.	Wage increases have remained moderate, but the labour cost dispersion continues to be very compressed.
Improve labour force skills and competencies	
Ensure that education and training meet evolving needs of labour market.	Finnish students have received very high marks for their educational achievements in international comparison, but studies still take too many years compared to other countries.
PRODUCT MARKETS	
Enhance product market competition. Strengthen competitive forces by reforming the Finnish competition authorities. The enforcement capabilities of the Finnish Competition Authorities should be strengthened.	The Market Court has replaced two existing agencies.
Examine scope for completing privatisation of partially privatised state enterprises to increase efficiency.	Following the widening of the privatisation mandates in 2001, little progress has been made.
Place full privatisation of most state-owned companies on the agenda and strengthen the impact of competition principles by organisationally separating the government's role as regulator from its role as owner.	A separation of the government's role as owner and regulator is under consideration.
Promote genuine level playing fields in the energy and telecommunication markets. Stimulate competition by development of alternative networks such as cable television network for telecom services.	The Finnish Competition Authority has launched various actions in the two sectors.
Promote genuine level playing field in the postal market. Review the universal service obligation in the postal market.	No action.
Reduce agricultural subsidies.	No action.

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Table 15. **Recommendations for further structural reform and actions taken** (*cont.*)
Based on previous and current *Surveys* and action taken

Previous and new recommendations	Actions taken since previous *Survey*
FINANCIAL MARKETS	
Enhance supervision. Merge the Financial Supervision Authority and the Insurance Supervision Authority.	Enforcement powers of the Financial Supervision Authority have been strengthened. The government decided against a merger of the two authorities.
Raise competition among employment pension insurers.	Various proposals to raise competition exist.
TAXATION	
Broaden the tax base on personal income and reduce statutory rates.	Statutory rates reduced somewhat.
Cut social security contributions especially for the low-paid.	Social security contributions will be raised in 2005.
Remove the age-dependency of social security contributions. Reduce tax incentives for people to retire early.	No action.
Keep the capital/corporate income tax rate competitive. Consider reducing imputation credit for dividends.	Proposals by a Working Group on taxation go in this direction.
SUSTAINABLE DEVELOPMENT	
Systematic evaluation of sustainable development is needed.	No action.
Increase the cost-effectiveness of environmental measures by a greater use of economic instruments.	No action.
Consistency between sectoral and environmental policies should be improved.	No action.

Source: OECD.

Although there is much discussion about its potential, at present biotechnology accounts for no more than about 10 per cent of R&D spending, even though there are claims that the size of the industry is nevertheless relatively large compared with that in other European countries.[83] In October 2001 a network was established including all ministries with the aim of tackling biotechnology issues in science and innovation but also industry, environment, agriculture and forestry, health and trade policies.

A feature of knowledge-based industries is rapid innovation with the implication that the competitiveness situation of particular firms can also change very quickly. There may be little policy-makers can do to prevent this, but they can provide the general economic framework conditions to foster rather than inhibit innovation. While such conditions are clearly generally favourable in Finland, the single most important weakness is the relatively high taxation of labour, which reduces Finland's attractiveness for international investors and highly-skilled personnel, thus weakening its competitiveness as a centre for high-tech industries.

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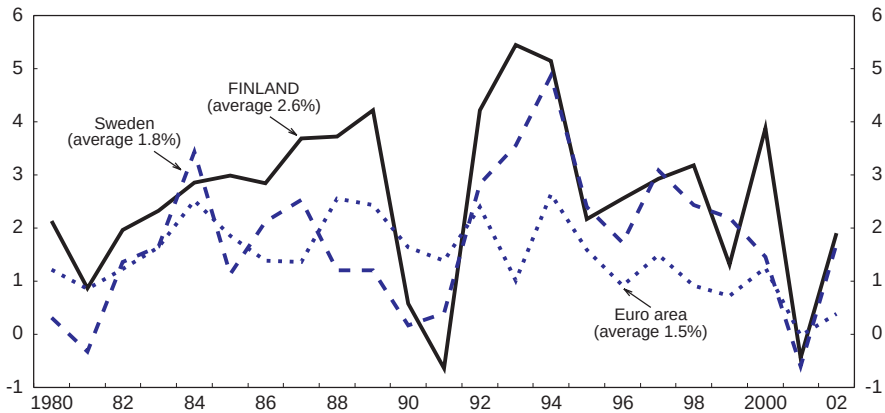
Figure 24. **Finland's productivity compared with Sweden and the euro area**
GDP in constant prices per employed person, percentage growth



Source: OECD.

Productivity growth has increased in some industries, which typically make intensive use of ICT. In particular, productivity growth in banking and insurance has been rapid,[84] with Finland being one of the leading countries for Internet banking. The greater part of the weaknesses in Finland's productivity probably lies in the sheltered sectors. Productivity growth in the public sector is apparently very low but this is because the national accounts measure value added using input indicators. Nevertheless, Statistics Finland has been quantifying output and productivity in the public sector using alternative output-based measures along the lines being advocated by Eurostat,[85] and for the most part these also indicate a poor performance. With population ageing the productivity performance of the public sector – particularly in areas such as health and social work – will become increasingly important to the whole economy. Indeed recent work by the Bank of Finland suggests that a modest but sustained improvement in the productivity performance of ½ per cent per annum in the ageing-related public services could outweigh the effect of the recent pension reform in improving long-term finances (Kinnunen, 2002).

Looking forward, the strength of Finland's productivity looks set to continue to rely heavily on the manufacturing sector and for the immediate future, the successful roll-out of UMTS (3G) is crucial (Box 9). As with all new products there is uncertainty as to the demand and optimal pricing strategy for UMTS products and the prices set for individual services, such as e-mail, will probably

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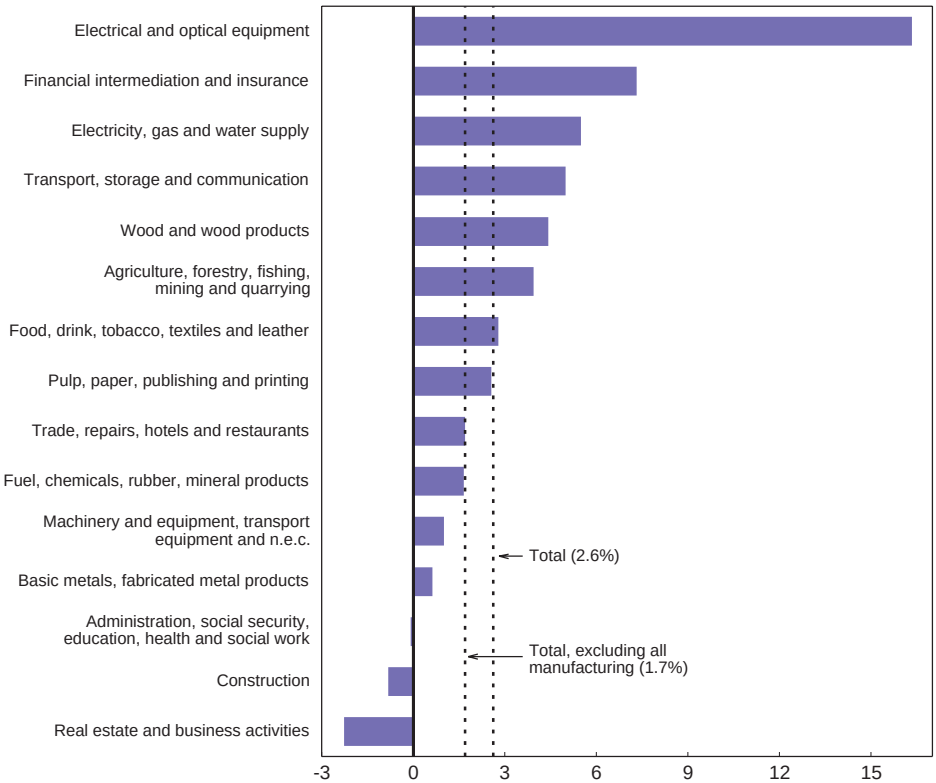
Figure 25. **Labour productivity by sector**

Value added (in constant 1995 prices) per hour worked, annual percentage change, 1995-2001



Source: Statistics Finland (2002), *National Accounts 1993-2001* and OECD.

require refining and adjustment as telecommunications companies discover the true nature of demand. A further complication is the threat of competition from wireless local area networks (WLANs).[86]

Policy towards R&D

Government policy towards R&D has a number of commendable features. As a general rule the policy approach has been to facilitate private sector R&D and for this reason the government has largely avoided the risks entailed in a strategy of "picking winners" backed up by large-scale public funding.[87] Although government-

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Box 9. What is UMTS (3G)?

Universal Mobile Telecommunications Systems (UMTS), also known as third-generation or 3G mobile telephony, is a communication infrastructure that aims to provide not only faster connections compared with existing technology but also a more developed integration with the world wide web. The widely used 2G network allows for vocal communications but permits only relatively short text messages or simple graphic images to be transmitted within the telephone network or between the telephone network and the World Wide Web. A growing share of the market is using "2.5G" technology that allows much larger messages to be transmitted and services such as web-browsing and maps and other navigational devices. UMTS is designed to allow much more comprehensive web-based services including large e-mail messages and audio or video streaming. However, the digital communication speed that users will access through UMTS is likely to be less than was initially estimated, limiting the potential range of services. Indeed, some claim the gap between UMTS and incumbent technology will not prove to be very large at all.[1]

Slots in the radio-wave spectrum to carry UMTS were allocated by international agreement in 1992 and a large number of governments, usually the *de jure* owners, have subsequently sold licences for their use to telecommunications companies.[2] In a number of countries, Finland included, there has been intense debate as to whether the price paid for the licences threatens the commercial viability of UMTS services. Finland's partially state-owned company Sonera has come under particular criticism for its purchase of a German UMTS licence.

1. Manufacturers are claiming that the first generation of UMTS devices will be communicating at 384 kilobits per second (kbps), well below the speed of 2 megabits per second (mbps) that was claimed as possible when UMTS was first announced. Some claim that even 384 kbps is an optimistic speed and in fact it might not be very different in speed from the 50 to 60 kbps available under current technology.
2. The 1992 World Administrative Radio Conference allocated 230MHZ of new radio spectrum to terrestrial and satellite services. Of this 155MHZ was set aside in the 2GHZ bands for terrestrial 3G services.

funded R&D is large by international standards, it is not a large proportion of overall R&D activity. In fact, Finland's mix of R&D is more tilted towards the business sector than in many other countries. Furthermore, there is substantial co-operation between business, public and higher-education research; businesses fund a fairly high share of government and higher-education research and, more broadly, a large share of firms has some sort of co-operation with non-business research institutes. Although these statistics paint a favourable picture, there is nevertheless room for improvement. As noted in the previous S*urvey*, greater use could be made of competitive tendering and R&D spending by ministries that is not chan-

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nelled through the research councils of the Academy of Finland or through the technology development centres (TEKES) of the Ministry of Trade and Industry should be better evaluated. A broad policy issue is the lack of diversification in R&D activity. Much of what makes Finland stand out in international comparison is largely attributable to the ICT sector, and arguably in some areas to just one firm – Nokia. As a result, much of what appears to be good about Finnish R&D and government policy towards it could disappear if key companies were to decide to carry out more of their R&D activities overseas.

The labour market: ambitious goals have been set

The employment of those aged 15 to 64 has risen from 60 per cent in 1994 to over 67 per cent in 2002, but remains well below the 70 per cent target specified in the 1999 Government Programme, which is also the same objective as specified in the EU-wide Lisbon target. At the same time the standardised unemployment rate at over 9 per cent remains well above the EU average[88] and on a broader definition, including those on unemployment and disability pensions or on training schemes, is about double that (Figure 26). Recently an official working group was set up to establish the economic and employment policy guidelines for raising the employment rate to 75 per cent and to lower the unemployment rate to less than 5 per cent over the medium term. These goals appear challenging and almost certainly require further structural reforms over and above those that are currently planned. However, such high employment rates can be found in other countries, for example in Sweden it is over 78 per cent, and the employment rate in Finland was over 74 per cent in the late 1980s. Similarly the objective of a 5 per cent unemployment rate is within the current experience of the better performing OECD economies as well as Finland prior to 1990.

The pension reform, discussed in Chapter II, addresses what appears to be one of the most obvious weaknesses of the labour market, namely the relatively low employment rate of older (especially male) age groups compared to many other OECD countries. In this respect, the more pronounced upturn in the employment rate of those aged 55 to 59 in recent years is a promising sign, and is partly attributable to the tightening of various early retirement schemes that have already taken place (Figure 27). Nevertheless, the estimates reviewed in Chapter II suggest that the resulting increase in the aggregate employment rate, from the pension reform and measures already taken to tighten access to early retirement schemes, is unlikely to exceed 3 to 4 percentage points, and this may take decades to materialise. At the other end of the working age spectrum, policy should address the length of time taken for students to earn degrees in tertiary education, as discussed in Chapter III. This period is typically longer than in many other countries and delays the moment when skilled workers enter the labour market. However, to approach the 75 per cent employment rate objective will

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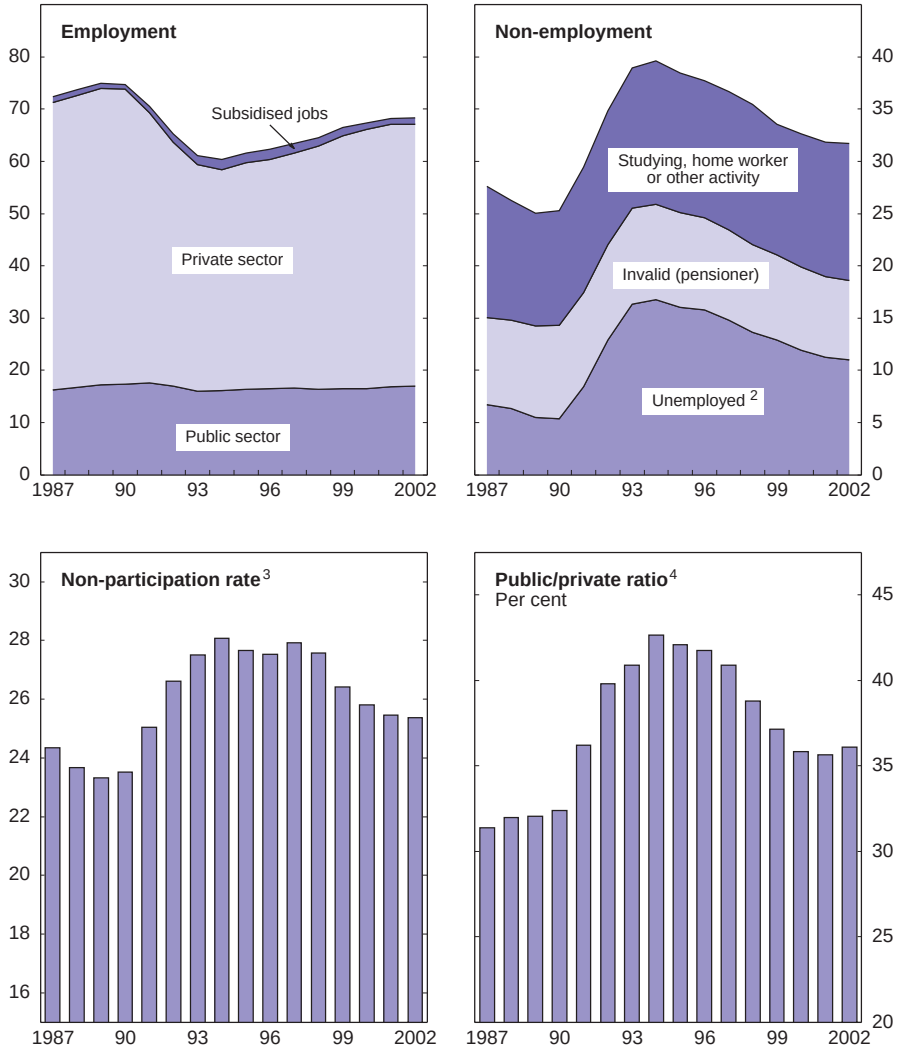
Figure 26. **Structure of the working-age population**
Per cent of population aged 15-64[1]



1. Estimates for 2002.
2. Unemployed receiving benefits or pensions, job seekers receiving no benefits and those on employment training.
3. Those not in the labour force as a per cent of the working age population (both aged 15-64).
4. Public sector and subsidised employment as a per cent of private sector employment.

Source: Ministry of Labour, Statistics Finland and OECD.

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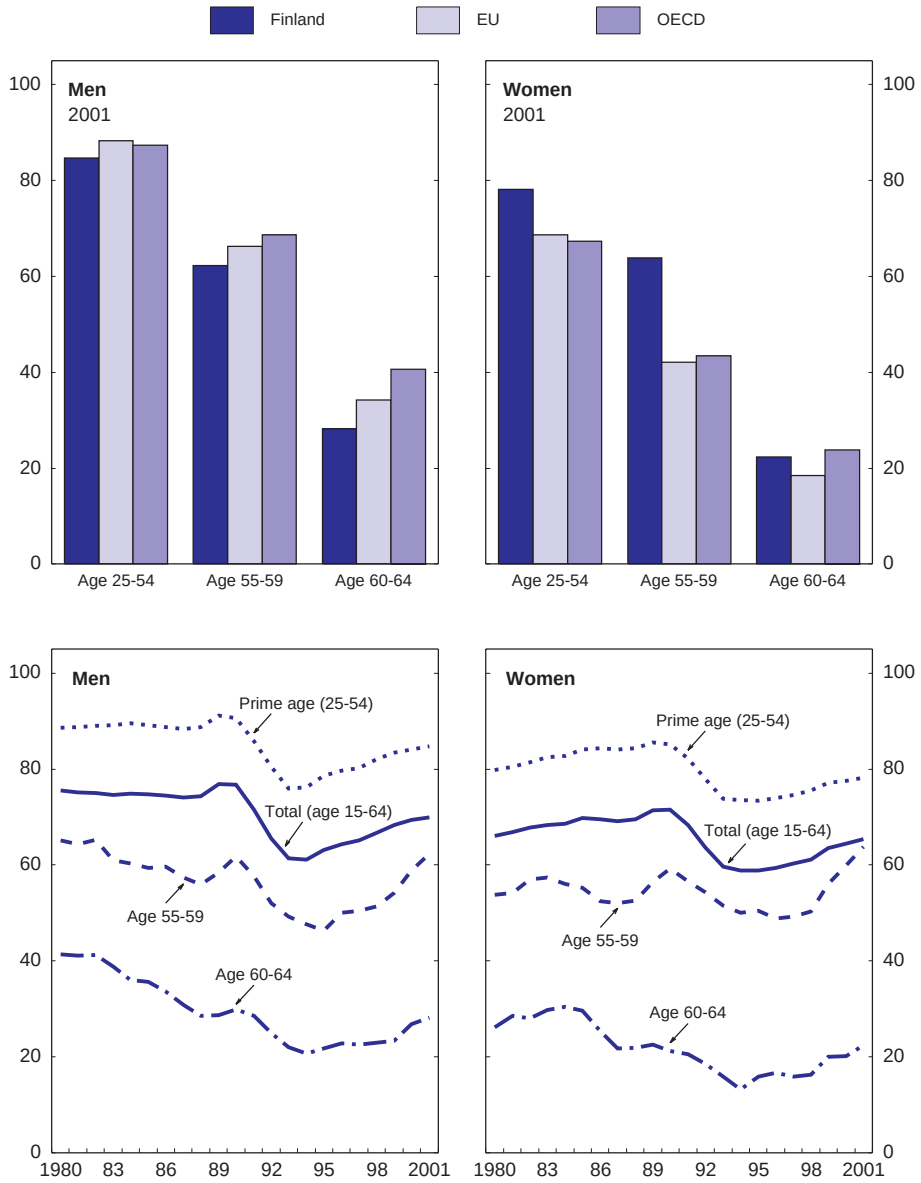
Figure 27. **Employment rates**

Employment as a per cent of population in the same age group



Source: OECD (2002), *Labour Force Statistics*, Part III.

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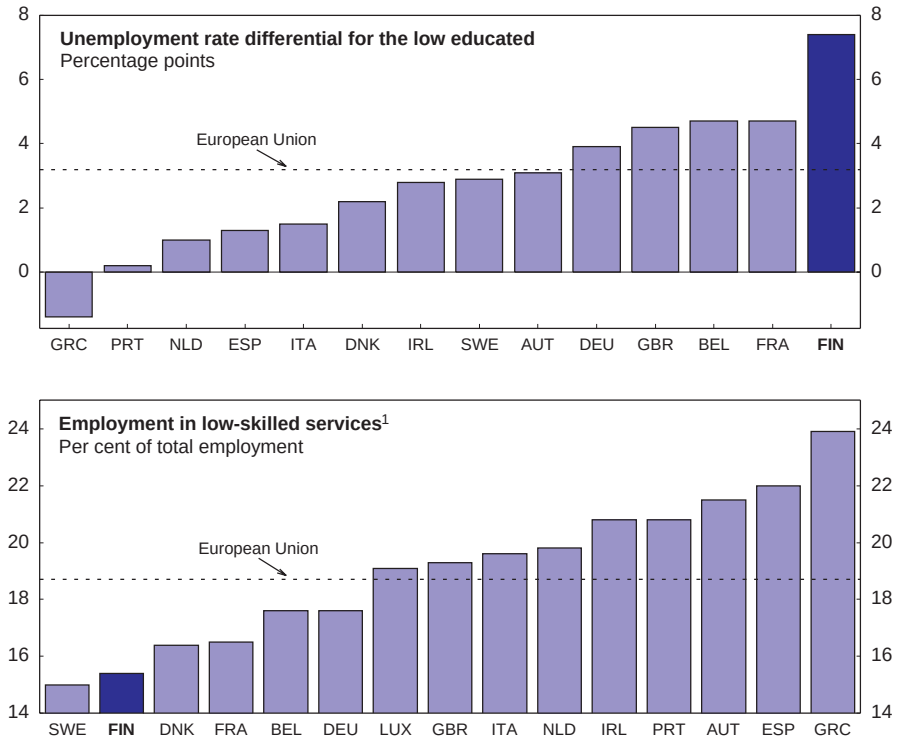
require increases across a broader range of age groups, which in turn will require measures that address the wage-setting, tax and benefit nexus.

Wage compression reduces low-skilled service sector employment

The difference between the unemployment rate of those with low educational attainment and the aggregate unemployment rate is higher in Finland than for any other EU country.[89] At the same time the share of employment in the low-skilled service sector is relatively small (Figure 28). Both are partly consequences of the centralised wage agreements[90] that typically cover the vast majority of workers and result in a highly compressed wage structure (Figure 29).[91] Given that

Figure 28. **Labour market outcomes for the low-skilled**
Age 15-64, 2001



1. Wholesale and retail trade, repair of motor vehicles, hotels and restaurants.
Source: Eurostat.

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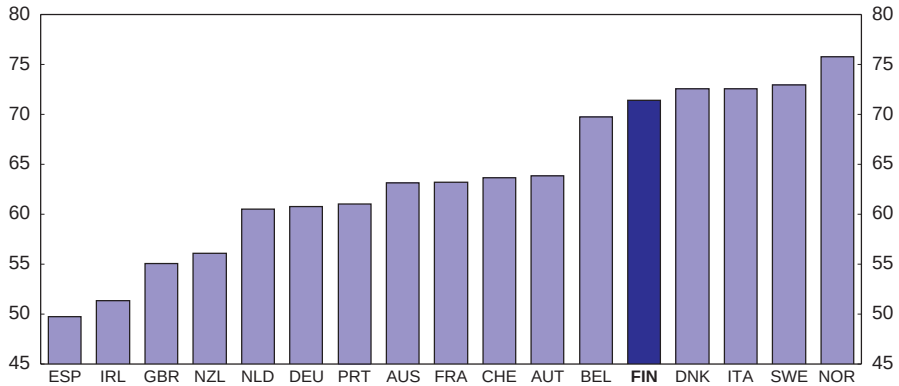
Figure 29. **Measure of earnings compression in selected OECD countries**[1]
Earnings of lowest decile as a percentage of median, 1998 or latest year available[2]



1. Data are based on gross earnings of full-time workers.
2. 1997 for Ireland, Netherlands and New Zealand; 1996 for Austria and Italy; 1995 for Belgium and Spain; 1993 for Portugal; 1991 for Norway; 1990 for Denmark.

Source: OECD Earnings database, December 2002.

aggregate productivity developments play an important part in such agreements, and that productivity in services typically rises by less than in manufacturing, insufficient flexibility in wages leads to a lack of demand for low-skilled labour in services.[92] There is also evidence that wage compression has increased over the last decade,[93] which is consistent with a decline in the extent of market-driven sectoral wage drift around the central wage agreement during the1990s compared to previous decades (Marjanen, 2002). Moreover, realising the full potential of these employment gains may be even more important in the future as demand for many low-wage personal services is likely to increase as the population ages. This would be helped by the introduction of greater flexibility in the wage agreements to reflect differences in sectoral productivity developments and other factors affecting different segments of labour and product markets. At the same time it would be desirable to preserve those aspects of the wage agreements that have been beneficial from a macroeconomic perspective in containing inflationary pressure.

Taxes on labour remain too high

Despite cuts since the mid-1990s, the average and marginal tax rates on labour income remain among the highest in the OECD. In 2001, the average tax

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rate on labour income (income tax plus employee and employer contributions less cash benefits) for a single person on average earnings was 46 per cent with a marginal tax rate of 57 per cent – ranking sixth and fourth highest across all OECD countries (OECD, 2002e).[94] Such taxes reduce the demand for labour, one aspect of which is the promotion of do-it-yourself practices instead of buying low-wage services from the market.[95] The previous *Survey* included an in-depth chapter on taxation, and one of its key conclusions was that high taxes, especially on labour income, are still hampering the growth of potential and distorting behaviour. It argued that there was some scope for shifting the tax burden from labour to property and consumption taxes, and that the earned-income tax allowance should play a smaller role, enabling cuts in statutory income tax rates. It also recommended cuts in social security contributions for the low paid, while recognising the risk that, in the absence of reforms to the wage bargaining framework, there might be offsetting increases in wages. Lowering taxes on labour income was also recommended by two Working Groups recently set up to review the tax system, although there were differing views as to how this should be implemented (see Box 10).

High replacement rates reduce the incentive to seek a regular job

Financial support for those becoming unemployed appears quite generous in international comparison, especially for those with a family. More importantly such support falls off more slowly as the spell of unemployment lengthens than elsewhere. On first becoming unemployed, a person can typically claim the earnings-related "unemployment allowance". Those not eligible, or who have reached the maximum length of the allowance (500 days) are eligible for KELA's "labour market subsidy" which, unlike the unemployment allowance, is means tested and therefore less generous for some households. As its name suggests, the labour market subsidy scheme also involves greater commitments to training and job search. While these are the principle sources of income for most of the unemployed, there are of course other factors to take into account. Other benefits may be received, such as housing and child allowances, and in some circumstances the unemployed may only be eligible for municipal-based social assistance. In addition, income from unemployment insurance in Finland is subject to taxation and this can make an important difference in comparisons with other countries. Table 16 shows the results of some such comparisons, for example the net income of a single unemployed person is estimated to be 65 per cent of that of the average production worker. This particular "net replacement rate" shows Finland to be middle ranking, eleventh highest out of 28 countries. However, for a worker earning two-thirds of the average wages, replacement rates can be very high and provide low-skilled workers with little incentive to find a regular job. More interestingly and largely reflecting the longevity of benefits, for three of the four family types shown in the table Finland moves up the ranking of countries

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Box 10. Working groups on tax reform

Two working groups – one set up by the Ministry of Finance, the other by the Economic Council – have recently published their conclusions on priorities for reforming the Finnish tax system.*

The Ministry of Finance group has a particular focus on company and capital taxation, emphasising pressures from international tax competition. The Economic Council group takes a broader view of the entire tax system, but with a particular emphasis on raising the employment rate. In common with the recommendations from the previous *Survey*, that featured in a special chapter on taxation, both groups attach a priority to reducing the tax rate on labour income. There is also agreement between the *Survey* and the Economic Council working group that reductions in income tax rates can be financed by shifting the burden to higher indirect and property taxes as well as by reducing earned income tax allowances, but that savings in general government expenditure will be required to finance deeper cuts.

Further details of differences and similarities in their main recommendations, both with each other and with those from the previous *Survey*, are summarised below.

- **Personal income tax**. The Ministry of Finance group proposes lowering income tax rates by the same percentage across all income levels, with a specific target of reducing the highest marginal tax rate from about 60 per cent currently to 50 per cent by 2007. The Economic Council group suggests tax cuts should be targeted at low-income groups. It also recommends consideration be given to increasing tax credits for hiring household personnel.
- **Social security contributions**. The Economic Council group proposes reducing payroll taxes to promote employment of the low-skilled, in accordance with the recommendations of the previous *Survey*.
- **Indirect tax**. The *Survey* proposed that the range of goods and services taxed at the normal value added tax (VAT) rate should be widened. Consideration of such a measure is supported by the Economic Council group, as well as raising the general rate of VAT, although it also proposes lowering the VAT rate for the service sector. They also suggest the lower limit for paying VAT should become a sliding minimum to promote the growth of small enterprises.
- **Capital and corporate tax**. While both Working groups and the *Survey* stress the need to ensure that capital and corporate income tax rates remain competitive, the Ministry of Finance group attaches greater priority to reducing rates. It specifically proposes reducing the corporate and capital income tax rate from 29 to 25 per cent and to tax all forms of capital in a neutral way. The group also proposes the abolition of the current imputation system, taxing dividend income at the capital income rate in the future. Sympathy for such a reform was also expressed in both the *Survey* and in the report of the Economic Council working group.

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Box 10. **Working groups on tax reform** (*cont.*)

– **Property tax**. The last *Survey's* recommendations for increasing property taxation gets some support from the Economic Council group, which proposes raising real estate, inheritance and gift taxes. The Ministry of Finance group, in common with the *Survey*, proposes the abolition of the wealth tax.

– **Environmentally related taxes**. The Economic Council group recognises that higher taxes may be required to achieve environmental goals. The *Survey* recommended reducing tax relief granted to heavy polluters as well as reducing the tax advantage of diesel.

* Economic Council (2002) and Ministry of Finance (2002f)

Table 16. **Net replacement rates for an unemployed worker**[1]

1999

Earning level relative to an average production worker	Benefit duration	Family type			
		Single	Married	Couple 2 children	Lone parent 2 children
100 per cent	First month	65	71	83	87
	OECD *ranking*[2]	11	7	5	3
	Long-term	53	71	89	62
	OECD *ranking*[2]	7	2	1	11
66.7 per cent	First month	79	83	88	92
	OECD *ranking*[2]	9	5	5	4
	Long-term	73	92	100	69
	OECD *ranking*[2]	5	4	4	18

1. After tax and including family and housing benefits.
2. Twenty-eight OECD countries; Mexico and Turkey excluded.

Source: OECD (2002e).

when the replacement rate for the first month of unemployment is compared with that for the long-term. Clearly incentives to intensify the search for work or undergo training that are driven by the prospect of declining income with continuing unemployment are weaker in Finland compared with other countries, and this might be worth re-assessing in any future evaluation of the unemployment benefit system.

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Active labour market policies and regional mismatch

Labour demand increasingly focuses on the young and skilled labour and the regional concentration of new jobs in Southern Finland has intensified, suggesting that a significant element of unemployment can be ascribed to regional and skill mismatch. Of particular concern is the group of unskilled who lost their jobs during the early 1990's recession. Special programmes of education or subsidised work in the public sector limit the duration of unemployment spells, but seldom lead to permanent employment in the open labour market. This explains why long-term unemployment is relatively low at about 2 per cent, well below the EU average, even though structural unemployment is high. Abolition of some of the early retirement schemes, particularly the unemployment pension, may lead to an increasing need for active labour market policy (ALMP) measures to focus on the less-educated older labour force.

In addressing regional unemployment problems ALMPs have concentrated on creating subsidised job opportunities for the long-term unemployed in their home areas. Those regions where unemployment is highest have up to four times more participants in ALMP programmes in proportion to the total labour force than other regions. After three months' training nearly half of the participants became unemployed again and after three months' subsidised work almost two-thirds were again unemployed (Ministry of Labour, 2002a). Due to the ineffectiveness of subsidised public sector jobs, the previous S*urvey* recommended a reduction in their number, whereas subsidising private sector employment seems to provide better employment prospects. Recent developments do indeed go in this direction.[96] Often measures that do not lead to permanent employment in the open sector, are nevertheless kept because they prevent exclusion from the labour market, which can be important especially in the case of young people. However, it would be better to promote labour mobility to areas where labour demand is higher. This requires efficient placement services, which should be further developed. Experimental projects outsourcing the placement services to the private sector are interesting and the results should be evaluated carefully.[97]

EU *enlargement will boost immigration*

Despite rising immigration during the 1990s, the share of foreigners is still less than 2 per cent of the labour force, which is the lowest in the European Union. Enlargement of the European Union is expected to lead to large flows of young people from Eastern to Western Europe. It is estimated that in the first year of enlargement there will be about 335 000 persons moving from the new to the existing member countries, but that only about 1½ per cent of them will go to Finland (Ministry of Labour, 2002b). That would mean annual immigration of only about 4 000 to 5 000 persons from the new member countries, mainly from Estonia, in the early phase of enlargement. These flows are expected to diminish

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gradually until 2030, when the share of foreigners is estimated to approach about 6 per cent of the population. Immigration on this scale could make a limited, but significant, contribution to relieving labour shortages, particularly in the service sector, which are likely to become increasingly apparent as the population ages.[98]

Product markets: enhancing competition and stepping up privatisation

A competition-friendly policy framework is important for sustaining a strong long-term growth performance. As noted above, Finland has indeed achieved rapid productivity growth, but this was confined to a few sectors only. Despite the success in bringing high-tech products to the market and the sizeable gains from the early and thorough liberalisation of network industries,[99] several indicators suggest that competition may not be very strong in a number of sectors. Relative price levels are high and import penetration is low, while sectoral indicators (concentration ratios and mark-ups) suggest that competition might be weak in sheltered sectors. It is crucial, therefore, that the competition authority and the sectoral supervisors remain vigilant against anti-competitive behaviour and that obstacles to competition be removed. Moreover, the decision by the current government to continue privatisation of the still sizeable state-owned enterprise sector has been followed through to only a limited extent.

Competition policy

Framework issues[100]

The Finnish Competition Authority (FCA) has broad powers of investigation, negotiation, and recommendation of enforcement action, largely based on the EU competition "toolkit" for horizontal and vertical agreements[101] and mergers. It also has an explicit, active role in regulatory reform. In this respect, the FCA has recently established a specific unit responsible for advocacy. The FCA is an independent agency, although for budgetary reasons it is administratively related to the Ministry of Trade and Industry (MTI), which is itself involved in legislation about competition policy and manages some of the state-owned enterprises. Despite the FCA's administrative position within the MTI, the MTI does not have any legal authority to interfere with the FCA's individual decisions and actions. However, the MTI's shareholder role could lead at least to the appearance of a conflict of interest. The MTI in its shareholder role might be expected to object to the application of competition law against a state-owned company that would reduce the value of its shares. Granting also budgetary independence to the principal first-instance competition watchdog, the FCA, would tend to reduce such concerns.

The FCA does propose, but not impose orders and fines. This is the role of a judicial body, the Market Court, which was created in March 2002 as a result of

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the merger of the earlier Market Court and the Competition Council.[102] It is an independent decision-maker, composed of judges and market experts and its resources are provided by the Ministry of Justice, rather than the MTI. As a true court it is expected to follow standard judicial processes more closely than the Council did and it has greater procedural authority than the Council, such as subpoena powers to compel attendance of witnesses.

While the competition policy framework is basically sound, and the institutional framework has been strengthened with the establishment of the Market Court, enforcement could be strengthened. There is an apparent lack of judicial support for enforcement against horizontal restraints. The size of the maximum fine – EUR 670 000 or in exceptional cases up to 10 per cent of the previous year's turnover – is modest by international comparison, and there have been no "exceptional" cases drawing the 10 per cent fine. Furthermore, courts have only twice exceeded the basic range of sanctions (EUR 84 to 670 000). Due to the limited number of cartel cases brought before the courts, far reaching conclusions should, however, not be drawn in this respect. In addition, the courts have a tendency to reduce the fines proposed by the FCA, sometimes to insignificant amounts. The highest fine imposed so far was EUR 4.2 million for the abuse of dominance through discriminatory denial of access to telecoms infrastructure. The basic range of sanctions should be expanded greatly, and it should be possible to compute fines based on turnover or gain over the whole period of violation. Moreover, some criminal liability or sanction on the individuals who are responsible for the violations could be considered. The risk of personal liability should discourage executives from engaging in illegal conduct. Detecting restraints of competition would also be made easier, if a leniency programme existed. Given the low level of actual fines, there is probably little incentive for witnesses to come forward. However, stiffer fines in conjunction with a leniency programme might help investigations and give teeth to competition policy.

The FCA is generally seen to be working well, and those who deal with the agency regularly find it responsive and fair (OECD, 2003). However, its workload and priorities reveal conflicting demands on its resources, which are smaller than in the other Nordic countries. A high proportion of FCA matters have dealt with abuse of dominance and mergers, and the thresholds for investigating mergers may be too low. However, cartel enforcement has received less attention in recent years. This has already changed, as the FCA has been restructured, including a new unit, focusing exclusively on eradicating cartels. Nevertheless, it is probably necessary to step up FCA resources in order to ensure more effective competition in Finland.

Sectoral issues[103]

Primary agricultural producers are excluded from the Competition Act,[104] but processing, distribution and marketing are not exempted. Enforcement action

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has targeted the abuse of dominance by the major dairy company, horizontal co-operation on purchase prices and market division in the meat processing industry, abuse of dominance by a grain trader and prohibited price fixing by egg producers. Abuse of dominant position, together with the fact that associations of producers often represent their members in negotiations with buyers and processors, are of particular concern in this sector. Concerning *retail trade*, regulatory constraints as well as high concentration and industry co-operation may be dampening retail competition. Land use controls, for instance, have discouraged the entry of large-scale operators. Foreign chain retailers have only become established in Finland within the last two years. Moreover, controls on opening hours can inhibit competition among outlets already in the market. These controls were relaxed somewhat in 2001, and some retail sectors are exempted from limits on opening hours. The explicit purpose of the legislation is to shield particular retail sectors from competition. This legislation is reviewed every three to five years; thus no change will take place before 2004-06. Entry of *pharmacies* is guided by a market size test (they are licensed by the National Agency for Medicines), while there is a tariff schedule of recommended maximum prices. The maximum prices are *de facto* the uniform prices found in the market. Already in the mid-1990s, the FCA had recommended the abolition of the market size test and of the recommended retail prices. The advice has so far not been heeded. Alko, a state-owned company, has a monopoly on retail sales of alcoholic beverages, with the exception of a few low-alcohol products. There have been some controversies about Alko's obviously dominant position, through discriminatory practices affecting wholesale and import markets. In the meantime, import and wholesale operations have been separated from retail trade, into Altia Ltd., which is supposed to compete with private firms. Plans to privatise Altia have not yet been followed through.

Network industries were liberalised early on in Finland. Competition issues have remained prominent, however. The FCA has concentrated considerable resources on investigating *telecoms services*. It has dealt with issues such as interconnection and access pricing by local exchanges or pricing of interconnections and has investigated several cases of abuse of dominant position. Many cases have involved "ownership discount" arrangements, which can discourage entry. And the FCA has resisted efforts by major players to acquire dominant positions in cable infrastructure, because cable is often the only source of alternative fixed-line services. The *electricity market* is among the few markets in the European Union that is fully liberalised, and it is integrated in the Nordic electricity market. The FCA and the Energy Market Authority both monitor grid access and pricing. However, vertical ties and concentration in generation still require vigilance. The Competition Act contains a special rule about acquisitions in this sector, aimed principally at preventing the recreation of vertically-integrated large firms. Moreover, the MTI is preparing legislation that would require more extensive vertical separation of network operations.

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In the transport sector, the FCA has launched a wide-ranging investigation on the market for *air transport* in concert with the other Nordic competition authorities (Report from the Nordic competition authorities, 2002). The FCA is also investigating Finnair's strong market position, as there is only one independent additional carrier on the domestic market that co-operates with Finnair. Moreover, Finnair's market position is also strong for international and charter flights. The investigation focuses on potential entry barriers, such as measures that create customer loyalty. A separate case focuses on Finnair's activity in the ground handling market, where it is also an important player. In the *railway sector*, competition will only come slowly. Railway services are still essentially a monopoly (the state-owned VR-Group) and liberalisation will be no faster than the EU schedule: international freight, amounting to only 3 per cent of traffic, will be opened to competition in 2003. The FCA has reiterated its view that the opening to competition should be faster and more wide-ranging and a working group of the Ministry of Transport has recently suggested to spin-off VR-Track, the arm of the company responsible for track construction and maintenance, with the final aim being its privatisation. The Ministry has also suggested separating the Road Traffic Administration from the Interior Ministry. After separation it would also be a potential candidate for privatisation. However, privatisation seems unlikely to come soon.

Agricultural subsidies

State aid to most sectors is low and considerably below the EU average. Agricultural subsidies, on the other hand, have remained very high. National spending amounted to EUR 1.4 billion between 1999 and 2001, which includes the fully nationally financed National Aid programme, the national spending that is co-financing EU programmes and outlays on general services. This compares to only EUR 0.4 billion in Sweden or somewhat above EUR 1 billion in Italy or the United Kingdom. Spending was only higher in France and Germany (OECD, 2002f). In 2001, support payments by the Ministry of Agriculture on agriculture and horticulture amounted to EUR 1.7 billion, 42 per cent of which were financed by the European Union. In addition to the direct aid granted via the Common Agricultural Policy and the National Aid programme, the sector benefits from EU environmental aid and the EU least favoured area support, while indirect aid is provided by consumers, who have to pay prices that are often far above world market prices. National aid and the least favoured area support aim at offsetting the impact of low agricultural productivity due to the harsh climate. When Finland joined the European Union, a large amount of national aid was agreed for a five-year transitional period ending in 1999, which has been replaced since then by the National Aid for Southern Finland. The report A *Strategy for Finnish Agriculture to Year* 2010 was released in 2001 and represented a joint effort led by the Ministry of Agriculture with representatives of producer organisations and the government parties. The strategy calls for reinforcing consumer-oriented action in the whole food chain,

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e.g. through securing the profitability and operating conditions of agriculture as well as making the Common Agricultural Policy better reflect the differences in competitiveness due to natural conditions. As highlighted in previous Surveys, regional and environmental goals of agricultural policy should be achieved in a less costly and distortionary way than under the current agricultural aid arrangements.[105] The Finnish agricultural sector has experienced rapid structural change. The number of farms has fallen from 100 000 in 1995 to about 75 000 in 2001. At the same time, the migration from the rural areas to the cities has accelerated. The further restructuring of the agricultural sector, where Finland generally does not have a comparative advantage would release resources for more productive use, reduce food prices for consumers and improve the budgetary situation.

Privatisation

According to the Ministry of Trade and Industry, the Finnish State does not have a privatisation programme, but pursues the gradual extension of the ownership base with the goal of strengthening the companies' operating conditions. In a welcome move, Parliament actually broadened the privatisation mandates in 2001. However, privatisation receipts (net of recapitalisations) were actually negative in 2001 and amounted to only EUR 1.3 billion between January and September 2002. Since the case-by-case privatisation process started in 1992, nearly three quarters of the privatisation proceeds have accrued from the partial sale of Sonera, the dominant telecommunications company, while the reduction in shareholdings in traditional industries has been limited, partly reflecting the decline in the stock market since 2000. State-ownership remains, therefore, substantial, with full and partly state-owned enterprises employing around 200 000 persons, almost 12 per cent of private sector employment. The market value of state shareholdings in listed companies amounted to EUR 10.4 billion in September 2002 and the unused privatisation mandates to around EUR 7 billion (Table 17), while in early 2000, the value of non-listed companies (for instance, Alko, Posti and VR-Group) was estimated at EUR 8 billion.

Employees of state-owned enterprises are employed under standard private sector legislation and collective agreements, rather than enjoying a civil servant status. This provides flexibility to the enterprises but raises employment risks for the employees. Concerns about these risks are a key constraint on the pace and extent of privatisation, as unions prefer the government to retain a significant minority interest in privatised companies, even where these do not pursue any obvious public policy objectives. This constraint operates in part through the parliamentary mandates for privatisation, which have been more limited than the proposals put forward by the government in some cases (OECD, 2003).

The state's shareholdings are administered by the ministry covering the sector in which the company operates. Nine ministries are currently involved in

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Table 17. **Main state-owned companies**

Company	Sector	Persons employed in 2001 (thousands)	In September 2002 (% of share)		Previous minimum	
			Ownership	Minimum	Ownership (% of share)	Year of change
Alko	Alcoholic beverage retail trade	2.2	100	100	. .	. .
Altia	Production and wholesale of alcoholic beverages	0.7	100	50.1	66.7	2001
Avena	Grain wholesale trade, handling and storage	0.4	100	0	100	2001
Avesta Polarit	Stainless steel	9.0	1.2	0	. .	. .
Edita	Printing and publishing	1.6	100	100	. .	. .
Finnair	Airline	10.8	58.4	50.1	66.7	1994
Fortum	Energy	14.8	60.8	50.1	66.7	1998
Inspecta	Technical inspection	0.3	100	0	33.4	2001
Kemijoki	Energy	0.4	67.0	51.0	66.7	1997
Kemira	Chemicals	10.2	56.2	15.0	33.4	2001
Kone	Lifts and escalators	23.0	4.7	0	. .	. .
Metso	Metal engineering	25.6	11.6	0	. .	. .
Outokumpu	Metals and technology	19.0	39.7	10.0	33.4	2001
Patria	Defence	2.3	73.2	50.1	100	1999
Posti	Postal services	23.3	100	100	. .	. .
Rautaruukki	Metals	13.7	40.1	20.0	33.4	2001
Sampo	Banking and insurance	10.2	40.3	0	20.0	2001
Sonera	Telecommunications	10.5	52.8	0	33.4	2000
Stora Enso	Forest products	44.3	10.8[1]	0	. .	. .
Vapo	Peat and timber	0.3	66.7	50.1	66.7	2001
Veikkaus	Lottery and football pools	1.2	100	100	. .	. .
VR Group	Railways	14.9	100	100	. .	. .
Yleisradio	Broadcasting	3.8	100	100	. .	. .

1. Government share in votes is 23.4 per cent.
Source: Ministry of Trade and Industry.

the management of state shareholdings, with the Ministry of Trade and Industry in a co-ordinating role. The spread of ownership functions between ministries means that regulation of a sector and ownership of a major firm in that sector often coincides. One objective of ownership is to earn profits and maximise the value of the firm, but insofar as the firm has market power this might be contrary to the interests of the community at large or other participants in the industry. There is thus the potential under present arrangements for conflicts of interest to arise within a ministry. While firewalls ensure a strict separation of regulatory and ownership functions within ministries, these potential conflicts must be resolved somewhere in government. Separating the ownership from the regulatory function would

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avoid such conflicts of interest and a centralisation of ownership function would make it easier to deal with ownership issues on a more consistent basis. This issue has come to the fore partly due to the large losses incurred by Sonera[106] on its UMTS license investment in Germany and the consequent need for recapitalisation. The government has submitted a policy statement to Parliament indicating that share-holdings of most state-owned enterprises should be centralised in one government body. However, there is no consensus yet as to where this centralised body would be located.

Financial markets: banking on prudence

As elsewhere, financial markets in Finland have suffered from the meltdown in stock markets, following the bursting of the ICT bubble and the confidence crises in the wake of the Enron scandal. Since early 1998, the Helsinki exchange has shown a much bigger movement than either in the United States or the euro area, reflecting the high weight of the ICT sector (Figure 30). While the decline since early 2000 was very large in Finland, the index is still above its value of four years ago, in contrast with the United States and euro area index. While also suffering, stock prices of banks have held up relatively well, but the fall for insurance companies has been substantial since early 2001, especially for the life-insurance sector.

Prior to the current downturn, Finnish financial companies had strong profitability and balance sheets reflecting high economic growth, substantial cost reductions due to restructuring and the introduction of new technologies. However, the profitability of banks and banking groups has been sliding since 2001 largely because of a reduction in non-interest income, while net interest income increased. Higher, though still low, loan loss provisions had a modest negative effect on profits until mid-2002, but they are likely to rise. Overall, the profitability of Finnish banks is still fairly good. Inspection visits by the Financial Supervision Authority (FSA, 2002) concluded that Information Technology (IT) sector risks did not pose a threat to bank solvency. While household indebtedness is still relatively low as a per cent of GDP, housing loans have expanded by more than 10 per cent annually since late 1998.

Insurance companies have suffered considerably more than banks from the steep drop in equity prices. The decline was to some extent offset by the steady flow of premium incomes and revenues from real estate holdings as well as money market and debt instruments. The biggest employment pension insurers (Varma-Sampo and Ilmarinen) and the biggest life insurer (Henki-Sampo) incurred large losses on their investment portfolios and reductions in their solvency margin. The solvency requirements are comprised of a solvency limit and a target zone, with the lower limit of the target zone double and the upper limit four times the solvency limit. The stock market losses have reduced solvency to the lower

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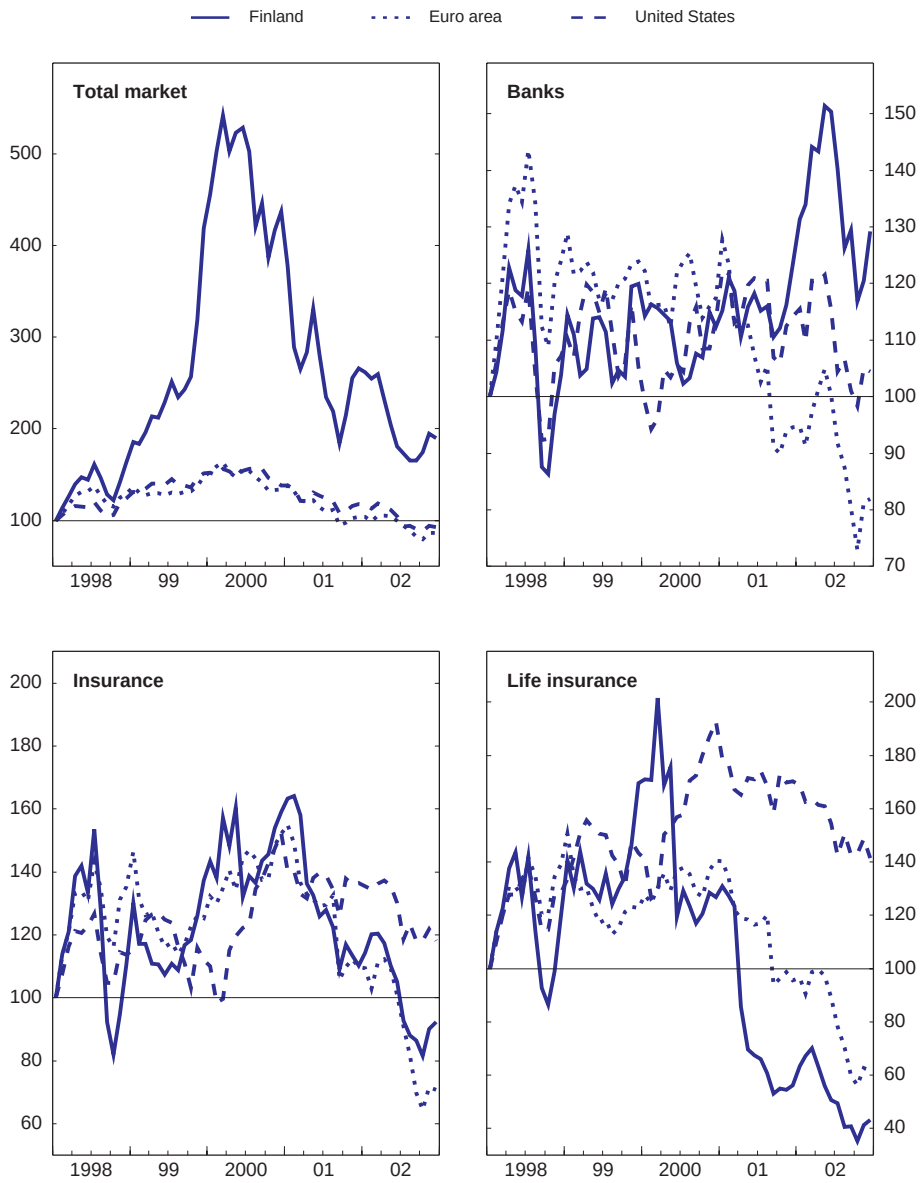
Figure 30. **Equity indices**
January 1998 = 100



Source: Datastream and OECD.

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limit of the target zone in the first half of 2002, which implies that pension funds can no longer provide rebates to an employer, even though there is still a considerable margin until the solvency limit will be hit. Continuing stock market weakness could imply that the cost of pension insurance will have to go up.

Regulation and supervision of the financial sector has been under pressure to adapt to the ongoing merger process between banks and insurance companies and internationalisation of the financial sector. The biggest Finnish bank, Nordea, for instance, has its head office in Sweden, which implies that the Swedish supervisory authority supervises the holding company and the FSA the subsidiary. There is now close co-operation between the Nordic supervisory authorities. In Finland, the merger of the FSA and Insurance Supervision Authority had been contemplated, but the government decided that the current set-up, including exchange of information and common board members, works well (FSA, 2002). While there has been a trend towards establishing a single supervisory authority in a number of countries, the Finnish situation is more complex, because of the core role that insurance companies play in the national pension system. On the other hand, the FSA's powers to take action will be strengthened. Legislation is under preparation that will provide the FSA with powers to impose certain administrative sanctions (cautions and warnings) and information requirements will be strengthened. Moreover, the responsibility for the licensing and revocation of licenses of banks and investment firms, now belonging to the Ministry of Finance, will be given to the FSA. The new act will stress the accountability of the FSA by demanding that the Board issue annual reports to the Parliamentary Supervisory Council on the objectives of the FSA and whether they have been reached. A new act on the Supervision of Financial Conglomerates had already entered into force, which makes the supervision of conglomerates clearer and more effective.

Sustainable development

There is growing concern that long-run sustainable development may be compromised unless measures are taken to achieve balance between economic, environmental and social outcomes. This section looks at three issues at the interface of the environmental and the economic dimensions of sustainable development, drawing out the social implications when relevant, that are of particular importance for Finland. In each case, indicators are presented to measure progress and the evolution of potential problems, and an assessment is made of government policies that affect the issue. The section also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (see Box 11). The section first considers Finnish policies in the realm of *climate change* policy then looks at *air pollution*. Finally, the section examines policies to ensure *sustainable use of natural resources*,

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Box 11. The integration of policies across sustainable development areas*

Since 1993, the National Council for Sustainable Development has been the main forum for government and other institutions to discuss sustainable development policy issues. It is chaired by the Prime Minister and includes representatives from the Parliament, central and local government, business, labour and non-governmental organisations, and scientists. The broad composition of the Council offers the possibility to integrate economic, environmental and social concerns in its deliberations about sustainable development issues. The Council does not have legislative power, but its sub-committees can influence policy through the preparation of background documents. Under the aegis of the Council, the 1998 Programme for Sustainable Development was developed. Ministries take the national programme into account in developing specific sectoral programmes.

Policy integration is institutionalised through widespread consultation during the drafting of new legislation. However, providing potential losers with a strong voice in the consultation process risks unduly weakening the eventual measures or could build in a bias for sector-specific policies that may be costly for society as a whole. For example, cost-effective pollution abatement measures (which are in the general interest) have occasionally not been introduced due to strong opposition from sectoral interest groups.

A lack of attention to cost-benefit analysis is a serious weakness in integrating the different dimensions of sustainable development in policy making. Environmental assessments are obligatory for projects that may have a significant environmental impact. Following the approval of EC Directive 2001/42, environmental assessments of policies, programmes or plans will also be undertaken when these set the framework for future projects in the following sectors: forestry, fishing, energy, industry, transport, waste management, telecommunications, tourism, town and country planning or land use. However, a cost-benefit analysis is not required. "Economic assessments" are required for all legislative proposals, including those for environmental and social policies, but these are generally not formal cost-benefit analysis. Greater use of such techniques would better identify the trade-offs between the various goals of sustainable development policy and thus allow policy makers to make more informed choices.

* The sections in this report dealing with climate change, air pollution and the sustainable use of natural resources are inputs into the OECD's follow up on Sustainable Development as mandated by the Ministerial Council decision in May 2001.

focusing on forest management. Of course, the reform of the pension system, assessed in Chapter II, is also important in improving sustainable growth in the long run.

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Table 18. **Main indicators: climate change**

Indicators of greenhouse gas emission intensity, grams of CO_2 equivalent per USD[1] of GDP, in 1995 prices

	Level of emissions, 1999				Average annual percentage change 1990-99			
	Total	CO_2 from electricity	CO_2 from transport	Other	Total	CO_2 from electricity	CO_2 from transport	Other
Finland	**652**	**181**	**105**	**366**	**–1.88**	**–0.02**	**–1.29**	**–2.83**
Denmark	549	194	94	261	–1.64	–1.43	–1.49	–1.85
Norway	487	4	113	369	–2.54	1.31	–1.53	–2.87
Sweden	358	41	112	204	–1.55	0.07	–0.65	–2.30
Austria	419	72	91	256	–1.87	–2.75	–0.52	–2.06
Belgium	617	97	101	419	–1.36	–2.12	0.16	–1.52
France	416	32	103	280	–1.69	–2.04	0.16	–2.26
Germany	536	169	96	271	–4.00	–3.86	–0.57	–5.05
Greece	813	275	130	408	–0.24	0.07	0.74	–0.73
Ireland	694	165	103	426	–4.27	–2.41	0.79	–5.75
Italy	439	105	92	242	–1.05	–0.82	0.37	–1.64
Netherlands	573	138	82	352	–2.38	–1.03	–0.94	–3.15
Portugal	540	149	106	285	0.41	2.58	3.37	–1.39
Spain	537	127	130	280	0.41	1.12	1.28	–0.26
United Kingdom	526	132	108	287	–3.66	–5.30	–1.38	–3.61
Australia	1 053	370	155	528	–2.07	–0.21	–1.93	–3.24
Canada	893	151	193	549	–0.98	–0.12	–0.36	–1.41
New Zealand	1 096	92	175	828	–2.28	4.58	0.65	–3.32
Switzerland	276	3	79	195	–0.62	–1.96	–0.28	–0.73
United States	792	278	196	318	–1.89	–0.60	–1.18	–3.28
OECD total	649	196	140	312	–1.80	–0.98	–0.38	–2.83
European Union	506	120	103	283	–2.36	–2.60	–0.16	–2.95

1. National currencies converted to USD using purchasing power parities.

Source: Greenhouse gas emissions from national submissions to the United Nations Framework Convention on Climate Change (UNFCCC) and national publications; carbon dioxide emissions for electricity and transport from the International Energy Agency and GDP from OECD, National Accounts database.

Climate change

Main issues

Man-made emissions of greenhouse gases (GHGs) are widely agreed to have been the principal cause of the increase in average temperatures over the past century. With global emissions continuing to grow, climate change may impose significant costs on OECD and, especially, non-OECD countries. Finland led the way in taxing carbon emissions and is currently participating in global efforts to limit GHG emissions through the Kyoto Protocol. Even if the EU Burden Sharing agreement stipulates that Finland only has to stabilise its emissions by

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2008-12 at their 1990 level (in contrast to the 8 per cent fall for the European Union as a whole), it will be a challenge for the authorities to attain this target without imposing large economic costs on society.

Performance

Greenhouse gas emissions per unit of GDP are close to the OECD average but very high compared with other EU countries, and they have fallen less than in these countries in the 1990s (Table 18).[107] The high GHG intensity is partly related to the high share of energy-intensive industries, notwithstanding the strong decoupling of emission from output trends in manufacturing industries in the 1990s (Table 19). Carbon dioxide emissions from electricity generation in 1999 were well above the EU average and emissions per kilowatt-hour have fallen less rapidly than on average in the European Union. Carbon-free energy accounted for 41 per cent of total energy consumption in 2001. Business-as-usual projections suggest that, in the absence of any corrective measures, Finland would miss its target under the EU burden-sharing agreement by 18 per cent.

Policy

The government undertook a major review of various policy options to meet international commitments when preparing the National Climate Strategy. It determined that the construction of a new nuclear electricity plant would be the most economic method of reducing emissions (Tarjanne and Luostarinen, 2001) (Figure 31).[108] A private company, whose shareholders include major industrial electricity users, has received permission to construct a new nuclear power plant, which should, by 2010, lower emissions by 9 per cent, about half the expected Finnish overshoot of the target.

The remaining carbon abatement to achieve the Kyoto targets is to come primarily from policies to promote renewable energy and energy saving and efficiency.[109] The government expects to achieve almost a quarter of its targeted abatement through renewable energy.[110] Biomass energy production will be granted investment subsidies of up to 40 per cent, rebates from the electricity tax, and exemption from the carbon dioxide tax on heating. The government will also continue to subsidise the forestry sector to produce energy wood, albeit on a small scale.[111] Emissions of greenhouse gases from landfill sites should be reduced through the already implemented landfill tax that is being periodically increased.[112] In addition, by 2005, all sites will have to comply with regulations that impose the recovery of methane emissions. Voluntary agreements and energy audits are important components of policies designed to attain gains in energy saving and efficiency. Over the period 1992 to 2001, energy audits are estimated by the government to have lowered total greenhouse gas emissions by around

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Table 19. **Greenhouse gas emissions and sectoral indicators**

Annual average percentage change, 1990-99[1]

	Total level in 1999[1]	Total emissions	CO_2 emissions				Indicators per unit of GDP	
			Per kilowatt hour of electricity	Manufacturing, per unit of industrial production	Residential, per unit of private consumption	Road transport, per vehicle	Electricity use	Industrial output
Finland	**76**	**–0.1**	**–1.0**	**–5.4**	**–7.7**	**–0.6**	**1.0**	**2.9**
Denmark	73	0.6	–3.7	–3.6	–5.5	0.5	2.4	0.8
Norway	56	0.9	4.8	–2.0	–7.8	0.9	–3.4	–1.5
Sweden	71	0.0	1.0	–4.0	–4.1	0.4	–0.9	1.5
Austria	80	0.3	–2.5	–3.1	–4.6	–1.8	–0.2	1.1
Belgium	151	0.6	–2.0	–1.0	–1.3	0.0	–0.1	–0.3
France	549	–0.1	–2.8	–1.5	–1.2	0.1	0.8	–0.2
Germany	994	–2.3	–2.2	–2.7	–5.0	–2.6	–1.7	–0.9
Greece	124	1.9	–1.7	–1.6	8.0	–1.1	1.8	–1.4
Ireland	65	2.2	–0.6	–11.2	–5.6	3.5	–1.8	5.6
Italy	540	0.4	–1.6	–1.8	0.4	0.4	0.8	–0.4
Netherlands	218	0.4	–0.3	–2.4	–4.1	–0.2	–0.7	–0.8
Portugal	86	3.1	0.6	1.2	1.9	–0.4	2.0	–1.2
Spain	371	2.9	0.1	–1.5	1.9	–0.2	1.0	–0.4
United Kingdom	647	–1.5	–4.7	–1.7	–0.7	–0.2	–0.7	–1.6
Australia	489	1.6	0.4	–0.5	–2.0	–1.7	–0.6	–1.9
Canada	699	1.6	0.4	–3.1	–3.4	1.4	–0.6	1.0
New Zealand	76	0.5	5.6	0.3	–3.4	–0.6	–0.9	–1.0
Switzerland	53	0.0	–3.8	0.1	–2.5	–0.7	1.9	1.0
United States	6 830	1.2	0.2	–5.3	–2.3	1.1	–0.8	1.2
OECD[3]	14 257	0.5	–0.7	–3.1	–1.6	0.1	–0.3	0.1
OECD excluding USA[3]	7 428	0.0	–1.8	–2.0	–1.1	–0.6	0.0	–0.6
European Union	4 048	–0.4	–2.4	–2.2	–2.0	–0.6	–0.2	–0.5

1. 1990-98 for road transport CO_2 emissions. 1991-99 for German CO_2 emissions.
2. In million tonnes CO_2 equivalent.
3. The OECD total covers 27 countries.

Source: Greenhouse gas national submissions to the United Nations Framework Convention on Climate Change (UNFCCC), national sources and UNFCCC; carbon dioxide data from the International Energy Agency; industrial production and private consumption from OECD.

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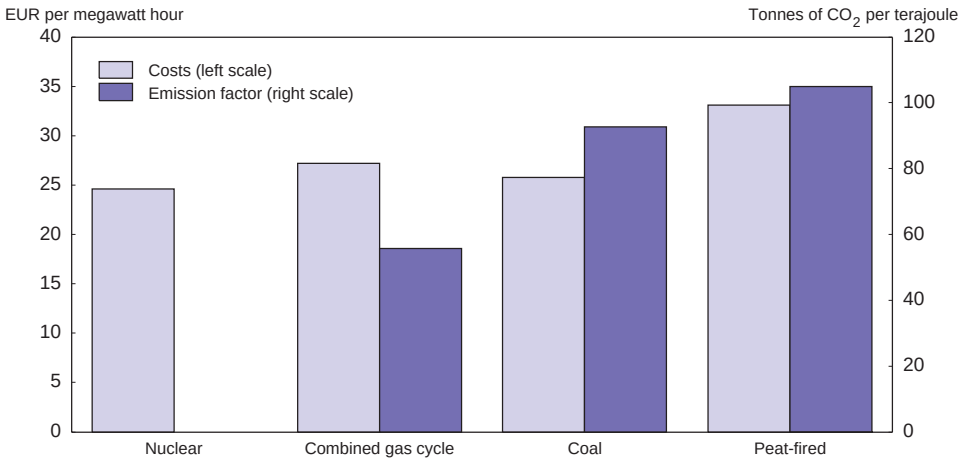
Figure 31. **Climate change options: capital and operating costs and emission factors**

Electricity generation[1]



1. 2000 data for capital and operating costs, and 1995 for emission factors.
Source: Tarjanne and Rissanen (2000), Lehtilä and Tuhkanen (1999).

0.2 per cent, while between 1998 and 2001 the reduction from voluntary agreements is estimated at about one per cent of emissions during that period.[113] The national climate strategy targets these programmes as important means of achieving part of the carbon abatement, though additional measures will be required to meet the abatement needed in 2008-12.

Since the 1990s, an aim of the Finnish government has been to shift the burden of taxation from labour towards environmental and energy taxes.[114] Finland was the first country to introduce a carbon tax in 1990, but this was reformed in 1997 following the opening of the Nordic electricity market. Domestic industry and the electricity sector were disadvantaged by the fact that energy-intensive sectors in the other Nordic countries were exempted from the carbon tax and, because of EU agreements, electricity imports could not be taxed at the average tax on domestic production (based on the carbon content of Finnish electricity).[115] As a result the carbon tax on electricity was scrapped and an electricity consumption tax was introduced with a lower rate for industrial use.[116] Moreover, the tax rate paid by energy intensive companies potentially falls 85 per cent relative to other industries, once energy taxes exceed both EUR 50 000 and 3.7 per cent of the value-added by the company. In practice, the majority of the companies that apply for this refund have to pay between one-half and three-fourths of the full

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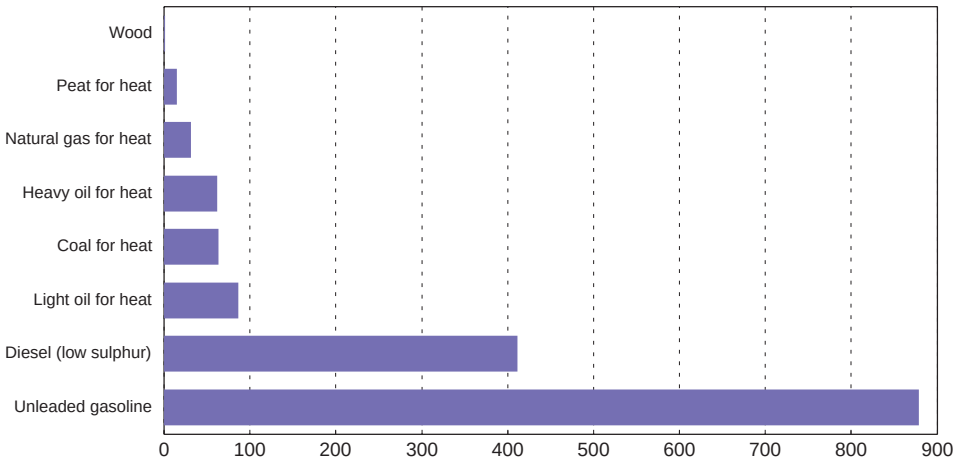
tax. Only in some exceptional cases does the refund amount to 65 per cent of the full tax. The carbon tax continues to be applied on heating fuels, with lower rates applied for natural gas and peat, and transportation fuels. Overall, tax rates vary considerably between fuels and sectors (Figure 32).

The climate change strategy emphasises the need to develop energy taxation to promote energy conservation. Calculations made when the national climate change strategy was being developed showed that energy taxation should be raised. Given the development of a new nuclear power plant, energy taxes would need to rise by between 30 and 50 per cent. The government is committed to reviewing further the role and efficiency of energy taxation in the climate change strategy, taking into account other prospective flexible measures, such as the EU-wide emission-trading scheme. The national climate strategy supports the use of such mechanisms if domestic measures are insufficient to meet abatement commitments or if they prove more cost-effective than domestic measures. As the present system does not target carbon dioxide emissions particularly effectively, a cross border trading scheme would help in equalising abatement incentives. Indeed, in December 2002, the Finnish government stated its support for an EU-wide emission-trading scheme.

An area where current policies have perverse results from the point of view of external costs is the failure to treat peat and other forms of carbon equally.

Figure 32. **Implied carbon tax for different fuels in varying uses**[1]
EUR per tonne of carbon, January 2002



1. The standard carbon tax rate is EUR 63 per tonne of carbon.
Source: Calculations based on data from the OECD Environmental Tax database.

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This fuel is even more carbon intensive than coal and accounted for almost 9 per cent of carbon-based fuel used in 2001. Currently available EU studies[117] suggest that the net emissions of carbon dioxide from new plants using this fuel source are 7 per cent higher than those from coal and 117 per cent higher than those from natural gas.[118] Such estimates take into account the likely absorption of carbon dioxide when peat mires are reforested, though these gains would be lower if the peat mires themselves were regenerated which takes around 2000 years.[119] The electricity co-generated in small-scale peat-fired heating installations receives tax rebates on the same scale as renewable energy sources. Furthermore, the carbon tax on peat is only one-fourth of that for coal. Despite current information on the external costs of generating electricity from peat, the national climate strategy assesses that further research is needed and supports for the time being the continued use of peat in heat generation and in co-generation of electricity and heat, unless its use threatens the competitiveness of wood in energy production. Peat is a competitive source of fuel in certain areas of Finland,[120] but in other parts of the country it requires a substantial tax rebate to be economic in new small-scale plants, though some combustion efficiency gains may be possible with the co-firing of wood-based fuels and peat. Eliminating the support for peat used in heat and electricity production would, assessed in the context of the current methodology, lower environmental costs.

The reluctance to change policy is related to perceived high social costs. Only a small fraction of the total Finnish labour force is engaged in peat production, but this is often located in remote areas and forms the economic base for whole communities. Reducing public support for such activity could thus risk upsetting the current regional settlement pattern. If the objective is to preserve employment in these regions then it would be better to move away from supporting peat production to providing financial assistance for start-ups in the regions in question. Security of supply is also advanced as reason for keeping the capacity to use peat for cogeneration of electricity and heat.

Conclusions

The energy-intensive nature of the economy has resulted in high and rapidly growing greenhouse gas emissions that the government is committed to curb. The decision to allow the construction of a new nuclear power plant will reduce the expected overrun of the Finnish Kyoto emission target by half. The energy conservation and renewable energy policies designed to achieve the remaining reductions need to be implemented as cost effectively as possible. Finland was a pioneer in taxing carbon and has gained considerable experience in developing tax instruments for curbing emissions. However, for competitiveness reasons, it was faced with the need to transform the former energy tax system, which from the environmental point of view was well grounded, to one similar to that adopted in

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other countries. It continues to exempt peat from such taxes, so raising the cost of adjustment elsewhere in the economy. The difficulties faced in implementing this approach emphasise clearly the need for concerted action especially in the case of the sectors operating in international commodity markets. In this light, Finland may benefit from participating in the EU-wide permit-trading scheme and in such a way may further increase the cost effectiveness of emission abatement policies. There is nonetheless scope for better equalisation of carbon taxation across sectors by reducing the extent of discrimination. Notably the favourable treatment accorded to peat needs to be reassessed, taking into account the question of whether the long-term benefits from reforesting peat mires are significant. Any undesirable consequences for the regional settlement patterns should be addressed by targeted policies, such as support for new start-ups.

Air pollution

Main issues

Air pollution can have serious adverse effects on health, and the associated costs may be high for society. For the European Union as a whole, it is estimated that these welfare costs were as high as 3 to 3½ per cent of GDP in the mid-1990s.[121] In recognition of such adverse effects, governments throughout the OECD have established ambitious abatement targets for air pollutants both on a unilateral and multilateral basis. In the case of Finland, the main issue is to achieve such reductions without imposing unnecessary costs on producers and consumers.

Performance

Air pollution has been reduced in the past decade. Sulphur dioxide discharges fell dramatically during the 1990s. As other countries have achieved similar results, the extent of cross-border pollution affecting Finland has also been reduced. These two factors have led to the proportion of the total land area suffering from acidity levels above the critical load falling from 14 per cent in 1990 to 5 per cent by 1995. Moreover, ambient concentrations of sulphur dioxide were well below the World Health Organisation (WHO) and EU guidelines, and significantly lower than in most other OECD countries in the same year (Figure 33). On the other hand, emissions of nitrogen oxides and volatile organic compounds (VOCs) have fallen by less than those of sulphur dioxide, with the excess over the EU average of emissions (per unit of GDP) growing (Table 20). There has been little improvement in ambient concentrations for these two pollutants. In addition, emissions of particulate matter have not shown signs of trending down in all urban areas and such pollution remains a cause for concern in some localities. However, due to technological changes and policies already in place, the future concentrations of nitrogen dioxide, VOCs and particulate matter are likely to fall considerably.

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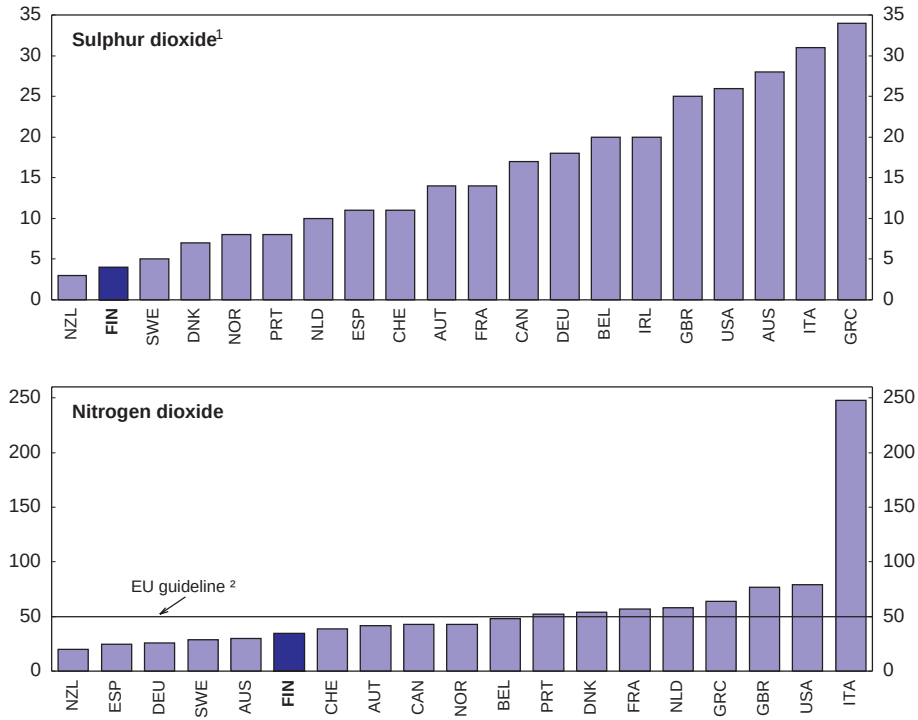
Figure 33. **Air pollutant concentrations in major cities of OECD countries**
Micrograms per cubic metre, 1995



1. The World Health Organisation guideline mean value for one year is 50.
2. Median of annual data of one hour values for concentrations.
Source: World Bank.

Policy

Within the framework of the UNECE Convention on Long-range Transboundary Air Pollution and the generally more demanding European Commission directives, the Finnish authorities are committed to achieve a reduction in emissions of various air pollutants by 2010 (Table 21). For sulphur dioxide, the sharp reductions in the 1990s have already pushed emission levels well below the target. For some other gases, substantial additional emission cuts will be required to reach the limit. Indeed, abatement efforts in the coming years will have to be intensified for nitrogen oxides, VOCs and ammonia to achieve EU targets. Given projected emission trends, existing policies and new policies envisaged in the Air

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Table 20. **Performance indicators: air pollution**

	Change in emissions in % of GDP[1] per year, 1990-99			Level of emissions in grams per dollar of GDP,[1] 1999			Improvement in productive efficiency, 1990-99	
	Sulphur dioxide	Nitrogen dioxide	Volatile organic compounds	Sulphur dioxide	Nitrogen dioxide	Volatile organic compounds	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
Finland	**–13.0**	**–3.8**	**–4.1**	**0.7**	**2.1**	**1.4**	**–65.3**	**–28.7**
Denmark	–14.2	–4.9	–5.2	0.4	1.6	1.0	–79.8	–38.9
Norway	–9.7	–2.8	–1.8	0.2	2.0	3.0	..	..
Sweden	–8.3	–4.3	–3.9	0.3	1.3	2.1	–26.0	..
Austria	–10.2	–3.5	–6.5	0.2	0.9	1.2	–77.2	–32.5
Belgium	–9.2	–3.6	–4.8	0.8	1.2	1.1	–64.4	–22.6
France	–8.3	–3.8	–5.1	0.5	1.2	1.4	–52.0	–37.8
Germany	–20.1	–7.1	–8.8	0.4	0.9	0.9	–85.2	–43.7
Greece	–1.4	–0.3	–0.2	3.6	2.5	2.6	–4.6	–22.3
Ireland	–8.1	–6.3	–7.9	1.7	1.3	1.0	–35.8	–19.0
Italy	–7.6	–4.3	–4.4	0.8	1.2	1.4	–52.8	–31.3
Netherlands	–10.1	–6.5	–8.8	0.3	1.1	0.7	–50.2	..
Portugal	–2.1	–0.9	0.1	2.4	2.3	3.1	–17.7	..
Spain	–5.8	–2.1	–3.5	2.2	1.7	3.6	–39.4	–14.4
United Kingdom	–13.9	–7.9	–6.6	1.0	1.3	1.4	–72.8	–51.0
Canada	–5.3	–2.8	–2.9	3.2	2.6	3.5	..	..
Switzerland	–6.0	–5.4	–6.3	0.1	0.5	0.9	..	..
United States	–5.5	–2.5	–4.7	2.0	2.7	1.9	..	..
European Union	–11.1	–5.0	–5.3	0.8	1.2	1.5	..	..
OECD Europe	–10.1	–4.7	–5.0	1.2	1.4	1.5	..	..
OECD	–7.3	–3.2	–4.6	1.6	2.0	1.8	..	..

1. Using 1995 purchasing power parities.

Source: Cooperative Programme for Monitoring and Evaluating of Long-Range Transmission of Air Pollutants in Europe (EMEP); World Health Organisation and OECD.

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Table 21. **Selected commitments for emissions of air pollutants**

Kilotonnes

	2010		Emissions	
	UNECE[1]	European Commission	1990	1999
Sulphur dioxide	116	116	260	87
Nitrogen oxides	170	152	300	247
Volatile organic compounds	130	110	209	168
Ammonia	31	31	35	35

1. United Nations Economic Commission for Europe.

Source: Directive 2001/81/EC of the European Parliament and of the Council of 23 October 2001 on national emission ceilings for certain atmospheric pollutants, UNECE Convention on Long-range Transboundary Pollution: The 1999 Gothenburg Protocol to Abate Acidification, Eutrophication and Ground-level Ozone.

Pollution Control Programme 2010, adopted by the government in late 2002, will allow Finland to fulfil its international obligations.

Since Finland's EU accession in 1995, air pollution control has relied heavily on issuing licences to new point source emitters based on best available technology and regulations on fuel content (OECD, 1997).[122] Unlike a number of other countries, Finland has not introduced a trading scheme based on pollution permits for nitrogen dioxide.[123] The continued use of the licensing approach will in principal make additional reductions somewhat more expensive. These costs will, however, be partially compensated by climate change policy that will reduce emissions of air pollutants in addition to those of greenhouse gases. Nonetheless, there may be a potential for increasing the cost-effectiveness through the use of suitable economic instruments.

A significant portion of emissions of nitrogen dioxide and particles comes from mobile sources that are harder to incorporate into trading schemes. The fall in pollution from cars has been restrained by three factors: the high car registration tax that doubles the price of new cars, a 28 per cent reduction in the annual vehicle tax applied to older vehicles and lower fuel taxes on diesel relative to gasoline. The first two factors extend the economic life of older, more polluting cars thus delaying the introduction of pollution-reducing innovations such as the three-way catalytic converter. This latter factor will be somewhat reduced when the current law that the government has introduced to reduce the level of the car tax from around 29 per cent to between 24 and 26 per cent takes effect. The tax advantage for diesel fuel, which is one of the largest in the OECD area, is despite emissions from diesel fuel imposing proportionally larger externalities, especially in urban areas (Figure 34).[124] There is, however, an additional annual motor vehicle tax on diesel cars, but this tax is more than offset by the lower fuel tax after 18 000 kilometres, which is 5 per cent less than the annual average distance travelled in Finland (Finnra, 2002). Taxation has been used as a means of lowering

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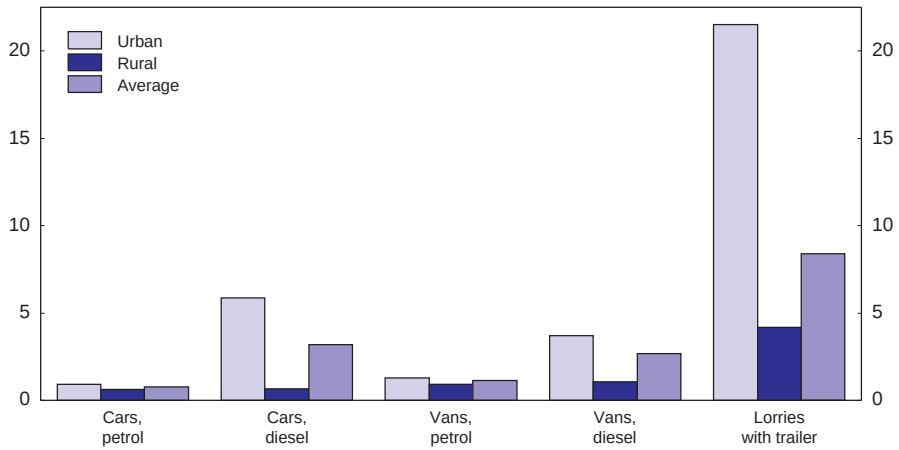
Figure 34. **Externalities of air pollution from different types of vehicles**
EUR per 100 kilometres, 1997



Source: Tervonen *et al.* (2001).

consumption of high sulphur content fuel oil. This policy may need to be widened to diesel to accelerate the introduction of very low sulphur diesel and gasoline, thus lowering particulate emissions and favouring the introduction of more efficient diesel particulate filters. Although the burning of wood in residential fires accounts for 40 per cent of emissions of very small particles in Finland, there is no policy to discourage such practises.[125] The Ministry of Environment is studying the possibility of introducing measures in this area.

Conclusions

Even though Finland will be able to fulfil its obligations through air pollution control measures already adopted or envisaged, there may be a potential for increasing the cost-effectiveness of policies through suitable economic instruments like a trading scheme for NO_x emissions. Other effective instruments to reduce particulate pollution would be to accelerate the replacement of existing diesel engines with new models that incorporate better filters and to ensure that the tax differential between diesel and gasoline adequately reflects externalities from, *inter alia*, air pollution and greenhouse gases. Imminent reductions in the tax on the purchase of new cars will facilitate a faster renewal of the fleet. At the same time road pricing should be considered in order to address the higher externalities from transport in urban areas. Finally, to contain the large emissions of fine

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particles from domestic fireplaces, the government should examine the introduction of appropriate policies to reflect the extent to which the burning of wood produces small particles.

Sustainable use of natural resources

Main issues

Finland is endowed with an abundance of natural resources and sustainable resource management is of key importance for the economy. The government plays an important role in resource management both as an owner of land and as a regulator. In some natural resource areas, such as water supply, there are no concerns over excessive exploitation. Given their significance to the Finnish economy, it is important that forestry resources are harvested in a sustainable way. This section focuses on the management of forestry resources.

Performance

Almost three-quarters of the land in Finland is covered by forests, one of the highest shares in the OECD area (Table 22). The annual harvest has never exceeded the new growth of wood, though the proportion harvested is one of the highest in the OECD area, while the production of roundwood has expanded appreciably. Nationwide performance figures mask significant differences across the northern and southern part of the country. The northern part of Finland has been characterised by a relatively low harvest as per cent of growth, reflecting a high share of conservation areas and an abundance of young forests. In the southern part of the country, on the other hand, forests have been harvested more intensively, raising concern over the loss of old woodland that serves as the traditional habitat for flora and fauna.[126]

Policy

The good performance in forest use has taken place against the background of predominantly private ownership of forest resources. Over half of all forests in Finland are the private property of almost one million small owners, 33 per cent are owned by the state, 8 per cent by business companies and 5 per cent by others. The policy of keeping forests as private and public property has avoided the problem of the commons, with owners managing their forestry assets on the basis of market-based timber prices. Management on the basis of market signals does not, however, take into account the various externalities associated with forests. These include storage of greenhouse gases, amenity or recreational services and the preservation of biodiversity.

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Table 22. **Forestry: performance indicators**

	Forest cover, %		Production of roundwood, thousands m³		Timber use, harvest as % of growth
	Latest available year	Annual change 1990-2000	1970	2000	
Finland	**75.5**	**0.0**	**45 130**	**54 263**	**83**
Denmark	10.5	0.2	2 293	3 086	59
Norway	39.2	0.4	8 542	8 173	41
Sweden	73.5	0.0	59 967	61 800	66
Austria	47.6	0.2	11 813	13 276	60
Belgium	22.2	–0.2	. .	4 510	85
France	31.4	0.4	37 853	50 170	68
Germany	30.1	0.0	37 246	49 106	37
Greece	22.8	0.9	3 046	2 171	60
Ireland	8.8	3.0	382	2 673	65
Italy	23.3	0.3	11 667	9 329	27
Netherlands	9.2	0.3	945	1 039	62
Portugal	37.9	1.7	6 370	9 450	77
Spain	32.3	0.6	13 653	14 810	52
United Kingdom	10.5	0.6	3 492	7 451	68
Australia	19.4	–0.2	12 033	30 493	. .
Canada	45.3	0.0	121 625	187 444	44
New Zealand	29.5	0.5	8 706	18 898	57
Switzerland	31.7	0.4	4 190	10 428	53
United States	32.6	0.2	327 945	500 434	60

Source: Food and Agriculture Organisation of the United Nations (FAO), Forestry information system (FORIS) database and OECD.

Policies governing forestry management have increasingly focused on dealing with these externalities. The current National Forest Programme now pays greater attention than its predecessors to issues such as protecting valuable habitats and promoting biodiversity. Nonetheless, about four-fifths of all public support to forestry is devoted to subsidising and promoting wood production and use.

In safeguarding valuable habitats, current policy determines areas of particular importance and either limits the ability of owners to endanger their natural characteristics or purchases it from private owners.[127] In 2000, landowners of such designated areas received EUR 0.7 million compensation for not exploiting designated timber stands, while land purchases amounted to EUR 17 million and the management of conservation areas cost a similar amount. In October 2002, the government adopted a resolution on an action programme to secure the diversity of forests in southern Finland, where there is some concern of a lack of protection. The programme includes voluntary protection of old forests through competitive bidding from forest owners to protect areas, and recommends the use of pilot projects to assess the costs of further protection.[128]

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To implement forest policy, an elaborate system of notification and monitoring has been established. The regional forestry agencies review over 100 000 notifications of felling and inspect a sample of over 5 000 sites annually. The annual cost of the supervisory structure has risen recently to over EUR 40 million, of which EUR 13 million is related to notification and inspection. The regional forestry agencies also support independent inspections conducted by the sustainable forestry certification schemes, which cover 95 per cent of Finnish forests.

Conclusions

The sustainable use of forest resources is well organised. The considerable expansion of harvesting has not led to overexploitation of forestry resources. In examining the expansion of protected forest areas in southern Finland, the government should weigh carefully the benefits and costs of this measure. The implementation of the proposal to use competitive bidding would be an innovative way to expand forest protection at least cost to society.

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Notes

1. Both over the last five years and the last ten years the standard deviation of the seasonally adjusted quarter-on-quarter GDP growth, as well as the change in this growth rate between quarters, is higher in Finland than for any other euro area country. This may partly reflect greater variation in climatic conditions; quarterly GDP growth is also relatively volatile in Iceland, Norway, Sweden and Denmark compared to other European countries. In more recent years the marked fluctuations in the output of the electronics industry, where the ending of the production of a particular model is only followed by production of the more up-to-date replacement model after a delay of several months, has been a major factor behind volatility.
2. In the year 2000 Asia accounted for 9½ per cent of Finnish goods exports which is a higher proportion than for any euro area country, and compares to the euro area average of 6 per cent. Asia accounted for 15 per cent of Finnish electronic exports. Non-OECD Europe accounted for 9 per cent of goods exports in 2000 (only exceeded by Greece in the euro area) compared to the area-wide average of 3 per cent.
3. This is a feature of an increasing number of OECD countries (OECD, 2002g).
4. The reduction in the saving ratio in 2000 and the bounce-back in 2001 are mainly statistical artefacts as steep revenue increases due to stock options and capital gains in 2000 are not included in the national accounts measure of income, while taxes paid on these revenues are.
5. Indeed, the unemployment rate of older workers is in some sense understated because, as discussed in Chapter II, it excludes those that draw the "unemployment pension" from age 57.
6. About 5½ per cent of those aged 20 to 24 were attending education at the same time as being classified as unemployed in 2000, the highest proportion in the OECD, and compared to an OECD average of only 1½ per cent (OECD E*mployment* O*utlook* 2002, Chapter I).
7. As argued in the OECD E*mployment* O*utlook* 2002, Chapter I, student unemployment is not usually regarded as a policy problem, and policies do not usually seek to influence it.
8. Measurement of the non-accelerating inflation rate of unemployment (NAIRU) and potential output are particularly problematic in the case of Finland. The OECD's standard method for calculating potential output would suggest that output was above potential in 2000. However, because much of the growth in this year was generated in the ICT sector, where productivity levels are very high, fast growth may be possible without generating much tension in the labour market. Estimates of the NAIRU suggest that there may have been some slack in 2000.
9. The agreement, reached in late 2000 covering some 90 per cent of Finnish workers, called for a 3.1 per cent wage increase in 2001 and a 2.3 per cent rise in 2002.

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10. There is some ambiguity in this formulation, as discussed later, given that relevant deflators are not clearly specified.
11. GDP growth over the period 1998 to 2002 is likely to have averaged just over 3 per cent per annum.
12. Based on the 2003 Budget proposal.
13. See Eschenbach and Schuknecht (2002).
14. Several reforms implemented during the 1990s will reduce spending on all pension schemes by an estimated 8½ per cent of the wage sum by 2030 (see Box 5 of the 1999-2000 OECD *Economic Survey of Finland*).
15. The 1999-2000 OECD *Economic Survey of Finland* included a special chapter on ageing.
16. See, for example, Blöndal and Scarpetta (1998) and OECD (2002b).
17. For further details of these calculations see OECD (2002b).
18. Pension payments are reduced by 0.4 per cent for each month of early retirement before the statutory age of retirement of 65 and increased by 0.6 per cent for each month of work beyond the age of 65.
19. The ceiling is currently equal to 60 accrued percentage points of pensionable wages. For a person starting work at 23 the ceiling discourages work after 61 as further employment will lead to additional pension contributions without accruing further pension rights. A second reason why the change in pension wealth is negative is that pension income is taxed.
20. Johnson (2000), examining experience of pension reforms across a number of OECD countries, finds "substantial" empirical evidence that both the replacement rate and the implicit tax on continued work influence the participation rate of male workers over the age of 60. He further argues that separating out the contribution from the two effects is difficult because of collinearity. However, on the basis of particular country episodes he suggests that in some cases the effect of the implicit tax rate on continued work has been more important and in other cases the replacement rate effect has been dominant.
21. This partly reflects policy changes in recent years that have tended to make pension schemes more actuarially neutral.
22. The calculation of the proportion of retirees on disability pension here includes those on "individual early retirement pension", which is a form of disability pension. The proportion of all pension recipients on early retirement schemes is much smaller when considered as a proportion of the "stock" of all pensioners (about 30 per cent in 2001), than taken as a proportion of the "flow" of new retirees (about three-quarters in 2001), as shown in Figure 12. This is because eventually most recipients of early retirement pensions end up taking the regular old-age pension at the age of 65.
23. In Figure 13 the replacement rate shown is that when the person first enters the unemployment or disability scheme. For some countries, although not particularly Finland, the level of unemployment benefit might diminish over time, which would reduce the replacement rate and the incentive to withdraw from the labour market.
24. The net replacement rate for a worker entering the unemployment pipeline at the earliest opportunity (at present 55, after the reform 57) and working to the earliest age possible to draw an old-age pension (at present 60, after the reform 62) are likely to be similar. Under the current and reformed systems, pension rights will accrue at about the same rate over this period. However, a major difference between the present and

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reformed systems is that there will be more incentive to continue work after the age of 62, because of the abolition of the pension ceiling and the higher accrual rate. Thus, the incentives to work after the age of 57 will be improved as a result of the reform, the longer the period of work history, and the more the intention is to work through into their mid-sixties rather than retire in their early-sixties.

25. In practice, employers' contribution rates only depend on the age of their employees for firms with over 50 employees, see the chapter on taxation in the previous *Survey* for details.

26. Alternatively, assessments of fiscal sustainability may focus on a measure of net general government financial liabilities (OECD *Economic Outlook* 2002, No. 72, Chapter IV). However, the precise definition of such measures needs to be carefully examined, particularly when making international comparisons. For example according to Annex Table 34 of this publication, the current position of Finland appears very favourable – being one of the few OECD countries with substantial general government net *assets* equivalent to 42 per cent of GDP in 2001. However, this figure includes the assets of pension funds, equivalent to 48 per cent of GDP in 2001, but not corresponding pension liabilities. Thus, a more meaningful comparison is between the combined net financial liabilities of central and local government of Finland (but excluding pension funds), with the figures reported for general government for other countries. On this basis the position of Finland is much less favourable with net financial *liabilities* equal to 6 per cent of GDP in 2001.

27. The Bank projections incorporate an assumption that as the population ages, the demand for labour declines more slowly than supply. This accounts, perhaps optimistically, for a rise in the average employment rate by 4 percentage points and a fall in the unemployment rate by about half that to a level of 6 per cent.

28. This chapter is the latest in a series of country-specific reviews that form part of the OECD's ongoing project on public expenditure. To-date the other countries covered in the reviews include: Canada, the Czech Republic, Denmark, Germany, Greece, Hungary, Italy, Japan, Mexico, New Zealand, Norway, Poland, Sweden, Switzerland and the United Kingdom. General assessments of public expenditure issues have also been made as part of the project, see Atkinson and van den Noord (2001) and Lundsgaard (2002).

29. The division between merit good expenditure and income transfers in is based on gross expenditure. In international comparisons made by the OECD of net social expenditure that take into account the tax treatment of transfers and tax expenditures, Finland and other Nordic countries continue to rank highly (Figure 16 and Adema, 2001).

30. For a broad evaluation of public-sector efficiency see Ministry of Finance (2002d, 2002e).

31. For example, examination of productivity developments in Finnish health centres by Luoma and Järviö (2000) found that half the health centres had improved productivity between 1988 and 1995 by at least 6 per cent and a quarter of them by at least 17 per cent.

32. Finland has a similar tax to GDP ratio as Austria, Belgium, Denmark, France, Norway and Sweden. In all of these countries the ratio is at least 45 per cent. Most other OECD countries have tax to GDP ratios well below 40 per cent (OECD, 2002h).

33. By the end of 2003 Finland must adjust to freer regulation on the importation of alcohol (and tobacco) products from other EU member states. Also, the current system of

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taxation applied to new and second-hand cars has been deemed illegal by the European Union as a result of a court case. Compliance with the decision will result in reduced import taxes on second-hand cars. Compliance on both alcohol and car taxes is estimated to reduce taxes by approximately 1 per cent of GDP, largely due to the loss of alcohol tax revenue.

34. The 2002 Article IV assessment by the International Monetary Fund also advocates a stronger multiyear approach in the budgeting process (IMF, 2002).

35. See the forthcoming OECD regulatory reform review (OECD, 2003) for further discussion on this issue.

36. In the past this division of responsibilities reportedly resulted in a revenue-saving "carousel" strategy by municipalities. It was apparently not uncommon for municipalities to wait until the long-term unemployed were close to the end of their KELA benefits and then offer them a job (typically subsidised by central government) until they re-qualify for KELA benefits, thus avoiding payment of municipality social assistance. This problem has since been rectified.

37. The principal layer of regional government consists of 20 regions (*maakunta*) each headed by a regional council (*maakuntaliitto*). Established in the late 1990s, they are based on traditional Finnish provinces and form part of the EU's regional structure. They have relatively small budgetary requirements, which are principally funded by annual membership fees, paid by the member municipalities. One of the principal roles of the councils is to represent the region internationally, most importantly with regard to EU regional funding and policy. The regional councils are also involved in certain areas of planning, such as roads and water services.

38. Moisio (2002) compares municipality finances before and after the change to block grants and indeed finds that under the new system the municipalities are making more careful economic decisions. For example, under the new system there is a stronger tendency for municipalities to simultaneously assess expenditure and revenue as opposed to a "spend then tax" approach that characterised the matching grant system.

39. Central government typically steps in with discretionary financial support for those municipalities in financial difficulties. In recent years the number of municipalities which have been unable to cover operating expenditure with operating income has increased. However, these negative accounts remain relatively small and the size of the discretionary state grants used to cover them does not pose a significant threat to the independence of municipalities or to overall expenditure control. According to Ministry of Interior figures, in 2001, 114 municipalities received discretionary state-aid grants whose valued totalled about EUR 60 million, a relatively small sum in relation to total municipal income of EUR 27 billion.

40. Some cuts in central government grants have been because of policy changes that have reduced municipalities' costs. For example there have been cuts in contributions municipalities have to make to the Social Insurance Institution of Finland, KELA.

41. Although there has been some improvement recently, there is still about a two-year lag between a given fiscal year and the equalisation adjustment. A more radical proposal by Pekkarinen (2001) for a real-time equalisation between municipalities was discussed, but not taken up by policymakers.

42. Also, despite a good deal of similarity in the systems, Finnish municipality funding has not exhibited the Danish situation of upwardly spiralling expenditures, funded by increases in municipal tax rates.

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43. Indeed in a meeting with OECD staff one municipality official stated that tax competition was "not a sport".

44. Norway has a similar structure of municipalities as Finland and it is interesting to note that Langørgen *et al.* (2002) estimate that it would be possible to save NOK 4.1 billion, or about ¼ per cent of GDP, if Norway's 434 municipalities were merged into 90 large-scale administrative units.

45. According to the Ministry of the Interior only about 30 mergers have taken place in as many years. About 100 municipalities, however, have initiated negotiations on possible mergers.

46. Indeed, such considerations, *inter alia* arguably motivated the European Union to incorporate the principle of "subsidiarity", *i.e.* that public policy and its implementation should be assigned to the lowest level of government with the capacity to achieve objectives. See the OECD framework paper on public expenditure, Atkinson and van den Noord (2001). For further discussion on the advantages to democracy in local government as well as general discussion on fiscal federalism see Oates (1999).

47. Recognising a high degree of inertia in moving away from a tradition of in-house provision, the government in 1998 launched the Government and the Market Project that aims to encourage competition between private and public producers. Recently the project has focused on the role of the purchasers, *i.e.* for the main part municipalities or joint municipality boards. A survey of 15 municipalities confirms that private provision has remained limited (see 2002 S*urvey*), although nearly all municipalities claimed to be working on the issue (FCA, 2002).

48. For further discussion of competition issues in the public sector see Chapter III of the OECD's forthcoming regulatory review of Finland (OECD, 2003).

49. In principle the public and private producers of services are treated in the same way in taxation, as it is not the type of producer but the type of service which determines the VAT-exemption. In the case of services subject to VAT both the municipalities and the private producers get deductions for VAT already paid on inputs so as to avoid double taxation. However differential treatment of private and public sector producers can occur in the case of VAT-free services. In this case public producers get a refund of 5 per cent which is an estimate of the average amount spent of VAT on inputs for services while private producers get no such refund. School meals are an important example. Due to the VAT on inputs included in the private sector prices it can be cheaper for municipalities to arrange the meal services in-house.

50. A survey monitoring the use of various management tools to improve the efficiency and quality of the public sector was conducted in 2000. The results, covering both the state and municipal sector showed, for example, quite extensive use of client feedback systems, whilst service charters were used rather less commonly.

51. Funding formulas for higher-education establishments that include the number of graduations have been introduced elsewhere, *e.g.* Norway and Switzerland. As in other countries, the universities in Finland also subject themselves to peer review. Given the small number of institutions the reviews are, commendably, conducted by non-Finnish institutions.

52. The guidelines for evaluating public-sector activities have been developed by the Association of Finnish Local and Regional Authorities, the Ministry of the Interior and the Ministry of Finance as part of the National Productivity Project (T*uottavuudella tulevaisuuteen*).

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53. Indication of a more open policy to information on public services may also be seen in a recent overview of social policy (Ministry of Social Affairs and Health, 2002a) in which it is reported that information on hospital waiting times may be made publicly available on the internet.

54. An index of the overall maturity of e-government services places Finland fourth behind Canada, Norway and the United States in a comparison of nine OECD countries (Ministry of Finance, 2002e). In other evidence the European Commission's *Flash Eurobarometer* survey indicates Finnish computer users to be among the most frequent users of e-government among European countries.

55. The new general portal for the Finnish government is located at *www.Suomi.fi*

56. One of the pioneering municipalities in this field is Espoo *www.Espoo.fi*, one of the largest cities in Finland.

57. For an overview of public-sector employment in Finland see Ministry of Finance (1999).

58. According to data collected in spring 2002 by the Ministry of Finance, only 14 per cent (15 700 employees) of central government employees were in the new wage system.

59. A recent OECD project looking at public employment issues across eleven member countries reported current and prospective shortages of staff in certain areas to be a common issue. As in Finland, demographic trends in the civil service are typically fuelling the problem (OECD, 2001d).

60. There are also regional considerations in assessing the future demand in public services. In growth centres (especially in the Helsinki area) there is a need to increase the number of child-care staff and teachers, whereas in the more peripheral areas which are losing working-age population there is a need to invest in elderly care (Vuorento, 2001).

61. The temporary contracts in the public sector are used for new employees but also where staff with a permanent post move to another position. About a quarter of public-sector employees hold temporary contracts. However, the share of these who also hold permanent contracts is unknown.

62. The Finnish Medical Association (*Suomen Lääkärilüitto*, SLL) did not sign the two-year centralised incomes policy agreement concluded in December 2000 and proceeded to negotiate separately with its employer, the Local Authority Employers Association (*Kunnallinen Työmarkkinalaitos*, KT). A 20-week strike ensued with a two-year wage deal being struck in August 2001. The pay increase is worth about 10½ per cent (nominal) over two years and is accompanied by improvements in working conditions.

63. There are approximately 270 health centres for the 448 municipalities.

64. In 2000 an important change to the structure of the hospital districts in the Helsinki region was made when the Helsinki and Uusimaa districts were merged along with the Helsinki University Hospital into a single hospital district, HUS.

65. There is no legislation on the price system or price levels arranged between hospital districts and municipalities. Typically municipalities come to an agreement, rather than a formal contract, with the hospital district on the provision and pricing of services on an annual basis (Järvelin, 2002).

66. Fees for health and social services are stipulated in the Act on Client Fees in Social Welfare and Health Care.

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67. A recent review of client-fee policy (Ministry of Social Affairs and Health, 2002a), underscores the complexity and lack of transparency of client fees in the Finnish health system.

68. The OECD's *Health at a Glance* (OECD, 2001b) estimates the average annual rate of growth in per capita pharmaceutical expenditure to be 3.7 per cent in Finland over the period 1970 to 1998, a rate of growth similar to other countries.

69. A paper on the budgetary implications of ageing (European Commission, 2001) has projections which show that Finland may have one of the largest increases in long-term care costs among EU countries, equivalent to about 2 per cent of GDP between 2000 and 2050.

70. There are a number of biases towards the use of service housing for the elderly. Municipalities have a freer hand in pricing service housing (and auxiliary domestic services) compared with institutional care. Also, individuals living in service housing can receive various benefits from the national social insurance fund (KELA) (housing allowance, pensioner's care allowance and health care reimbursements) that are not paid to people in long-term institutional care. And, until recently service housing has received support from the Slot Machine Association and the Housing Fund of Finland.

71. The initiatives to improve elderly care services fall under the broad strategy outlined in the Target and Action Plan for Social Welfare and Health Care.

72. The possibility of significant costs and undesirable social outcomes from negative competition at the local government level leads some experts in fiscal federalism to favour the centralisation of social welfare.

73. For a comparison of pre-primary and primary educational enrolment patterns see OECD (2002d). The widely used system of day care and a tradition of a one year pre-school in Finland means that most children are exposed to public education for a considerable time before beginning compulsory schooling. Note that Norway recently lowered its school-entry age to six.

74. OECD (2002d) shows Finland as having one of the highest rates of entry to tertiary education among OECD countries.

75. The 1998 Universities Act obliges universities and polytechnics to engage in systematic evaluation and introduced an overseeing body, the Finnish Higher Education Evaluation Council for this purpose (Ministry of Education, 2000).

76. For evaluations of school efficiency see, for example Häkkinen *et al.* (2000) and Kirjavainen and Loikkanen (1998).

77. Source: Association of Independent Schools, *www.ykl.pp.fi/index.html*

78. The government has been promoting vocational education as part of its campaign to overcome labour shortages in certain professions. For example, vocational training is a prominent topic in the 1999-2004 *Development Plan* of the Ministry of Education (Ministry of Education, 1999). See also the National Board of Education's promotional Web site entitled "Occupy your time" *www.viisikko.fi/ammatillinengb/*

79. In 2002 Finland was second in the World Economic Forum competitiveness rankings (having been displaced from top position in the previous year by the United States). The same result was reported in 2002 by the International Institute for Management Development.

80. The large swings in Finland's productivity and output growth, particularly those associated with the recession of the early 1990s, make it difficult to separate cycle and trend.

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For this reason, more than is the case for other countries, estimates of current "trend" growth should be treated with caution.

81. For example, recent estimates by Jalava (2002) show that multifactor productivity growth has made by far the largest positive contribution to growth in non-residential market production in Finland since at least the mid-1970s. As a rule, the contributions from capital stock, capital quality, labour hours and labour quality have each been comparatively small; the main exception is the strongly negative impact of reduced working hours during the recession of the early 1990s.

82. Various other statistics highlight Finland's strengths in science and technology. The OECD *Science, Technology and Industry Scoreboard* (OECD, 2001c), shows Finland to have one of the highest shares of scientists and engineers in the labour force, a high number of patent applications per capita, and a large share of scientific and technical articles in engineering and technology.

83. In an evaluation of biotechnology activity, Statistics Finland estimate that about 7 per cent of business R&D was in biotechnology, with corresponding figures of 5 and 11 per cent in public-sector and higher education research (*www.stat.fi/tk/yr/ttbio_en.html*). Articles by the industry lobby group Finnish Bioindustries claims Finland's biotechnology industry to rank 6th in Europe with 10 per cent of Europe's biotechnology industry being Finnish (*www.finbio.net/articles/nature.htm*).

84. While high productivity growth in the financial intermediation and insurance sector may be partly due to the application of new technology, it may also reflect measurement problems. For example, one reason for high productivity growth may be the use of interest margins as an indicator of financial intermediation services, or the use of an output measure for the insurance industry based largely on the difference between premiums and claims.

85. Eurostat's approach to service sector measurement is outlined in its *Handbook on Price and Volume Measures* (Eurostat, 2001). Methods and results of output and productivity using alternatives to the national-accounting approach are discussed in Statistics Finland's annual productivity report (Statistics Finland, 2002).

86. Wireless local area networks (WLANs) provide access to the World Wide Web and in principle can be much faster than UMTS but currently only within a range of about 100 metres. So called "Wi-Fi" is not only being set up by individuals (who often allow free access to those within the range) but also by some major fixed connection network companies such as British Telecom (BT), Telia of Sweden and Korea Telecom which aim to provide subscription-based access. They are being set up in a variety of circumstances such as an alternative to fixed lines in offices or as a service in hotels and cafes. While WLANs do not allow truly mobile access, proponents argue that users are typically static when they are using services such as e-mail and Internet.

87. Interestingly, however Finnish policy does not provide special tax-treatment for R&D (OECD, 2001c). This could be considered surprising given the importance of the R&D-rich ICT sector.

88. The extent to which the standardised unemployment exceeds the EU average is somewhat exaggerated by the large number of students that are counted as unemployed, as discussed in Chapter I.

89. The ratio of the employment rate of those with low-educational attainment to the aggregate employment rate is also much lower in Finland (71 per cent in 2001) than the EU average (77 per cent), although there are some countries (Germany, Belgium and Austria) where this ratio is even lower (European Commission, 2002).

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90. First the central confederations agree on the general principles applicable to wage increases and other changes to working conditions. Afterwards individual trade unions and employers' federations negotiate collective agreements for each branch. Finally agreements are made at the firm level.

91. Nevertheless there are some countries that have a highly compressed wage structure, for example Sweden and Denmark, where the difference between the unemployment rate of those with low educational attainment and the aggregate unemployment rate is relatively small.

92. At the same time, by international comparison, the relative after-tax wage of skilled service sector jobs requiring many years of education is low relative to that in industrial sector jobs (Kouvonen, 2002).

93. The degree of wage compression as measured by the ratio of the full-time earnings of the lowest decile to the median (the measure used in Figure 29) has increased since the early 1990s.

94. Average and marginal tax rates are high across all wage levels and family types. In 2001, for a single person on two-thirds average wages the average tax rate (income tax plus employee and employer contributions less cash benefits) was 41 per cent, whereas the marginal rate was 53 per cent (ninth and fifth highest, respectively, across the OECD). For a married person with two children the average and marginal tax rates were 39 and 57 per cent (sixth and fifth highest across the OECD). See OECD, 2002e.

95. There is also evidence that the elasticity of labour demand with respect to wages is highest among the least educated (Tuomiaro, 2001).

96. The number of people employed by municipal employment subsidies has decreased during the last six years from 30 000 to 14 000 while subsidised employment in enterprises has doubled from around 7 000 to 14 000.

97. The pilot reform started at the beginning of 2002 and will last until the end of 2004.

98. Almost one-third of foreigners are currently working in health, education and social services, one-quarter work in wholesale, retail and accommodation and almost one-fifth in other services. Their employment in industry is relatively small, particularly compared to other European countries such as Germany (OECD, 2001a).

99. Nicoletti *et al.* (2001) estimated that the liberalisation of the gas, electricity, postal, telecoms and transport sectors between 1978 and 1998 boosted business sector employment by close to 2 per cent, which is among the largest gains in the OECD.

100. This section largely draws on Chapter III of OECD *Reviews of Regulatory Reform: Finland* (OECD, 2003).

101. The FCA cannot apply the EC Treaty's Articles 81 and 82 directly, but Finland's competition policy complies with Community legislation in practice. Finland is prepared to carry out the necessary legislative changes to apply Articles 81 and 82.

102. The Competition Council was not formally a court and only its chair was a judge.

103. This section is based on FCA (2002) and OECD (2003).

104. The exclusion is not absolute, but phrased in a way to cover, and thus exempt, the operation of laws to support producer incomes.

105. A recent investigation of the State Audit Office revealed that subsidies distort the profitability of farms. For example, milk production should be the most profitable sector, but shows the poorest profitability. Without subsidies, dairy farmers would be

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more profitable than pork or grain-producing farmers, but the subsidies reverse this order.

106. Sonera has recently merged with Telia, the Swedish telecoms company. The Finnish state owns about 20 per cent of the merged company and the Swedish state 45 per cent.

107. These estimates are based on the current methodology of the United Nations Framework Convention on Climate Change (UNFCCC) that takes into account only the emissions from the combustion of peat and does not analyse the entire life cycle of the greenhouse gas balance of peat. The Finnish authorities have launched a comprehensive research programme with the view to provide a better knowledge of this complicated issue and, if justified, to influence the further development of the methodology used under the UNFCCC.

108. The costs include a fee to the state nuclear waste management fund. This fund should contain sufficient money to meet the costs of decommissioning, but until the first major plants are decommissioned there remains uncertainty about actual costs and the long-term implications.

109. The projected reductions in carbon dioxide emissions are 3.5 million tonnes from energy saving and efficiency and 4.5 million tonnes from the renewables programme.

110. A central goal is to increase the share of renewable energy in primary energy use to 27 per cent in 2010 from 21 per cent in 1995 (Ministry of Trade and Industry, 2000). Electricity generated from renewable resources, with the exception of hydropower, receives a tax refund. The refund is EUR 6.9 per megawatt hour for wind-power (95 per cent of the standard tax rate on electricity), EUR 2.5 per megawatt hour for waste fuel and EUR 4.2 per megawatt hour for other sources. On the basis that this electricity replaces coal-generated capacity with an emission factor of 0.77 tonne of carbon dioxide per megawatt hour, the cost of this measure for wind energy is EUR 33 per tonne of carbon emissions avoided.

111. The programme is extremely small. Land eligible for the subsidy amounts to 52 km^2 (0.02 per cent of the forest area) and total payments are around EUR 2 million annually. The subsidy is paid in three parts: for managing the forest area, harvesting of young trees, and chipping the wood. The latter two subsidies are equivalent to EUR 11 per m^3 of wood, equivalent to over EUR 40 per tonne of carbon abated.

112. On 1 January 2003, the tax rose from EUR 15 per tonne to EUR 30 and will increase again in 2005 to EUR 50 per tonne.

113. While energy audits are included in the voluntary agreements they remain separate programmes, preventing the aggregation of results. By 2001, the estimated cumulative annual abatement reached between 614 and 789 thousand tonnes of carbon dioxide for the voluntary agreements and between 156 and 202 thousand tonnes of carbon dioxide for energy audits, depending on the carbon intensity of the fuels being replaced. Total expenditure on the voluntary agreement programme (1998-2001) was EUR 5.4 million and on the energy audit programme (1992-2001) EUR 13.1 million.

114. Between 1996 and 1999, tax revenues from labour had been reduced by EUR 2.3 billion, which was partially counter-balanced by higher energy tax revenue of EUR 0.6 billion.

115. EU single market legislation prevented the imposition of a border tax on imported electricity.

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116. Following tax rises introduced at the beginning of 2003 the tax on electricity rose to EUR 4.4 per megawatt hour for industry and EUR 7.3 per megawatt hour for other sectors.

117. Current estimates of the externalities caused by the use of peat as fuel are based on Pingoud *et al.* (1997).

118. In addition, the public health externalities of peat are 80 per cent higher than those of coal, due to higher emissions of other air pollutants (*op. cit.*).

119. The absorption of carbon dioxide is based on a study by Mälkki and Frilander (1997).

120. Due to the low heating value and large volumes required, the transport of peat to a power generation plant is uneconomic for distances greater than 100 kilometres.

121. See for example Sommer *et al.* (1999) or MIRA-T (2001).

122. Improved air quality in Finland has also been pursued by concluding bilateral agreements with Estonia and Russia to reduce cross-border pollution in particular hotspots in Finland.

123. By 2003, the United Kingdom, the Netherlands and the United States will have such programmes operational. Slovakia will start a sulphur dioxide programme in 2004.

124. In addition, due to the higher carbon content of diesel fuel, greenhouse gas emissions per litre of fuel are 14 per cent higher than for gasoline. However, due to the greater fuel efficiency of diesel engines, emissions per kilometre travelled are lower than for gasoline engines. This latter point should give incentives for individuals to purchase diesel engine vehicles up to the point when the investment is justified by the saving in external costs, without the need for additional fiscal incentives.

125. Small particles can also be generated from the gritting of roads. So far policy moves have been limited to reduce the production of particulates associated with the spring thaw. Heightened levels of particulate matter in spring arises due to the gritting of roads during winter.

126. The government has completed an inventory of forests that are important in terms of conservation.

127. Around 11 per cent of forestland is protected.

128. There is a need to ensure that the further expansion of protected forest areas will be cost-effective. Willingness-to-pay studies show this declines as the area under protection increases (Mäntymaa *et al.*, 2002).

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Glossary of acronyms

ALMP	Active labour market policy
CO_2	Carbon dioxide
EU	European Union
EUR	Euro
FCA	Finnish Competition Authority
FSA	Financial Supervision Authority
GDP	Gross domestic product
GHGs	Greenhouse gases
ICT	Information and communication technology
KELA	Social Insurance Institution of Finland
km	Kilometre
MTI	Ministry of Trade and Industry
NOK	Norwegian kroner
PTR	Potential tax revenue per capita
R&D	Research and development
UMTS (3G)	Universal mobile telecommunications systems (third generation mobile telephony)
UNECE	United Nations Economic Commission for Europe
USD	United States dollar
VAT	Value added tax
VOC	Volatile organic compound
WLAN	Wireless local area network

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Annex I

Examples of the pension reform effects on individual pension levels

The Central Pension Security Institute (2002) has estimated that the effect of the pension reform will raise the level of pensions in the private sector by 15 per cent on average compared to the present system. However, there are substantial differences between individuals depending on a variety of factors. To illustrate these differences the Institute has provided a number of representative examples (Table A1). It is important to note that these estimates do not include the effect of the life expectancy adjustment. To the extent that life expectancy increases, the generosity of the new system relative to the current 2001 system would be reduced.

The length of working career and its timing during the life cycle affect the pensions both in the present and in the new system. In these examples the career span varies between 44 years (case 1) and 28 years (case 6). In the case of a long working career the new system gives a clearly better pension level than the present system. The increase in the pension due to the reform is smallest in the case of a career in which the salary is "rising" more strongly than normal and largest when the salary is "declining" over the course of a career. This is because the present system uses the last ten years of any employment relationship as the basis for calculating the pension rather than lifetime earnings.

Comparing cases 1 and 2 shows that working between age 63 and 65 brings a bigger increase in pension in the new system than in the present system. This is due to the higher accrual rate from the age of 63. Also a normal career of about 35 years clearly gives a better pension level in the new system compared to the present system. But compared to cases 1 and 2 a shorter working career (case 3) gives a lower pension.

In case 4 the effect of the pension reform is much smaller than in the previous cases. There are two reasons for this: retirement takes place before the higher accrual rates start having an effect; also having just one job during a career is an advantage in the present system, but not in the new system.

Case 5 has the same career length as case 4, but the pension reform has a much more positive effect. This is because the new system will penalise interruptions to a working career by much less because: pension rights still accrue during some non-work periods (*e.g.* childcare); and, as in case 4, having more than one job in a career is not a disadvantage in the new system.

In case 6 the working career is much shorter than in the other cases due to the non-working periods and early retirement on disability pension. Even in this case the new pension system gives a better pension level than the present system.

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Table A.1. **Pension payments under various circumstances**

EUR per month

Wage development	Last salary	Pension			
		2001 system	New system	New system as % of the last wage	New system as % of 2001 system
Case 1. Working age 21-65, 3 jobs					
Normal	2 306	1 360	1 678	73	123
Declining	1 770	1 203	1 515	86	126
Rising	2 767	1 482	1 744	63	118
Case 2. Like case 1, but retirement at age 63					
Normal	2 310	1 341	1 447	63	108
Declining	1 788	1 173	1 330	74	113
Rising	2 672	1 431	1 467	55	103
Case 3. Like case 2, but also non-working periods (35 years of work)					
Normal	2 310	1 139	1 239	54	109
Declining	1 788	941	1 122	63	119
Rising	2 672	1 202	1 267	47	105
Case 4. Studying for 5 years, working at age 27-63, 1 job					
Normal	2 310	1 310	1 302	56	99
Declining	1 788	1 116	1 185	66	106
Rising	2 672	1 398	1 331	50	95
Case 5. Working age 21-63, childcare for 2 times (lasting 6 years altogether), 36 years of work					
Normal	2 308	1 167	1 364	59	117
Declining	1 779	1 021	1 247	70	122
Rising	2 719	1 233	1 393	51	113
Case 6. Working age 21-58, 3 jobs, non-working periods (28 years of work)					
Normal	2 320	912	1 040	45	114
Declining	1 833	802	945	52	118
Rising	2 450	921	1 048	43	114

Source: Central Pension Security Institute (2002).

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Annex II

Calendar of main economic events

2002

January

Euro notes and coins are introduced.

All marginal central-government income tax rates are reduced by one percentage point while the lowest income bracket is increased considerably. In local-government taxation, the maximum deduction for earned income is raised further.

The new Occupational Healthcare Act comes into force.

Taxes and social security contributions are cut for shipping companies at an estimated cost of EUR 28 million.

Nokia Chairman and CEO Jorma Ollila sparks a new tax debate by expressing the need to cut taxes, especially the highest marginal income tax rates.

February

The Ministry of Trade and Industry decides to remove chief executives in state-owned enterprises from the boards of directors before the end of 2003 in order to improve corporate governance.

The Ministry of Trade and Industry announces that it is no longer searching for a strategic minority shareholder for the fully state-owned alcoholic beverage group Altia, the producer of Finlandia Vodka. At the end of 2001, Parliament blocked plans to sell a majority stake and the Ministry could not find buyers for a minority stake.

March

The earnings-related part of unemployment benefits increases from 42 to 45 per cent of the previous earnings in excess of the basic benefit.

The new Market Court is created as a result of the merger of the previous Market Court and the Competition Council.

The government agrees on spending limits for 2003-06. Central government outlays will fall by 0.4 per cent in volume terms over the same period but a central-government budget deficit is expected for 2003.

The Finnish Competition Authority approves the partnership of the fully state-owned company Vapo (wood-based fuels and peat) and the private company Metsäliitto Osuuskunta, under certain conditions. In November, the Ministry of Trade and Industry made

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a prior agreement on the sale of a third of the shares in Vapo to Metsäliitto.

The Finnish Competition Authority launches a dawn raid on asphalt companies after receiving information on price fixing.

The Finnish and Swedish partly state-owned telecommunication companies Sonera and Telia agree to merge. In June, the European Commission conditionally approves the merger.

The German energy group E.ON Energie AG becomes the majority owner of the City of Espoo electricity utility Espoon Sähkö. It acquires the 27.6 per cent stake of the partly state-owned electricity company Fortum and half of the 68 per cent holding of the municipality.

April

The Finnish state sells 2 per cent of its shares in Stora Enso (pulp and paper) to institutional investors for EUR 326 million, reducing its stake to 12.7 per cent representing 24.1 per cent of the shareholders' votes.

The government proposes to Parliament to extend, under certain conditions, a father's entitlement to paternity leave by two weeks from the present 18 days.

May

The Finnish state decides to sell its stake of 30.2 per cent in Partek (engineering) to elevator manufacturer Kone.

Elisa Communications decides to withdraw from the German mobile phone market after three loss-making years.

Ascension Day becomes a national paid public holiday as agreed by social partners in the 2001-02 central wage settlement.

The formal agreement between the Nordic competition agencies on notifying and co-ordinating actions comes into force.

The government decides to reduce its stake in the fully state-owned Avena (grain wholesale trade, handling and storage) by ceding shares to Lännen Tehtaat for EUR 47 million. Avena will be split and the State only keeps a 65 per cent share in Avena Siilot Holding. In July, the Finnish Competition Authority announces its investigation of the competition consequences of the acquisition.

Parliament approves the construction of a fifth commercial nuclear reactor. After this decision, the two ministers of the Green League withdraw from the five-party "rainbow" government.

A tripartite working group does not reach agreement on the replacement of the current temporary sabbatical leave scheme by a permanent one.

June

Through a public offering, the state reduces its stake in the energy company Fortum from 70.7 per cent to 60.8 per cent.

July

The partly state-owned telecommunication company Sonera writes down EUR 4.3 billion on its European UMTS investments. The value of its international UMTS investments after the write-down is EUR 72 million.

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August

Nokia announces a 5 per cent job cut for its networks division (900 employees worldwide including 60 in Finland).

In a report, the Minister of Transport and Communications concludes that Sonera's decision in 2000 to invest heavily in German UMTS licences was not discussed in the government or Ministry of Transport and Communications during 2000. The report launched a major discussion and the Chancellor of Justice started his own investigation into the role of the government in the decision of this majority state-owned company.

The partly state-owned Finnish-Swedish forest company Stora Enso booked a one-time impairment charge of EUR 1.15 billion to offset the large premium it paid in 2000 for the takeover of the American paper company Consolidated Papers.

In Transparency International's Corruption Perception Index for 2002, Finland is the least corrupt country of the 102 countries surveyed, for the third year in a row.

September

The social partners reached an agreement on the pension reform with the exception of the Confederation for Academic Professionals in Finland (Akava), which did not approve the proposal. The reform aims to raise the average age of retirement by 2-3 years by *e.g.* abolishing some forms of early retirement and by introducing a step-wise rise in accrual rates.

The budget proposal was sent to parliament. Government receipts for 2003 will amount to EUR 35.6 billion and spending to EUR 35.7 billion, with a deficit of EUR 155 million.

The EU Court of Justice decided that Finland must lower the taxes levied on used cars imported from other EU member countries.

Nokia and Sonera released their new third generation products in Helsinki. Nokia introduced its camera-mobile phone and Sonera its multimedia services (messages and calls with pictures).

October

The key trade union confederations announced their targets for collective wage bargaining. The Central Organisation of Finnish Trade Unions (SAK) announced it will be aiming for a 3.8 per cent wage increase for 2003 and 3.4 per cent for 2004. The Finnish Confederation of Salaried Employees (STTK) was targeting a 3.7 per cent increase for 2003 and a 3.5 per cent increase for 2004; and the Confederation of Unions for Academics Professionals (Akava) was aiming at similar increases.

The Bank of Finland governor Matti Vanhala criticised the collective bargaining system, which set off a bilateral discussion with the Prime Minister Paavo Lipponen, who had been supporting the ongoing wage negotiations.

November

Two official working groups published their reports on taxation. The Ministry of Finance's group, chaired by Mr. Lasse Arvela, proposed lowering the corporate tax rate from 29 to 25 per cent to meet the pressures of international tax competition. Also a 25 per cent dividend tax was proposed. The other group that was set up by the Prime Minister and chaired by the Economic Council secretary general emphasised more employment aspects and cuts in income taxes, especially for low-wage employees.

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The Ministry of Finance raised its GDP growth projection for 2002 from 1.3 to 1.6 per cent.

Finland was awarded second place in the annual Global Competitiveness Report of the World Economic Forum, displaced from last year's first position by the United States.

The new two-year wage agreement was reached providing a 2.9 per cent wage increase in 2003 and a 2.2 per cent increase in 2004.

The merger between Telia and Sonera was confirmed as shareholders of both firms accepted the merger and about 95 per cent of the Sonera shares were exchanged against Telia shares by the end of the extended offer period.

December

The new wage agreement was signed by most unions. It covers over 90 per cent of wage earners. The government announced that it will send its promised employment package to parliament. The package includes tax cuts and employment incentives worth EUR 112 million.

Trade in TeliaSonera shares started on the Helsinki Stock Exchange.

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